NETGEAR ENTERPRISE

User Manual

Insight Cloud Portal

202-12915-01
August 2026

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Introduction

NETGEAR Insight is a cloud-based management platform designed to simplify the setup and configuration of Insight-managed access points, switches, routers, security gateways, and mobile hotspots. This manual explains how to use the Insight Cloud Portal to manage your devices efficiently. Insight centralizes device management in the cloud and provides remote access, monitoring, maintenance, and scalable network management.

This chapter contains the following sections:

- Overview
- Insight subscription model
- Insight Cloud Portal and Insight App
- Insight Cloud Portal dashboard
- How Insight and the device UI interact with each other

Overview

NETGEAR Insight allows you to deploy, monitor, and manage supported devices from anywhere through the Insight Cloud Portal. It is intended for small businesses, IT managers, and managed service providers who require centralized control without relying on onsite IT resources.

Insight provides the following features:

- Cloud-based remote management of switches, access points, and routers
- Centralized visibility and control of your entire network
- Support for multiple organizations and locations
- Simplified device registration and setup
- Monitoring and alerts, including downtime notifications and WiFi activity tracking
- Automated device discovery to reduce configuration time
- Remote firmware updates and troubleshooting

To check if you can manage your device in the Insight Cloud Portal, see Supported Devices on page 266.

To check if Insight is available in your country, see Countries with Insight support on page 261.

Insight subscription model

Insight uses a single unified subscription model that supports Insight subscription bundles, monthly, one year, three year, and NETGEAR Partner subscription options. This model provides a simplified and consistent subscription experience across all Insight accounts. For information on how to configure subscriptions, see Manage Subscriptions on page 24.

Subscription Framework

The Insight unified subscription system supports the following subscription durations:

- **Insight Subscription bundle**: Devices purchased with a bundled Insight subscription include a 1, 4, or 6-year subscription term. You only need to purchase an additional subscription if you require more device licenses or your initial subscription has expired.
- **Monthly subscription**: You can purchase a monthly postpaid subscription.
- **Annual subscriptions**: You can purchase a one or three year prepaid subscription.
- **NETGEAR Partner (License Based)**: The NETGEAR Partner (License Based) subscription is a pre-paid license key subscription for Insight. You can only activate an LMS key when a monthly or annual subscription has fully expired. You can also activate an LMS key without having any previous subscriptions when you begin your free trial. Once activated, the NETGEAR Partner (License Based) subscription prevents further purchase of annual or monthly subscriptions.

NOTE: Legacy subscription types are discontinued. All pre-existing subscriptions are honored until their original expiry date.

NOTE: When you purchase a subscription, the Insight free trial ends immediately and the paid subscription becomes active.

NOTE: Prepaid annual and postpaid monthly subscriptions are not available in all countries. To view the list of supported countries, see Supported Countries in Insight on page 261. If Insight is not supported in your country, you must purchase these subscription types through a third-party vendor. You can also purchase a NETGEAR Partner (License Based) subscription through a third-party vendor.

Supported user types

The unified subscription model applies to Insight accounts and all associated user roles:

- Read-write users
- Read-only users
- Managers
- Secondary administrators
- Support mode users

Grace periods

When you decide to end a subscription, it is important to be aware of the following Insight grace periods for your specific plan. A grace period is a short window after your subscription expires during which your devices remain in the Insight system but are flagged for attention. This gives you time to renew before losing remote management access. If you renew within the grace period, your devices are automatically restored to managed status. If the grace period ends without renewal, your devices move to an expired state and remote management is no longer available.

- **Monthly Subscription**: 14 day grace period
- **Annual Subscriptions**: 30 day grace period.

> **NOTE:** Both monthly and annual subscriptions offer an auto-renewal (AR) service. Ensure you deactivate this setting before you terminate your subscription, otherwise no grace period is provided. For more information, see Manage Insight Subscriptions on page 24.

- **NETGEAR Partner (License Based)**: A 30 day grace period applies after NETGEAR Partner (License Based) expires.

Change subscription

If you want to change your Insight subscription, the following guidelines apply:

- **Yearly (prepaid) and monthly (postpaid) subscriptions**: You can change between plans (yearly, monthly, or NETGEAR Partner) only after your current subscription expires. You can upgrade from a one-year to a three-year subscription with a pro-rated charge. For other subscription types, changes can only be made after the current subscription expires.
- **NETGEAR Partner subscriptions**: If you want to change from the NETGEAR Partner subscription to another plan, you must contact NETGEAR support.

Insight Cloud Portal and Insight App

This manual refers specifically to the Insight Cloud Portal, but you can access the Insight cloud-based management platform in two ways, either through the Insight Cloud Portal using a web browser on your Windows-based computer, MAC, or tablet, or through the Insight App on a smartphone or tablet.

- **Insight Cloud Portal**: The Insight Cloud Portal lets you access and manage your Insight devices online from a web browser. The Insight Cloud Portal supports the following features for Insight-managed devices:
  - Guided installation and configuration
  - Secure remote access
  - Access to logs and traffic history
  - Feature parity with the Insight App for device configuration and management.
  - A granular dashboard on which you can customize how your Insight diagnostics display.
  - A layout that takes advantage of your computer's screen size to display more information at one time.
- **Insight App**: The Insight App is an application that is available for iOS and Android devices and supports the following features for Insight-managed devices:
  - Guided installation and configuration.
  - Four different ways to add a device to a network location, including scanning your network, scanning the device QR code, scanning the device barcode, and entering the

device serial number.
- Secure remote access
- Instant alerts for critical events
- Access to logs and traffic history
- Self-help and click-to-connect support portal

The Insight App and the Insight Cloud Portal are different interfaces into the same cloud-based management platform. The cloud-based management platform applies the configuration changes in the order that it receives them. However, we do not recommend that different users configure the same Insight network simultaneously, one using the Insight App and the other using the Insight Cloud Portal.

Insight Cloud Portal dashboard

General summary section

The **General Summary** section of the dashboard displays the following tiles:

Field	Values	Description
Organizations	Healthy, Critical	Shows the overall status of all organizations. Select the number in the **Healthy** or **Critical** field to view the status of individual organizations. An organization displays as Critical when one or more of its managed devices are offline or experiencing issues like connectivity, restarting, IP conflicts, or similar problems.
Devices	Online, Offline, Critical	Shows the overall status of all devices. Select the number in the **Online**, **Offline**, or **Critical** field to view the status of individual devices. A device displays as Critical when it is offline or experiencing issues like connectivity, restarting, IP conflicts, or similar problems.
Subscription Status	Active, In Grace, Expiring soon	Shows the overall status of all subscriptions. For more information about Insight subscriptions, see Manage Insight subscriptions on page 24.
What's New / Subscription Update	N/A	The top-right tile displays What's New in Insight and Subscription Update information.

Organizations summary section

The **Organizations Summary** section displays the following views:

View	Description
Top Organizations by Devices	Lists organizations ranked by total device count.
Top Organizations by New Devices	Lists organizations ranked by recently added devices.
Top Organizations by Clients	Lists organizations ranked by client count.
Top Organizations by Critical Alarms	Lists organizations ranked by number of critical alarms.
Top Organizations by Traffic	Lists organizations ranked by traffic. Use the menu in the top right of this section to filter by device type: AP (access point), PR (router), or SW (switch).

Dashboard controls

The dashboard includes the following controls:

Control	Location	Description
Search	Top-right	Finds any device, client, or feature in your environment. You can search for devices using their serial number, MAC address or device name.
Language	Top-right	Select a language for the user interface. For more information, see Select a language on page 15.
Notifications	Top-right	View alarms in your Insight environment. For more information on alarms, see Manage alarms on page 120.
Theme customization	Top-right	Select a light or dark theme, and font size for the Insight UI. You can also reset the Insight UI theme to its default settings. For more information, see Customize the UI settings on page 14.
Refresh button	Dashboard	Manually refreshes the dashboard. The dashboard also refreshes automatically every 30–60 seconds.
Help icon	Top-right	Opens the help menu. See Help menu options on page 6 for available tabs.
Account icon	Top-right	Opens the account menu. See Account menu options on page 6 for

Control	Location	Description
		available tabs.

Help menu options

Selecting the question mark icon in the top right corner displays the following tabs:

Tab	Description
FAQs	Displays answers to frequently asked questions about Insight.
Insight Knowledge Base	Displays information on common tasks in Insight. Select the question mark icon, then select **Insight Knowledge Base**.
Insight guided tour	Launches a guided tour of Insight navigation.

Account menu options

Selecting the account icon in the upper right corner displays the following tabs:

Tab	Description
Update Profile	Select this tab to view or edit your account information and login settings.
Support	Request technical support or grant access to technical support. For more information, see Grant technical support access to your Insight account on page 19.
About	Displays the current version of Insight, terms and conditions, and Open Source Licenses.
Logout	Logs you out of Insight.
Switch to Classic	Reverts to the legacy Insight user interface.

How Insight and the device UI interact with each other

For smart switches and routers: You can manage the device both through Insight and the device UI. That is, these management methods are not mutually exclusive but complement each other. Changes to Insight are synchronized to the device UI, and the other way around, changes to the device UI are synchronized to Insight. By default, Insight-mode is enabled in these devices. Bidirectional Insight is supported on the most recently released smart switch models, all Pro Routers and security gateways. For information on specific switch models, see kb.netgear.com/000066230.

NOTE: Synchronization between the device UI and Insight might take up to 15 minutes. During the synchronization period, do not make the same changes on both the device UI and Insight or conflicting changes on the device UI and Insight.

For all other devices: Insight provides a traditional, local device user interface that functions independently of the Insight cloud-based management platform. This hybrid model lets you manage your device either with the local browser interface or with Insight. However, if you intend to use Insight, we do not recommend that you set up a device in "offline" mode because any configuration changes are not pushed to the Insight cloud-based management platform and are therefore not reflected in the Insight App and Insight Cloud Portal. To manage a device through the Insight Cloud Portal, log into the device UI, and select the Insight-management mode.

Getting started with Insight

To begin to set up your Insight environment, you can create an Insight organization, network location, and add devices. You can also configure the user interface settings to your preference, and login settings.

This chapter contains the following sections:

- Create an Insight account
- Log in to the Insight Cloud Portal
- Add an organization
- Add a location
- Add multiple locations
- Add a device
- Interpret the green, red, orange, and gray circles next to a device
- Customize the UI settings
- Customize table settings
- Customize data filters
- Select a language
- Set up two-step verification for logging in

Create an Insight account

You can use one account for all NETGEAR apps and for the Insight Cloud Portal. If you already set up a MyNETGEAR account for another NETGEAR app, you can use that account to access Insight.

To create an Insight account and sign in to your new account:

1. Visit insight.netgear.com.
2. Select **Account Log In**.
   The Account Log In page displays.
3. Click the **Create one** link.
4. In the **Email Address** field, enter your email address.
5. In the **Password** field, enter a password.
6. In the **First Name** field, enter your first name.
7. In the **Last Name** field, enter your last name.
8. From the **Country** menu, select a country.
9. In the **Phone Number** field, enter your phone number.
10. Select the check box to accept NETGEAR's terms and conditions.

11. Click the **Sign Up** button.
    A confirmation email is sent to you. If you did not receive an email, click the **Resend Confirmation Email** link.
12. Verify the email to complete Insight registration.
    After you verify your email, your Insight account becomes active, and you are prompted to create an organization and location.
13. Create an organization. For more information, see Add an organization on page 10.
14. Create a location. For more information, see Add a location on page 11.
15. Click the **Next** button.
    A page displays to advise your Insight account is successfully created.
16. Optional: Click the **Start Guided Tour** button to learn more about the Insight Cloud Portal.

> **NOTE:** A 90-day free trial for Insight automatically starts once you add your first device.

Log in to the Insight Cloud Portal

To log in to the Insight Cloud Portal:

1. Visit insight.netgear.com
   The Account Log In page displays.
2. **(Optional)** If you do not have an Insight account, click the **Create one** link.
   For more information, see Create an Insight account on page 8.
3. In the **Email** field, enter your email address.
4. In the **Password** field, enter a password.

> **NOTE:** If you forgot your password, click the **Forgot Password?** button.

5. Click the **Sign in** button.
6. Alternatively, you can log in using a one-time code:
   a. Click **Login with one-time password** link.
   b. Enter your email address and click the **Send One-Time Password** button.
   c. Enter the verification code that was sent to your email, and click the **Verify Code** button.
   If you did not receive a code, click the **Resend Message** button. The code is valid for 30 minutes.

   If you want to switch another verification method, click the **Try Another Verification Method** button.

Add an organization

The Insight Cloud Portal is the main place where you group your company's locations, devices, and users so you can manage everything together in one account.

To add an organization using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Select the **Organizations** tab.
   The Organizations page displays.
3. Click the **Add** button.
   The Add New Organization page displays.
4. In the **Organization Name** field, enter an organization name.
5. From the **ISO Country** menu, select the country that the organization is in.
6. In the **Location Address** field, do one of the following:enter an address for the location or click the navigation icon to detect the location automatically.
   - In the **Location Address** field, enter any part of the address (such as street, building, area, or postal code) to detect the full address.
   - Click the navigation icon to detect the location address automatically.
   - Drag and drop the red pin on the map. The address automatically displays in the address fields based on the location of the pin.

   NOTE: The location must be within the ISO country you chose in step 5.
7. In the **Organization Image** section, do the following:
   a. Click the **Upload** button to add an organization image.
   b. Select the image you want to upload.

   NOTE: The following image formats are supported: JPG, JPEG, PNG, and GIF. The maximum supported file size is 5 MB.
8. Click the **Save** button.
   Your organization is created.
   A pop-up displays to ask if you want to create locations automatically.
9. Click the **Yes** button.
   For more information on creating locations, see Add a location on page 11.

Add a location

A network location is a collection of devices in the same physical location that use the same administrator password and can be monitored simultaneously in Insight. If you want to monitor and manage devices in more than one physical location, you must create a new network location for each physical location.

To add a location:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Click the **Add** button.
   The Add New Location page displays.
4. Select the **Single Location** radio button.
5. From the **Organization** menu, select an organization.
6. In the **Location Name** field, enter a location name.
7. In the **Device Admin Password** field, enter a new password for the device.
   This password replaces the existing admin password on each device that you add to the network location.
8. From the **ISO Country** menu, select the country that the location is in.
   Your browser automatically populates this field.

   The Time Zone field automatically displays that of the ISO country you select.
9. If there are multiple time zones in that ISO country, from the **Time Zone** menu, select the correct time zone.
10. In the **Location Address** section, do one of the following:
    a. In the **Location Address** field, enter any part of the address (such as street, building, area, or postal code) to detect the full address.
    b. Click the navigation icon to detect the location address automatically.
    c. Drag and drop the red pin on the map. The address automatically displays in the address fields based on the location of the pin.

    NOTE: The location must be within the ISO country you chose in step 8.

11. In the **Location Image** section, do the following:
    a. Click the **Upload** button to add a location image.
    b. Select the image you want to upload.

> **NOTE:** The following image formats are supported: JPG, JPEG, PNG, and GIF. The maximum supported file size is 5 MB.

12. Click the **Save** button.
    The new location is created.

Add multiple locations

To add multiple locations:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Click the **Add** button.
   The Add New Location page displays.
4. Select the **Multiple Locations** radio button.
5. From the **Organization** menu, select an organization.
6. In the **Number of Locations** field, enter how many locations you want to create.
7. In the **Device Admin Password** field, enter a new password for the device.
   This password replaces the existing admin password on each device that you add to the network location.
8. From the **ISO Country** menu, select the country that the location is in.
   Your browser automatically populates this field.

   The Time Zone field automatically displays that of the ISO country you select.
9. If there are multiple time zones in that ISO country, from the **Time Zone** menu, select the correct time zone.
10. Click the **Next** button.
    The Add Location pop-up page displays.

11. In the **Review Locations** section, do one of the following to set the address for each location:
    - In the **Location Address** field, enter any part of the address (such as street, building, area, or postal code) to detect the full address.
    - Click the navigation icon to detect the location address automatically.
    - Drag and drop the red pin on the map. The address automatically displays in the address fields based on the location of the pin.

    NOTE: The location must be within the ISO country you chose in step 8.

    NOTE: You can also update the following location settings in this section: Location Name, Time Zone and ISO Country.

12. Click the **Select** button.
13. Click the **Add** button.
    A pop-up displays to ask if you want to copy configuration from an existing location after adding the new locations.
14. Click the **Yes** or **No** button.
15. If you select the Yes button, do the following:
    a. In the **Add Location** pop-up page, select the source location to copy the configuration from.
    b. Select one or more of the new locations where you want to apply this configuration.
    c. Click the **Apply** button.

    The new locations are created.

Add a device

You can add a single device to a location in an organization using the Insight Cloud Portal.

To add a device:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Select **Devices > Add**.
   The Add Device page displays.
3. Select the **Single Device** radio button.

   NOTE: You can also add multiple devices at the same time to an organization. For more information, see Add multiple devices on page 49.

4. From the **Select Location** menu, select a location you want to add the device to.

5. In the **Serial Number** field, enter the serial number of the device.
6. In the **Device Name** field, enter a device name.
7. In the **MAC Address** field, enter the MAC address of the device.
8. Click the **Add** button.

   The device is added to the organization.

NOTE: When you onboard a device to Insight, or enable the Insight management mode, any pre-existing device configuration or changes to the device settings through the device UI is overwritten by the configuration in the Insight Cloud.

Interpret the green, red, orange, and gray circles next to a device

In the Insight Cloud Portal, the colored circle next to each device indicates the current status of the device as follows:

- **Green**: The device is connected to Insight.
- **Red**: The device is disconnected from Insight.
- **Orange**: The device is unmanaged.
- **Gray**: Waiting for the first connection of the device.

Customize the UI settings

You can modify the Insight Cloud Portal interface by selecting either a light or dark theme and adjusting the font size to suit your preferences. These configuration options enable a more tailored and efficient Insight experience by enhancing clarity, comfort, and overall usability.

To customize the UI settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. In the top right corner of the page, click the theme icon.
   The Theme Customization pop-up window displays.
3. In the **Theme Mode** section, select the **Light Theme** or the **Dark Theme** button.
   The page adjusts to display the theme you select.
4. In the **Font Size** section, slide the toggle to adjust the font size.
   The page adjusts to display the font size you select.
5. To reset the UI theme and font size, click the **Reset** button.
   Your settings are saved.

Customize table settings

You can customize table settings in the Insight Cloud Portal by choosing which columns to display, adjusting their order, and configure row settings. These options let you shape tables into a clearer and more personalized view.

To customize table settings using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Select any of the tabs on the left of the page that contains a table.
3. In the top right of the table, select the **Settings** icon.
   The Table Settings pop-up window displays.
4. In the **Column Settings** tab, do the following:
   a. In the **Show Column** column, select what table columns you want to display.
   b. In the **Left Pin** column, select the column you want to pin to the left of the table.
5. Click the **Done** button.
   A pop-up displays to advise that your settings are successfully saved.

Customize data filters

You can use data filters to display only the table rows that match specific column values.

To customize data filters using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Select any of the tabs on the left of the page that contains a table.
3. In the top right of the table, select the **Filter** icon.
   The Data Filter pop-up window displays.
4. From the **Columns** menu, select a column heading.
5. In the **Value** field, enter a value based on the selected column.
6. Click the **Apply** button.
   Your settings are saved.

Select a language

To select a language to use in the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. In the top-right corner of the page, click the flag icon, and select one of the following languages:
   - **English**
   - **French**
   - **German**
   - **Japanese**
   - **Chinese (Simplified)**
   - **Chinese (Traditional)**

   The page adjusts to display your chosen language.

Set up two-step verification for logging in

With two-step verification, you can log in to Insight with an extra verification step. That is, you not only must enter your password, you must also enter a login verification code that you receive as an SMS message on the phone number that you must specify as the primary number, or as an email message at your account email address.

For easier verification without the requirement to enter a login verification code after initial verification, you can set up push notifications to a trusted device.

The following security measures apply:

- **Primary number**: If you set up a primary phone number, you receive an SMS text message with a login verification code when you or someone else tries to log in to your account from another phone number. A one-time password (OTP) is sent to the primary phone number so that you can approve the login attempt, for example, by forwarding the OTP to the other phone number.
- **Trusted device**: A trusted device is a device that is already verified by Insight. If you set up a trusted device, you receive a push notification if you or someone else tries to log in to your account from a non-trusted device so that you can approve the login attempt.

To set up two-step verification for logging in using the Insight Cloud Portal

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **account** icon in the upper right corner of the page.
   A pop-up menu displays.
3. Select **Update Profile**.
   Your profile page displays.

4. Select **Login Settings > Two-step Verification**.
   The Two-Step Verification page displays.
5. Click the **Enable** toggle so that the toggle is positioned to the right.
   The Select verification method page displays. You can use both email notifications and SMS text messages. By default, the primary verification method is email and defaults to the registered email on your Insight account.

   NOTE: If you previously enabled the Two-Step Verification and you want to disable the setting, click the toggle so that it is gray and positioned to the left, and in the pop-up window, click the **Disable** button
6. **(Optional)** To add SMS verification, do the following:
   a. Click the **SMS Text Message** button.
      The Add SMS Verification page displays.
   b. Enter your phone number and click the **Next** button.
      Your country code is automatically detected. The Before We Continue security pop-up page displays.
   c. Click and hold **Press & Hold** until the entire button displays blue.
   d. Click the **Next** button.
   e. Enter the code that was sent to your phone number and click the **Verify** button.
      If you did not receive a code, click the **Resend Message** button. The code is valid for 30 minutes.

      SMS Verification is successfully added.

Manage accounts and subscriptions

You can manage your Insight user account, user roles, access policies, and subscription settings to control permissions, organize users, and oversee account usage across your network.

This chapter contains the following sections:

- Manage your Insight account
- Manage users
- Manage Insight subscriptions

Manage your Insight account

From your Insight account, you can manage your profile, update your login credentials, adjust security and notification settings, and share your Insight account with support for troubleshooting.

Update your Insight account credentials

You can update your personal details on your Insight account at any time. These details use the same credentials as your My NETGEAR Account, so any changes you make are synchronized to both platforms.

To edit your Insight account credentials:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. In the upper right corner of the page, click the **account** icon.
   The account menu displays.
3. Click the **Update Profile** button.
   The My NETGEAR Account displays.
4. Select the **Edit Profile** tab.
   The Edit Profile page displays.
5. Update the existing credentials, and click the **Save** button.

Update your Insight login settings

Use your Insight account settings to manage your Insight login credentials and enhance account security. From this page, you can update the email address and password used to sign in to Insight and configure additional protection, such as two-step verification (two-factor authentication).

To update your Insight login settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. In the upper right corner of the page, click the **account** icon.
   The account menu displays.
3. Click the **Update Profile** button.
   The My NETGEAR Account page displays.
4. Click the **Login Settings** button.
   The Login Settings page displays.
5. To update the Two-Step Verification, click the **Two-Step Verification** tab. For more information, see Set up two-step verification for logging in on page 16.
6. To change your registered email, click the **Change Email** tab, and update the settings.
7. To change your registered password, click the **Change Password** tab, and update the settings.
8. Click the **Submit** button.
   Your settings are saved.

Grant technical support access to your Insight account

Granting access allows the technical support team to navigate through and view your Insight account in read-only mode, including information related to your organizations and locations. This information is used for troubleshooting purposes only.

To grant technical support access to your Insight account:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. In the upper right corner of the page, click the **account** icon.
   The account menu displays.

3. Select **Support > Grant Access to Support**.
   A pop-up window displays.
4. Click the **Yes, Grant Access** button.
   A pop-up window displays at the bottom of the screen to advise that your account is shared with technical support.
5. **To stop sharing your account with technical support, click the End Technical Support Access button.**

Manage users

If you are an Insight administrator, you manage user accounts and permissions. You can create, edit, and delete users, and you can set and adjust their access levels as necessary.

For each organization that is set up under the Insight account, the following roles are supported:

User role	Description
Admin	The admin is the Insight account holder and can perform all Insight functions, including setting up locations for the organization, adding devices to a location and managing those devices at the location, and inviting managers and users for the location. The VAR, MSP, or business that establishes the Insight account always performs the role of admin for an organization. By default, an admin can perform all functions in Insight. The secondary-admin has the same permissions as the admin, but does not handle payment methods.
Manager	The manager of devices for one or more organizations. Generally, a manager receives read and write credentials in Insight from the admin and can add, configure, monitor, and maintain locations and devices for these locations. The most likely scenario is that a manager, not the admin, performs the day-to-day management of the networks and storage systems for locations at an organization. However, a manager can also receive read-only credentials and monitor, not manage, an organization and its network locations.

As an administrator, you can configure a Secondary Admin, Insight Manager (Read/Write), and Insight Manager (Read Only).

- Add a user account
- Edit a user account
- Delete a user account
- Configure organization manager policies

This chapter contains the following sections:

Add a user account

To add a user account using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Select **Administration > Users**.
   The Users page displays.
3. Click the **Add** button.
   The Add User pop-up window displays.
4. In the **Name** field, enter a user name.
5. In the **Email Address** field, enter an email address.
6. In the **Phone Number** field, enter a phone number.
7. From the **Select Role** menu, select one of the following roles:
   - **Secondary Admin**
   - **Insight Manager (Read/Write)**
   - **Insight Manager (Read Only)**

   NOTE: To see access rights for individual roles, click the information icon to the right of the Select Role menu.
8. The following steps apply only to Insight Manager (Read/Write) and Insight Manager (Read Only) user roles:
   a. Select the check boxes for the organizations to assign the user to. To assign the user to all organizations, select the **All** check box.
   b. If you select the **All** check box in the previous step, enable the **Grant access to all future organizations** toggle to add the users to all future organizations.

9. Click the **Invite** button.
   An invitation email is sent to the user. If the user accepts the email, the Status in the User table displays as Active.

   NOTE: The invitee must not already have the email registered with any Insight account.

10. **(Optional)** To view if the user has accepted the invitation, check the **Status** column in the **Users** table (**Administration** > **Users**).
    - **Active**: The user has accepted the invitation and can now access all configured resources.
    - **Pending**: The user has not accepted the invitation and cannot access the configured resources. To resend the user invitation:
      a. In the **Users** table, click the three dots on the right of the user row, and **Resend Invitation**.

         A pop-up window displays.
      b. Click **Yes**. The invitation is resent.

Edit a user account

To edit a user account:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Select **Administration > Users**.
   The Users page displays.
3. From the **Users** table, select a user and click the **Edit** button.
   The Edit User page displays.
4. Edit the user configuration, and click the **Update** button.
   A pop-up window displays to indicate that the user configuration is updated successfully.

Delete a user account

To delete a user:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Select **Administration > Users**.
   The Users page displays.

3. From the **Users** table, select a user and click the **Delete** button.
   A pop-up window displays.
4. Click the **Delete** button.
   A pop-up window displays to advise the user account is deleted. The user receives an email to advise them that their account is deleted.
5. If you do not want to delete the user account, click the **No, Don't Delete** button.

Configure organization manager policies

You can configure policies that define how information is communicated and managed within an organization. These configurations help streamline communication, improve oversight, and ensure the right stakeholders receive timely and relevant information.

NOTE: This task applies to organization administrators only.

To configure organization manager policies:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Select the Organizations tab, and from the organizations table, select an organization.
   The organization dashboard displays.
3. Select **Administration > Policy**.
   The Policy page displays.
4. To apply the policy to all organizations, click the **Apply to All Organizations** check box.

5. Configure the following organization manager policies:

Policy	Description
User Notifications	Configure how users receive email notifications about organization activity, including specific events and device-related updates. Select whether notifications apply to Admin, Manager, or both.
Mobile App Push Notification	Manage real-time push notifications sent to mobile applications, ensuring timely alerts for critical actions. Select whether notifications apply to Admin, Manager, or both.
Email Reports	Configure automated reports sent via email, summarizing key performance metrics, usage data, or alerts. Select whether reports are sent to Admin, Manager, or both.
Device Ownership	Define rules and assign ownership for devices within the organization to maintain accountability and streamline management. Select whether ownership is assigned to Admin or Manager.
Scheduled Reports	Set up periodic reporting to be automatically generated and sent to designated recipients. Select the reporting frequency: Off, Daily, Weekly, or Monthly.

6. Click the **Save** button.
   Your settings are saved.

Manage Insight subscriptions

Once you decide what Insight plan you want, you can configure the plan in your Insight environment and view the subscription information across all organizations in your Insight environment.

NOTE: You can manage Insight subscriptions through the Insight Cloud Portal only. You cannot manage subscriptions through the Insight app.

This chapter contains the following sections:

- Subscription dashboard
- Purchase a monthly or yearly Insight subscription
- Activate a NETGEAR Partner subscription
- Configure Auto Allocation subscription settings
- Allocate or deallocate an Insight subscription

Subscription dashboard

To view subscription information in the Insight Cloud Portal, click **Administration > Manage**

Subscriptions.

The Manage Subscriptions page displays subscription information in the following tables:

Table 1.Existing subscriptions

Card	Field	Meaning
Subscription	Allocated	Total subscriptions allocated to the organizations.
	Unallocated	Subscriptions that are available to allocate to an organization for use.
	Scheduled	Subscriptions that are planned to activate at a future date.
	Bundled	Devices directly purchased with bundled licenses included for a 1, 4 or 6 year subscription term.
Expirations	Expired	Devices whose subscriptions have expired.
	Grace Period	Devices with expired subscriptions, and are within the 30 day grace period.
	Expiring Soon	Devices whose subscriptions are due to expire within the next 30 days.

Table 2.Organization Pools

Column	Meaning
Organizations	Name of the organization
Status	This column displays the overall subscription health of the organization: • **All Active**: All devices have active subscriptions. • **Mixed status**: One or more devices are in Expiring Soon, Grace, or Expired status. The count of such devices is displayed.
Allocated Subscriptions	Subscriptions that are allocated to the organization.
Used Subscriptions	Subscriptions that were previously assigned to the organization and are no longer in use.
Managed Devices	Devices with active subscriptions including hardware bundled.
Auto Allocation	Indicates Auto or Manual subscription assignment

Purchase a monthly or yearly Insight subscription

Before you begin Before you can purchase a monthly or yearly plan, you need to first add a device to your Insight account. For more information, see Add a device on page 13.

If you choose to purchase a monthly or yearly subscription, you can purchase and activate it directly in the Insight Cloud Portal.

NOTE: Devices purchased with a bundled Insight subscription include a 1, 4, or 6-year subscription term. You only need to purchase an additional subscription if you require more device licenses or your initial subscription has expired.

To purchase a monthly or yearly Insight subscription:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click **Administration > Manage Subscriptions**.
   The Manage Subscriptions page displays.
3. Click the **Choose Plan** link.
   The Choose Subscription Plan pop-up window displays.
4. On the **Monthly**, **1 Year**, or **3 Years** tile, click the **Choose Plan** button.
   You are redirected to the payment page.
5. Enter your payment details and complete the payment process.
   Once payment is complete, your subscription plan is activated.

   NOTE: When you purchase your first Insight subscription plan, your Insight free-trial ends.

Activate a NETGEAR Partner subscription

If you choose to purchase a NETGEAR Partner subscription, you can purchase it through a company partner, and enter the LMS key they provide to you directly in the Insight Cloud Portal to activate the subscription.

To activate a NETGEAR Partner subscription:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click **Administration > Manage Subscriptions**.
   The Manage Subscriptions page displays.
3. Click the **Choose Plan** link.
   The Choose Subscription Plan pop-up window displays.
4. On the **NETGEAR Partner** tile, click the **Choose Plan** button.
   The Choose Plan pop-up window displays.

5. In the **Subscription Key** field, enter the LMS subscription key.
6. Click the radio button next to one of the following options:
   - **Shared (subscriptions can be used by multiple organizations)**
   - **Dedicated (subscriptions can only be used by one organization)**: Select an organization from the menu.
7. In the **Activation Time** section, select the radio button next to one of the following options:
   - **Activate Now**
   - **Activate Later**: If you choose this option, select a date and time for activation.

   NOTE: When you activate or schedule a subscription key, your Insight free-trial ends. Ensure you have enough keys for activation to keep all your devices managed before proceeding. If there are not sufficient keys, some devices might move to an unmanaged state.
8. Click the **Add** button.
   Your settings are saved.

Configure Auto Allocation subscription settings

The Auto Allocation setting automatically manages Insight subscriptions for you, automatically assigning them as necessary, without needing you to manually allocate or deallocate. When the Auto Allocation setting is deactivated, you must explicitly allocate or deallocate subscriptions at the organization level. You can activate or deactivate this setting for each organization using the Insight Cloud Portal.

To apply the Auto Allocation setting to an organization:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click **Administration > Manage Subscriptions > Organization Pools**.
   The Organization Pools tab displays.
3. Click the **Auto Allocation Settings** button.
   The Auto Allocation pop-up window displays.
4. Do one of the following:
   - Select the check box next to an organization to apply the Auto Allocation setting.
   - Deselect the check box next to an organization to remove the Auto Allocation setting.
5. Click the **Save** button.
   The Auto Allocation settings are applied to the organization.

Allocate or deallocate an Insight subscription

After you purchase an Insight subscription, you can allocate it to an organization. If the subscription is no longer necessary for that organization, you can also deallocate it.

To allocate an Insight subscription:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click **Administration > Manage Subscriptions**.
   The Manage Subscriptions page displays.
3. In the **Organization Pools** tab, in the **Actions** column of the table, click the **Allocate** link.
   The Allocate pop-up window displays.
4. In the **Subscription** section, configure how many subscriptions you want to add to the organization.
5. Click the **Allocate** button.
   Your settings are saved.

Manage organization and location settings

You can manage organization and location settings in Insight to define and control configurations for specific locations. This allows you to apply consistent settings, organize devices, and tailor network behavior to meet the needs of each organization and location.

NOTE: For information on organization and location dashboards, see Dashboard information and supported countries on page 251.

This chapter contains the following sections:

- Edit an organization
- Delete an organization
- Edit location settings
- Delete a location
- Configure floorplans and heatmaps
- View and configure an organization map
- View connected clients
- Configure RADIUS servers for a location
- Configure an mDNS Gateway for a location
- Configure syslog settings for a location
- Configure LED settings for devices in a location
- Manage network configuration backups
- Configure SNMP settings in a location
- Configure a routing VLAN for a location
- Configure DHCP L2 Relay for switches in a location
- Configure DHCP Snooping for switches in a location
- Configure Secure Access Service Edge security for an organization

Edit an organization

Edit an organization's settings to keep its details, access, and configuration up to date in the Insight Cloud Portal.

To edit the configuration of an organization using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Select the **Organizations** tab.
   The Organization page displays.

3. From the Organizations table, select an organization, and click the **Edit** button.
   The Edit Organization page displays.
4. Edit the organization parameters, and click the **Save** button.

Delete an organization

You can delete an organization to remove obsolete or incorrect organizational configuration from Insight Cloud Portal.

To delete an organization using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Select the **Organizations** tab.
   The Organization page displays.
3. From the **Organizations** table, select an organization, and click the **Delete** button.
   A pop-up window displays.
4. Click the **Delete** button.
   A pop-up window displays to advise the organization is deleted.

Edit location settings

You can edit the configuration of a location at any time to update details such as the name, address, contact information, or other settings that affect how the location is used in the system.

To edit location settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Click the **Edit** button.
   The Edit Location page displays.
4. Edit the configuration of the location.
5. Click the **Save** button.
   Your settings are saved.

Delete a location

You can delete a location to remove obsolete or incorrect organizational configuration.

To delete a location:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the check box of the location you want to delete.
4. Click the **Delete** button.
   A pop-up window displays.
5. Click the **Delete** button.
   The location is deleted.

Configure floorplans and heatmaps

In Insight, you can upload a scaled image of a physical space, such as an office, warehouse, or campus building as a floor plan, calibrate it with real-world dimensions, and place access points onto it. The heatmap then overlays a color-coded signal strength (RSSI) map onto that image, modeling how WiFi signal degrades with distance from each AP based on its placement, transmit power, antenna characteristics, selected frequency band, and height relative to client devices.

There are two primary ways to use the heatmap:

- **Pre-deployment planning**: Place planned (model-based) APs anywhere on your floor plan without needing physical devices installed. The heatmap simulates the expected coverage for the AP models you select, so you can experiment with different placements, quantities, and frequency bands before committing to an installation. This helps you identify gaps, avoid over-provisioning, and confirm that your layout meets coverage requirements.
- **Live coverage validation**: Use assigned (installed) APs that are already online in your network. The heatmap reflects their actual configuration, including real channel assignments and transmit power, so you can confirm coverage is performing as expected and identify any areas that need attention.

You can also combine both AP types on the same floor plan, for example to visualize how adding a new planned AP would improve coverage alongside your existing installed devices.

NOTE: The heatmap is available only for floor plans that have at least one assigned or planned AP configured.

The heatmap uses the following color scale to represent WiFi signal strength:

Signal Strength (RSSI)	Color
–20 dBm to –50 dBm (Excellent)	Green
–51 dBm to –70 dBm (Good to Fair)	Yellow
Below –70 dBm (Poor or no coverage)	Red

Add a floor plan to a location

Before you can visualize WiFi coverage with the heatmap, you need to create a floor plan in the Insight Cloud Portal. This involves uploading an image of your physical space, setting the real-world dimensions and scale. Once you configure a floor plan, Insight uses this information as the basis for all signal strength calculations and coverage modeling. You can have multiple floor plans within one location.

To add a floor plan to a location:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click the **Floor Plan** tab.
5. Click the **Add Floor** button.
   The Add Floor page displays.
6. In the **Floor Name** field, enter a floor name.
7. In the **Upload Map Image** section, click the **Open file** link to upload an image. Alternatively, drag and drop a file into the section.
8. Define the floor dimensions:
   - **Width and length**
   - **Units (feet or meters)**
   - **Height (meters per pixel)**
9. Click the **Add** button.
   The new floor plan tile displays in the Floor Plan tab.

Configure advanced floor plan and heatmap settings

Before you begin

You must have an existing floor plan configured for the location. For more information, see Add a floor plan to a location on page 32.

The Floor Plan view provides several display controls that help you inspect your wireless network layout. You can filter the access points shown by frequency band (2.4 GHz, 5 GHz, or 6 GHz), pin AP detail cards to the canvas, show or hide device labels, and overlay a signal heatmap to visualize coverage. You can also add walls, rooms, and obstacles to the floor plan.

To configure advanced floor plan settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click the **Floor Plan** tab.
   The Floor Plan page displays.
5. Select the tile of the existing floor plan you want to configure.
   The Floor Plan page displays.
6. Click the **Edit** button.
7. From the **Device Library** tab, drag and drop APs in desired positions across the floor plan.

   NOTE: You can reposition the APs at any time across the floor plan.
8. Move the **Heatmap opacity** slider to adjust the heatmap overlay intensity.
   Refer to the RSSI legend to interpret the heatmap colors. The legend displays signal strength in dBm on a color scale from red (weak signal, –90 dBm) to green (strong signal, –50 dBm).
9. Add a wall to the floor plan.
   a. Click the **Wall** button.
   b. Click and drag anywhere on the floor plan, then right-click to add the wall.
   c. In the **Edit Wall** section, from the Material menu, select the material of the wall.
   d. Move the **Thickness** slider to adjust the wall thickness.
   e. Move the **Height** slider to adjust the height of the wall.

10. Add a room to the floor plan.
    a. Click the **Room** button.
    b. Click and drag anywhere on the floor plan, then right-click to add the room.
    c. In the **Edit Room** section, add a name for the room.
    d. From the **Apply wall type to all sides** menu, select a material for all walls within the room.
    e. Click the **Done** button.

11. Add an obstacle to the floor plan.
    - Click the **Obstacle** button.
    - Click and drag anywhere on the floor plan, then right-click to add the obstacle.
    - In the **Draw Obstacle** section, from the Obstacle menu, select an obstacle type.
    - Move the **Height** slider to adjust the height of the obstacle.

12. **(Optional)** To edit the basic floor plan configuration, click the **Upload Floor Plan** tab.

13. Click the **Save** button.
    The Floor Plan page displays.

14. Filter the access points that display on the floor plan.
    a. From the **Show** menu, select the set of access points to display (for example, **All** to show all APs).
    b. From the **Band** menu, select one or more frequency band check boxes to limit the display to APs operating on those bands:
        - **2.4 GHz**: Displays APs broadcasting on the 2.4 GHz band.
        - **5 GHz**: Displays APs broadcasting on the 5 GHz band.
        - **6 GHz**: Displays APs broadcasting on the 6 GHz band.

    Only the access points matching the selected bands appear on the floor plan canvas.

15. Activate or deactivate the **Pin AP details** toggle.
    Activate this setting to keep AP information cards persistently visible on the canvas. When deactivated, AP detail cards display only on when you hover the mouse over the AP.

16. Activate or deactivate the **Device labels** toggle.
    Activate this setting to display or hide the name label beneath each AP icon on the floor plan.

17. Activate or deactivate the **Display heatmap** toggle.
    Activate this setting to overlay a color-coded signal strength visualization on the floor plan. The floor plan canvas updates in real time to reflect the heatmap overlay.

18. **(Optional)** Export the configured floor plan view.
    This action downloads the current floor plan, including any active filters and overlays, as an image file.
    Your settings are automatically saved.

View and configure an organization map

The organization map provides a geographic overview of all locations within an organization, displaying real-time device status across sites.

To view and configure an organization map:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Select the **Organizations** tab.
   The Organization page displays.
3. Click the **Map** tab.
   The organization map displays. If locations within your organization do not have address information, a pop-up displays.
4. Click the **View List** link.
   The Locations pop-up displays.
5. To update the location address, click the **Update Address** link.
   The Edit Location pop-up page displays.
6. Update the address, and click the **Save** button.
   The location address is saved and you can view the location on the organization map.
7. To isolate specific conditions and identify locations that require attention, in the **Status** section, select one of the following check boxes:
   - **All**
   - **Online**
   - **Offline**
   - **Critical**
   - **License expiring**
8. To view basic location information, on the organization map, hover the mouse over the location.

View connected clients

You can view connected clients in the Insight Cloud Portal to monitor all devices currently joined to your network. The Clients page provides a detailed overview of each connected device, including its status, connection type, associated device, IP address, and signal strength, giving you full visibility into your network.

To view connected clients:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Clients** tab.
   The Clients page displays.

   All the connected clients display with the following information:

Field	Description
Clients	The hostname or name of the connected client device
Status	The current connection status of the client, such as connected or disconnected
Connection	The connection type, either wired (Ethernet) or wireless (WiFi)
Associated Device	The Insight-managed access point, switch, or router the client is connected to
Device Model	The model of the associated Insight-managed device
SSID	The wireless network name the client is connected to
Port	The switch port the client is connected to, if wired
OS	The operating system of the client device
MAC Address	The unique hardware identifier of the client device's network interface
IP Address	The IP address assigned to the client device on the network
VLAN ID	The VLAN the client device is assigned to
User Name	The username associated with the client, if authentication is configured
RSSI	The received signal strength indicator, reflecting the quality of the client's wireless signal
Resume/Pause	Controls whether the client's network access is active or paused

5. To filter clients by status, select one of the following tabs:
   - **All**
   - **Connected**
   - **Disconnected**

Configure RADIUS servers for a location

The following features use 802.1X access authentication with RADIUS servers on Insight-managed devices:

- WPA2 Enterprise WiFi security (supported on access points)
- MAC ACLs with RADIUS authentication (supported on access points)
- Port-based network access control (supported on switches)

RADIUS server configuration is shared across your Insight location or organization, meaning access points and switches use the same RADIUS server settings. Changes made to the RADIUS configuration apply to all supported devices in that location or organization.

If your network uses *one* of these features, you must set up RADIUS servers. You can set up primary and secondary RADIUS servers. By default, accounting is enabled, but you cannot set up separate RADIUS accounting servers. You can also disable accounting.

To configure RADIUS servers for a location:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Settings > RADIUS**.
   The RADIUS page displays.
5. Activate the **802.1x Access Authentication** toggle.
   When this toggle is on, the location uses 802.1x with the configured RADIUS servers to authenticate devices and users before allowing network access. When it is off, 802.1x access authentication with RADIUS is not enforced for this location.
6. In the **NAS Identifier** field, enter a unique identifier for this location that matches the NAS identifier expected by your RADIUS server.
7. Specify the primary and secondary RADIUS servers and the reauthentication time. Be sure that the IP addresses that you specify are reachable from your network.
   By default, the reauthentication time is 3600 seconds.

8. Activate or deactivate the **Accounting Interval** toggle.
   This value specifies, for each session, the frequency accounting packets are sent to the RADIUS server. By default, this setting is activated.
9. Click the **Save** button.
   Your settings are saved.

Configure an mDNS Gateway for a location

A multicast DNS (mDNS) Gateway in Insight forwards local service discovery traffic between network segments, allowing client devices to discover printers, TVs, and other services across VLANs. You can configure an mDNS Gateway to centrally manage and control this discovery, improving visibility and ensuring services are available where users need them.

To configure a mDNS Gateway for a location:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Settings > mDNS Gateway**.
   The mDNS Gateway page displays.
5. From the Gateway menu, select the mDNS gateway you want to configure.
6. Activate the **mDNS Gateway** toggle.
7. In the **Policy** tab, click the **Add Policy** button.
   The policy defines which mDNS service types are allowed or blocked, and how they are forwarded between VLANs at this location.
8. In the **Policy Name** field, enter a name for the policy.
9. From the **Shared Service** menu, select the shared mDNS service you want to apply.
10. In the **IP Address** field, enter the IP address of the device.
11. From the **Service VLAN** menu, select the VLAN where the service is hosted.
12. In the **Allowed Wireless Network(s)** section, select the check box for each wireless network that should be allowed to access this mDNS service.

13. Click the **Add** button.
    Your settings are saved.
14. **(Optional)** Alternatively, to reset the policy configuration, click the **Reset** button.

Configure syslog settings for a location

You can configure syslog settings in Insight to send device logs to an external syslog server for monitoring and troubleshooting. These settings are applied at the location level, and devices in the location inherit them. However, syslog support and log detail might vary by device type and model.

NOTE: The Syslog Level setting is applicable to access points only.

To configure syslog settings for a location:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Settings > Syslog Configuration**.
   The Syslog Configuration page displays.
5. Activate the **Enable Syslog** toggle.
   By default, this setting is activated.
6. In the **Syslog Server IP Address** field, enter the IP address.
7. In the **Port Number** field, enter a port number for the syslog server.
8. In the **Syslog Level** section, move the slider to choose one of the following log levels:
   - Emergency
   - Error
   - Warning
   - Info
   - Debug

   NOTE: To capture WiFi Probing Client logs, set the Severity Level to Debug or Info.
9. Click the **Save** button.
   Your settings are saved.

Configure LED settings for devices in a location

You can configure LED settings so that devices in a location display consistent LED behavior.

NOTE: This setting is applicable only to access points and routers in a location.

To configure LED settings for devices in a location:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Settings > LED Settings**.
   The LED Settings page displays.
5. From the **LED Settings** menu, select one of the following:
   - **Default**
   - **Off**
   - **Off except Power LED**
6. Click the **Save** button.
   Your settings are saved.

Manage network configuration backups

You can manage network configuration backups in the Insight Cloud Portal to save and restore the network settings at a location. By creating backups, you can quickly recover your network configuration if a change causes issues or your settings need to be rolled back. You can add, edit, restore, and delete network configuration backups at the location level.

To manage network configuration backup and restore:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.

4. Select **Settings > Configuration Backup and Restore**.
   The Configuration Backup and Restore page displays.

5. **To add a new network configuration backup, do the following:**
   a. Click the **Add** button.
      The Add Configuration Backup page displays.
   b. In the **Name** field, enter a name for the network backup.
   c. In the **Description** field, enter a description for the network backup.

      NOTE: Insight supports a maximum of three configuration backups.

      NOTE: The network backup does not include device-level settings.

   d. Click the **Add** button.
      A network backup is created.

6. **To edit a network configuration backup, do the following:**
   a. From the **Configuration Backup and Restore** table, select the check box of a network backup.
   b. Click the **Edit** button.
      The Edit Configuration Backup page displays.
   c. Update the backup configuration as necessary.
   d. Click the **Save** button.
      A network backup is updated.

7. **To restore a network configuration backup, do the following:**
   a. From the **Configuration Backup and Restore** table, select the check box of a network backup.
   b. Click the **Restore** button.
      The Restore Backup pop-up window displays.
   c. Click the **Yes** button.
      A pop-up notification displays to advise that the configuration backup is successfully initiated.

8. **To delete a network configuration backup, do the following:**
   a. From the **Configuration Backup and Restore** table, select the check box of a network backup.
   b. Click the **Delete** button.
      The Delete Configuration Backup displays.
   c. Click the **Delete** button.

Configure SNMP settings in a location

Simple Network Management Protocol (SNMP) lets you monitor what happens on your network from a central location. Insight supports SNMP v1/v2c and SNMP v3. SNMP v1/v2c uses plain-text community strings for authentication and offers no encryption. SNMP v3 introduces user-based authentication and optional encryption, making it the more secure option. Settings are configured once per location and automatically applied to all compatible devices in that location.

To monitor your Insight-managed devices using SNMP, you must activate SNMP and download the management information base (MIB) files for your devices. You must also know the SNMP community string that you are using with your Insight-managed devices.

To configure the SNMP settings for devices in a location:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Settings > SNMP**.
   The SNMP page displays.
5. Activate the **Enable SNMP v1/v2** toggle so that it is positioned to the right.
   By default, this setting is deactivated.
6. In the **Community Name** field, enter a descriptive name for the SNMP community.
7. **(Optional)** To keep the same name for the trap community, activate the **Use the same name for Trap** toggle so that it is positioned to the right.
8. **(Optional)** To configure a new name for the trap community, enter a name in the **Trap Community Name** field.
9. In the **SNMP v3 settings** section, do one of the following:
   - Activate the **Enable SNMP v3** toggle so that it is positioned to the right.
   - Deactivate the **Enable SNMP v3** so that it is positioned to the left. SNMP v3 provides user-based authentication and optional encryption for network management traffic, making it more secure than SNMP v1/v2c. Configuration is applied at the location (network) level and pushed automatically to all compatible devices.
10. If you activate SNMP v3, in the **Username** field, enter a name for the SNMP v3 user account.

11. From the menu, select one of the following security levels:

Security level	Description
noAuthNoPriv	Provides no authentication and no privacy. Messages are neither authenticated nor encrypted.
authNoPriv	Provides authentication only. Messages are authenticated but not encrypted.
authPriv	Provides both authentication and privacy. Messages are authenticated and encrypted.

12. **(Optional)** To keep the same name for the trap community, activate the **Use the same name for Trap** toggle so that it is positioned to the right.
13. In the **SNMP Trap** section, in the **IP Address** field, enter the IP Address of the SNMP trap receiver, which is the device that receives monitoring information about your device.
14. In the **Trap Port** field, enter a port number for SNMP traps.
    Port number 161 and 162 are supported.
15. Click the **Save** button.
    Your settings are saved.
16. **(Optional)** To reset the SNMP configuration, click the **Clear** button.

Configure a routing VLAN for a location

Routing VLANs allow Layer 3 switches within a location to route traffic between different VLANs on that network. This allows devices in separate VLANs at the same site to communicate while maintaining segmentation. By configuring routing VLANs on switches in an Insight location, you can manage traffic flow locally without requiring an external router. By default, all ports on the switch are assigned to their default VLAN with VLAN ID 1.

To configure routing VLANs for a location:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.

4. Click **Settings > Routing**.
   The Routing page displays.
5. From the VLAN table, select the check box of a VLAN.
6. **(Optional)** To edit the configuration of a routing VLAN, click the **Edit** button.
7. At the bottom of the **Routing VLAN** page, in the **Gateway** section, enter an IP address.
   This IP address serves as the default gateway for devices in the VLAN.
8. Click the **Save** button.
   Your settings are saved.

Configure DHCP L2 Relay for switches in a location

DHCP L2 Relay is a Layer 2 feature on Insight-managed switches that forwards DHCP requests across VLANs to a DHCP server. When activated, the switch inserts DHCP Option 82 (Relay Agent Information) into each packet, identifying the ingress port and relay switch so the DHCP server can apply policies based on client location. In the Insight Cloud Portal, you configure DHCP L2 Relay at the location level, and the configuration is automatically pushed to all supported switches at that location. Admin Mode controls whether the feature is active at two levels: globally across the location and individually on an individual VLAN, so you can activate DHCP L2 relay only on the VLANs that require it.

To configure DHCP L2 relay:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wired > DHCP L2 Relay**.
5. Select the **Global Configuration** tab.
6. Do one of the following:
   - Activate the **Admin Mode** toggle so that it is positioned to the right and DHCP L2 Relay is activated at a global level.
   - Deactivate the **Admin Mode** toggle so that it is positioned to the left and DHCP L2 Relay is deactivated at a global level.
7. In the **VLAN Configuration** table, locate one or more VLANs to configure.

8. Do one of the following:
   - Activate the **Admin Mode** toggle so that it is positioned to the right and DHCP L2 Relay is activated at an individual VLAN level.
   - Deactivate the **Admin Mode** toggle so that it is positioned to the left and DHCP L2 Relay deactivated at an individual VLAN level.
9. Click the **Port Configuration** tab.
   Graphics of all switches in the location or within selected VLANs display.
10. On the switch graphic or graphics, select one or more ports.
11. **(Optional)** To select all ports on a switch, click the **Select All** button.
12. Click **Admin Mode** to activate DHCP L2 Relay on the selected ports.
13. Click **82 Option Trust Mode** to activate Option 82 trust on the selected ports.
14. **(Optional)** To clear configuration, click the **Delete** button.
15. Click the **Save** button.
    Your settings are saved.

Configure DHCP Snooping for switches in a location

DHCP Snooping is a security feature on Insight-managed switches that filters untrusted DHCP messages and maintains a binding table of trusted clients, helping to prevent rogue DHCP server spoofing, client spoofing, and denial-of-service attacks. In the Insight Cloud Portal, you can configure DHCP Snooping for switches at an individual VLAN and at the location level.

To configure DHCP snooping:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wired > DHCP Snooping**.
5. Select the **Global Configuration** tab.

6. Do one of the following:
   - Activate the **DHCP Snooping Mode** toggle so that it is positioned to the right and DHCP Snooping is activated at a global level.
   - Deactivate the **DHCP Snooping Mode** toggle so that it is positioned to the left and DHCP Snooping is deactivated at a global level.
7. Do one of the following:
   - Activate the **MAC Address Validation** toggle so that it is positioned to the right and MAC Address Validation is activated at a global level.
   - Deactivate the **MAC Address Validation** toggle so that it is positioned to the left and MAC Address Validation is deactivated at a global level.
8. In the **VLAN Configuration** table, locate one or more VLANs to configure.
9. Do one of the following:
   - Activate the **DHCP Snooping Mode** toggle so that it is positioned to the right and DHCP Snooping is activated at an individual VLAN level.
   - Deactivate the **DHCP Snooping Mode** toggle so that it is positioned to the left and DHCP Snooping is deactivated at an individual VLAN level.
10. Click the **Port Configuration** tab.
    Graphics of all switches in the location or within selected VLANs display.
11. On the switch graphic or graphics, select one or more ports.
12. **(Optional)** To select all ports, click the **Select All** button.
13. Click the **Trust Mode** button to designate the selected ports as trusted, allowing DHCP messages to be forwarded without filtering.
14. Click **Invalid Packets** button to view the count of DHCP packets dropped on the selected ports due to failed snooping validation.
15. **(Optional)** To clear configuration, click the **Delete** button.
16. Click the **Save** button.
    Your settings are saved.
    DHCP Snooping is configured for the selected VLANs. Untrusted DHCP messages are filtered according to the settings you apply.

Configure Secure Access Service Edge security for an organization

You can configure NETGEAR Exium Secure Access Service Edge (SASE) security for your organization using the Insight Cloud Portal and Exium Security Cloud to achieve a unified approach to network protection and management. This protects your organization with advanced safeguards, including secure remote access, traffic inspection, and policy-based control for users and applications across all connected locations. By centralizing configuration and monitoring in the cloud, you can consistently enforce security policies, gain visibility into network activity, and quickly respond to emerging threats.

To configure SASE security for an Insight organization:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Select the **Organizations** tab, and click the name of the organization.
   The organization dashboard displays.
3. Select the **Security Center** tab.
   The Security Center page displays.
4. Click the **Enable SASE Security** button.
   A pop-up message displays to advise that SASE Security is now activated for the organization and to finish the setup, open each device's details page and activate SASE security for the routers you want to protect. For more information, see Configure Secure Access Service Edge security on a router on page 185.
5. **(Optional)** Click the **Security Configurations** button to open the Exium Security page.

Manage devices

Once you have set up a device in Add a device on page 13, you can update existing device configurations, or remove devices that no longer apply.

For detailed device dashboard information and supported devices, see Dashboard information and supported countries on page 251.

This chapter contains the following sections:

- Edit device settings
- Delete a device
- Add multiple devices
- Update device firmware
- Schedule device firmware updates
- Set target firmware versions
- Reset a device to factory default settings
- Restart a device
- Schedule device restarts
- Move a device
- Manage device configuration backups
- Share Device Diagnostics
- Device Groups

Edit device settings

You can edit the configuration of a device using the Insight Cloud Portal.

To edit device settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Devices** tab.
   The Devices page displays.
3. Select the check box next to the device you want to edit.
4. Click the **Edit** button.
   The Edit Device page displays.
5. Edit the device configuration.
6. Click the **Save** button.
   The device configuration is updated.

Delete a device

You can delete a device to remove obsolete or incorrect organizational configuration from Insight Cloud Portal

To delete a device:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Devices** tab.
3. Select the check box next to the device you want to delete.
4. Click the **Delete** button.
   A pop-up window displays asking if you want to delete the device.
5. Click **Yes**.

   The device is deleted.

> **NOTE:** When you delete a device, the Insight subscription is automatically returned to the organization pool. For more information on subscriptions, see Manage Insight subscriptions on page 24.

Add multiple devices

You can add multiple devices to Insight in a single operation.

To add multiple devices:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Select **Devices > Add**.
   The Add Device page displays.
3. Select the **Multiple Devices** radio button.

> **NOTE:** You can also add a single device to an organization. For more information, see Add a device on page 13.

4. From the **Select Organization** menu, select an organization you want to add the device to.
5. In the **Device List** file section, click the **Click here** button to download the device template.

6. Complete the following details for each device in the device template:

   NOTE: The device upload limit is 1000.

   - **Serial number**
   - **Device name**
   - **MAC address**
   - **Location**
   - **Profile Name** (If adding a mobile hotspot)
   - **IMEI number** (If adding a mobile hotspot)

7. Click the **Open file** link to upload the completed device list file or drag and drop the file.
8. Click the **Next** button.
9. Select the check boxes next to the new devices you want to add.
10. Click the **Update Device List** button.
    The devices are added to the organization and location.

Update device firmware

If firmware updates are available for managed devices, Insight detects and lists the updates, and lets you update the firmware on individual devices. If you enable notifications, email notifications, or both, you receive notification of the new firmware. You can also schedule automatic firmware updates for a location, and you can target the specific firmware versions you want the device to use.

To update the device firmware:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Click on the name of the location in which you want to update device firmware.
   The dashboard page of the location displays.
4. From the **Location Menu**, select **Settings > Firmware**.
   The Firmware page displays.
5. In the **Firmware Management** tab, select the check box next to each device you want to update.
   The Upgrade Status column advises if there is a firmware upgrade available.

6. Click the **Update** button, or select the three dots next to the device, and select **Update Firmware**.
   The Update Firmware pop-up window displays.
7. From the **Upgrade To Version** menu, select a firmware version.
8. Click the **Update Now** button.
   The firmware update process starts. A progress bar shows the progress of the update. The process takes a few minutes.

   When the firmware update is complete, the device automatically restarts, causing it to temporarily disconnect from Insight.

Schedule device firmware updates

You can update the firmware for all devices in a location within a scheduled window.

NOTE: During an update, the Internet access of another device may stop, which can cause its firmware update to fail. To reduce this risk, Insight updates the firmware on all edge devices (access points, Orbi Pro, routers) at the same time, then updates switches, and then business routers. Review the status of all devices and manually update the firmware of any devices that were not included during the scheduled window.

To schedule firmware updates for all devices in a location:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Click on the name of the location in which you want to update device firmware.
   The dashboard page of the location displays.
4. From the **Location Menu**, select **Settings > Firmware**.
   The Firmware page displays.
5. In the **Schedule Upgrade** tab, activate the **Enable** toggle.
   By default, this setting is deactivated.
   If you activate this option, the manual update option is deactivated.
6. In the **Start Date & Time** field, click the calendar icon, and select a date and time for when you want the firmware update to start.
7. In the **End Date & Time** field, click the calendar icon, and select a date and time for when you want the firmware update to end.

8. From the **Repeats** menu, select one of the following repeat intervals:
   - None
   - Daily
   - Weekly
   - Monthly
9. Click the **Save** button.
   Your settings are saved.

Set target firmware versions

You can configure a target firmware version for each device in a location. You can select the three most recent firmware versions of the product. Once a target firmware version is set, that version becomes available for both manual updates and scheduled firmware updates. You can upgrade a device to any target firmware version, but you can downgrade to an earlier version on access points only. If you set a target firmware version before you add a device, the device upgrades or downgrades directly to that target version instead of the latest available firmware.

To set target firmware versions:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Click on the name of the location in which you want to update device firmware.
   The dashboard page of the location displays.
4. From the **Location Menu**, select **Firmware**.
   The Firmware page displays.
5. Click the **Target Firmware** tab.
   The Target Firmware tab displays.
6. To show all models in the location, click the **Show All Models** toggle.
   The page adjusts to display all product models in the location.
7. Locate the row of the product model and from the **Target Version** menu, select a firmware version.

   NOTE: You can upgrade a device to any target firmware version, but you can downgrade to an earlier version on access points only.
8. Click the **Save** button.
   Your settings are saved.

Reset a device to factory default settings

You can remotely reset an Insight-managed device to the factory default settings. If you use the Insight Cloud Portal, you can reset the device through the portal and do not need physical access to the device.

NOTE: After you return the device to factory default settings, it is removed from Insight. You need to manually re-add the device to Insight. For more information, see Add a device on page 13.

NOTE: Returning your device to factory default settings erases all configured settings. Do not follow this procedure unless you are sure that you want to return all settings to their factory defaults

NOTE: We recommend that you export and save the configuration file before you reset the device to factory default settings. For more information, see Share Device Diagnostics on page 59.

To reset a device to a factory default settings using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Devices** tab.
   The Devices page displays.
3. Click the device link of the device you want to restart.
   The device page displays.
4. In the **Summary** tab, in the upper right corner of the page, click the **More** menu.
   The More menu displays.
5. Select **Factory Reset**.
   The Factory Reset pop-up window displays to advise this action cannot be undone.
6. Click the **Factory Reset** button.
   The device is reset to factory default settings. Any previous Insight configuration no longer exists.

Restart a device

You can restart a device remotely using the Insight Cloud Portal.

To restart a device using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Devices** tab.
   The Devices page displays.
3. Click the device link of the device you want to restart.
   The device page displays.
4. In the Summary tab, in the upper right corner of the page, click the **Reboot** button.
   The Reboot pop-up window displays.
5. Click the **Reboot** button.
   As the device restarts, the device status displays "Device is rebooting".

Schedule device restarts

You can set up a one-time restart or a recurring schedule (daily, weekly, or monthly) for one or more devices using the Insight Cloud Portal. You can configure scheduled restarts at an individual device level, or for specified devices in a location or organization.

To schedule a restart for one or more devices:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Do one of the following:
   - **Organization-level**: Click the **Organizations** tab, click the name of an organization, and select **Schedules > Add**.
   - **Location-level**: Click the **Locations** tab, select a location, and select **Schedule > Add**.
   - **Device-level**: Click the **Devices** tab, click the name of a device, and select **Schedule > Add**.

   The Reboot page displays.
3. Select one of the following radio buttons:
   - **Schedule One-Time**
   - **Recurring**

4. If you select the **Recurring** radio button, from the **Recurring Type** menu, select one of the following schedules:
   - **Daily**: Allows you to restart devices every day.
   - **Weekly**: Allows you to restart devices on selected day(s) of the week. If you select this option, in the **Repeat on** section, select a one or more days the alarm repeats in the week.
   - **Monthly**: Allows you to restart devices every month. If you select this option, in the **Repeat on** section, you can choose either the same date every month or a specific day of the month.
5. In the **Schedule Name** field, enter a name for the restart schedule.
6. **(Optional)** In the **Description** field, enter a description for the restart schedule.
7. In the **Start Date & Time** field, click the calendar icon to set a schedule start time and date.
8. If you select the **Recurring** radio button and the Daily or Weekly recurrence, in the **End Date & Time** section, enter a schedule end date.
9. If you select the **Recurring** radio button and the monthly recurrence, in the **Ends** section, select one of the following radio buttons:
   - **On date**: The restart schedule ends on a specific date.
   - **After**: The restart schedule ends after a specific amount of recurrences.
10. From the table, select one or more devices to apply the schedule to.

    NOTE: Overlapping restart schedules for the same device are not allowed.

11. Click the **Schedule Reboot** or **Create Recurring** button.
    A pop-up notification displays to advise the restart schedule is set.

    NOTE: Each selected device becomes unreachable for one to two minutes. Connected clients might disconnect and might need to reconnect.

Move a device

You can move a device from one location to another using the Insight Cloud Portal.

To move a device:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Devices** tab.
   The Devices page displays.

3. Select the check box next to the device you want to move.
4. Select **More > Move Device**.
   The Move Device page displays.
5. From the **Location menu**, select a location to where you want to move the device.

   NOTE: To move an access point, the new location must be in the same wireless region (RF domain) that the AP is already in. You can move all other devices to different wireless regions.

   NOTE: When you move a device to another location, it wipes all existing configuration, then inherits the new location's configuration after it resets and restarts.
6. Click the **Move Device** button.
   A pop-up notification displays to advise the device is successfully moved to the new location.

Manage device configuration backups

You can configure and manage configuration backups directly on an Insight-managed device using the Insight Cloud Portal. Configuration backups capture the specific settings of your device at a point in time, giving you a reliable way to protect your network configuration and recover quickly when problems occur.

With configuration backup management, you can:

- Back up or restore the settings of device, with support for up to three saved backups per device.
- Replace a faulty or end-of-life device and restore its previous configuration to the new device.
- Upload a saved configuration backup file to replicate a standard configuration across multiple devices or recover from an unintended change.

Back up or restore device settings

You can back up the settings of your devices on the Insight Cloud Portal. This lets you restore settings that you know worked before, or save a copy to upload onto another device later. You can have up to three backups per device at a time.

To back up or restore device settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.

3. Select the name of a location.
   The location dashboard displays.
4. From the table, click the name of the device.
   The device dashboard displays.
5. Do one of the following:
   - If you are managing a router, click the **Backup & Restore** tab.
   - If you are managing a switch, click the **Operations > Backup and Restore** tab.
6. To create a new backup, do the following:

   NOTE: The new backup must be from a device of the same model.

   a. Click the **Add** button.
   b. In the **Add Configuration Backup** page, enter a name and description for the backup.
   c. Click the **Add** button.

Upload a device configuration backup file

If you previously downloaded a device backup configuration file, either from this device or another device of the same model, you can upload it to quickly restore the configuration or replicate settings across devices. Uploading a configuration file overwrites the current device-specific settings with those stored in the file. This is useful when replacing a faulty device, deploying a standard configuration to multiple locations, or recovering from an unintended configuration change.

To upload a device configuration backup file:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. From the table, click the name of the device.
   The device dashboard displays.
5. Do one of the following:
   - If you are managing a router, click the **Backup & Restore** tab.
   - If you are managing a switch, click the **Operations > Backup and Restore** tab.
6. Click the **Upload** button.
   The Upload Backup page displays.

7. Enter a name and description for the backup.
8. Click the **Choose A File** button, and upload the backup file from your local device.
9. Click the **Upload Backup** button.
   Your settings are saved.

Replace a device and restore previous device settings

Before you begin Back up the settings for your old device to the Insight Cloud before replacing the device.

After replacing a device on your network, you can restore its backed up settings from the original device as long as the two devices are identical models.

To restore a backup of an existing device configuration:

1. Back up the settings for your old device to the Insight Cloud Portal. For more information about how to back up your settings, see Back up or restore device settings on page 56.
2. Replace the old device with the new device, connecting it to the same cables and locations.
3. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
4. Click the **Locations** tab.
   The Locations page displays.
5. Select the name of a location.
   The location dashboard displays.
6. Select the **Devices** tab, and add the new device. For more information, see Add a device on page 13.
7. From the table, click the name of the device.
   The device dashboard displays.
8. Do one of the following:
   - If you are managing a router, click the **RMA** tab.
   - If you are managing a switch, click the **Operations > RMA** tab.
9. Click the **RMA** tab.
   The RMA page displays.
10. Select the **Restore an existing backup** radio button, and click the **Next** button.
    The Select Devices pop-up page displays and a list of available backups displays.
11. Select the old device that you want to use a backup of, and click the **Next** button.
    The Restore Backup pop-up page displays.

12. Select a back up to restore and click the **Restore** button.
    Insight restores the settings from the old device. Before you continue, check that your settings transferred successfully.
13. On the RMA screen, select **Replace an existing device** and click the **Next** button.
    The Select Devices pop-up page displays and a list of available devices display.
14. Select the old device, and click the **Next** button.
    The Replace Device pop-up page displays.
15. Select a backup to replace.
16. Click the **Replace** button.
    Your settings are saved.

Share Device Diagnostics

To share device diagnostics:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Devices** tab.
   The Devices page displays.
3. Click the device link of the device you want to restart.
   The device page displays.
4. In the **Summary** tab, in the upper right corner of the page, click the **More** menu.
   The More menu displays.
5. Select **Share Diagnostics**.
   The Share Diagnostics pop-up displays.
6. Enter the email with which you want to share the device diagnostics.
7. Click the **Send** button.
   A pop-up window displays to advise the device diagnostics are successfully sent.
   The email contains a `.tar.gz` detailing the device diagnostics.

Device Groups

A device group is a level of hierarchy below an Insight location, where you can add a group of devices with certain common configurations. Devices in a device group inherit some settings from their location, and also have other unique settings which are specific to the device group.

The following settings can be configured in a device group:

- WiFi
- WiFi Schedules
- Load Balancing
- Fast Roaming
- VLAN/Network Setup
- Advanced
  - Wireless Settings
  - Network Settings (Including Energy Efficiency Mode)
  - Multi PSK Settings

The following settings are inherited by a device group from the parent location:

- Locations settings
- RADIUS
- Syslog Configuration
- LED Settings
- Routing
- SNMP v1/v2
- Instant WiFi (Auto RRM)
- Firmware Update
- Troubleshoot

The following settings are not applicable to a device group and apply only to the parent location:

- Copy configuration
- Bulk onboarding
- Organization wide SSID

> **NOTE:** This setting is only applicable to a device group if the SSID existed before the group was created.

- Custom reports
- Backup and restore
- Instant Mesh
- URL filtering
- Multicast DNS Gateway (mDNS Gateway)
- Management VLAN Settings
- Wired Settings
  - LAG
  - Spanning Tree
  - PoE Downtime Schedules
  - RADIUS Configuration

Add a device group

To create a device group in a location:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Click on the name of the location where you want to create the device group.
   The dashboard page of the location displays.
4. Click the **Settings > Device Groups**.
   The Device Groups page displays.
5. Click the **Add** button.
   The Add Device Group displays.
6. In the **Name** field, enter a name for the device group.
7. In the **Description** field, enter a description for the device group.
8. To add devices from the location to the device group, click the **Add From Location** button.
   The Add Devices page displays.
9. Select the check box next to each device you want to add, and click the **Add** button.
10. Click the **Add** button.
    Your settings are saved.

Edit a device group

To edit a device group:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Click the name of the location where you want to create the device group.
   The dashboard page of the location displays.
4. Click the **Settings > Device Groups**.
   The Device Groups page displays.

5. Click the **Edit** button.
   The Edit Device Group page displays.
6. Edit the device group configuration, and click the **Save** button.
   Your settings are saved.

Delete a device group

To delete a device group:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of the location where you want to create the device group.
   The dashboard page of the location displays.
4. Click the **Settings > Device Groups**.
   The Device Groups page displays.
5. Select the check box next to the device group you want to delete.
6. Click the **Delete** button.
   The Delete Device Group(s) pop-up displays.
7. Click the **Yes** button.
   A pop-up displays to advise the device group is successfully deleted.

NOTE: If you delete a device group, the devices inside the device group are deleted from the Insight Cloud Portal account.

Add a device to a device group

To add a single device to a device group:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Click the name of the location where you want to create the device group.
   The dashboard page of the location displays.

4. Click the **Settings > Device Groups**.
   The Device Groups page displays.
5. Select the device group that you want to add the device to.
   The Devices page displays.
6. Click the **Add** button.
   The Add Device page displays.
7. Select the name of the location where you want to add a device.
8. Click the **Add** button.
   The Add Device pop-up displays.
9. Select the **Single Device** radio button.
10. From the **Select Location** menu, select a location.
11. From the **Device Group** menu, select a device group.
12. In the **Serial Number** field, enter the serial number of the device.
13. In the **Device Name** field, enter a name for the device.
14. In the **MAC Address** field, enter the MAC address of the device.
15. Click the **Add** button.
    Your settings are saved.

Add multiple devices to a device group

To add multiple devices to a device group:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Click the name of the location where you want to create the device group.
   The dashboard page of the location displays.
4. Click the **Settings > Device Groups**.
   The Device Groups page displays.
5. Select the name of the location where you want to add a device.
   The Devices page displays.

6. Click the **Add** button.
   The Add Device pop-up displays.
7. Select the **Multiple Devices** radio button.
8. Select an organization.
9. In the **Device List** file section, click the **Click here** button to download the device template.

   NOTE: The device upload limit is 1000.
10. Click the **Open file** link to upload the completed device list file or drag and drop the file.
11. Click the **Next** button.
12. Select the check boxes next to the new devices you want to add.
13. Click the **Update Device List** button.
    The devices are added to the device group.

Manage VLANs and VLAN-based features for a location

In Insight, you can centrally manage VLANs across all supported devices from a single cloud-based interface. You can create new VLANs, assign VLAN IDs, and configure VLAN membership on individual ports or SSIDs without needing to log in to each device separately. Insight also lets you apply VLAN profiles to multiple devices or locations at once, monitor which VLANs are active on each port, and adjust settings remotely as your network changes.

This chapter contains the following sections:

- VLAN concepts
- Plan the VLANs in your Insight location
- VLAN membership and tagging
- Management VLAN concepts
- How a VLAN works on an Insight-managed switch
- Create a custom VLAN
- Configure port VLAN IDs for switch ports
- Configure multiple VLANs

VLAN concepts

You can define a local area network (LAN) as a broadcast domain. Hubs, bridges, switches, and WiFi access points in the same physical segment or segments connect all end nodes. End nodes can communicate with each other without a router. Routers connect LANs, routing the traffic to each appropriate port.

A virtual LAN (VLAN) is a local area network that maps devices on a basis other than geographic location, for example, by department, type of user, or primary application. Traffic that flows between different VLANs must go through a router, just as if the VLANs are on two separate LANs.

A VLAN is a group of network devices that behave as if they are connected to a single network segment, even though they might not be. For example, the marketing personnel might be located throughout a building, but if they are all assigned to a single VLAN, they can share resources and bandwidth as if they are connected to the same segment. The resources of other departments can be invisible to the marketing VLAN members, accessible to all, or accessible only to specific individuals, depending on how you set up the VLAN.

VLANs provide a number of advantages:

- VLANs let you easily segment your network. You can group users who communicate most frequently with each other in a common VLAN, regardless of physical location. Each group's traffic is contained largely within the VLAN, reducing extraneous traffic and improving the

efficiency of the whole network.
- VLANs are easy to manage. You can quickly add or change network nodes and make other network changes through the Insight Cloud Portal.
- VLANs provide increased performance. VLANs free up bandwidth by limiting node-to-node and broadcast traffic throughout the network.
- VLANs enhance network security. VLANs create virtual boundaries that can be crossed only through a router. Therefore, you can use standard, router-based security measures to restrict access to a VLAN.

Plan the VLANs in your Insight location

Before you set up VLANs, we recommend that you plan the entire physical and logical setup for the network (known as network topology) carefully. VLAN configuration mistakes can cause serious connectivity and security problems on your network. If you are not experienced setting up computer networks, consider hiring an IT or network professional.

We also recommend that you plan your network's logical topology (which devices must be connected to each other) before you plan the physical topology (where each device must go, how to run the Ethernet cables).

When multiple VLANs exist on your network, decide which ports must be members of each VLAN. All ports that are members of a VLAN receive traffic that is sent on that VLAN. Then, you must decide whether each port must be a tagged member or an untagged member of the VLAN. A port is tagged for a VLAN when traffic that leaves the switch through that port includes an IEEE 802.1Q header with that VLAN's numerical identifier (VLAN ID) on it. If a port is an untagged member of a VLAN, the switch removes the existing 802.1Q header before sending traffic through that port.

VLAN membership and tagging

The following are basic principles of VLAN membership and tagging:
- Each port can be a member of an unlimited number of VLANs, but traffic on that port will be slow if it is a member of several busy VLANs. If you plan to make a port a member of multiple VLANs, consider setting up a link aggregation group (LAG) for increased bandwidth and throughput over that connection.
- Each port can be an untagged member of a single VLAN only. If a port is already an untagged member of a VLAN, you cannot add it as an untagged member of any other VLANs.
- All untagged traffic that enters the switch is assigned to the default or native VLAN, which is VLAN 1. VLAN 1 is also the management VLAN on switches that support management VLANs.
- If a port is a member of a LAG or you plan to add it to a LAG, do not add it to a VLAN or tag it

individually. You must add the LAG to the VLAN as a single unit.

- We recommend that you classify each port as either an access port or a trunk port. An access port is a member of a single VLAN and connects to a computer, printer, or other device on the edge of a network. A trunk port connects the switch to a router, to other switches, or to access points. A trunk port must participate in multiple VLANs because all traffic that passes between the switch and the rest of the network must go through that port.
- Some networked devices recognize 802.1Q tagging, and some do not. If a device does not recognize tags, it rejects any tagged traffic that it receives, so it can be only an untagged member of a VLAN.
- If you are not sure whether a device supports 802.1Q tagging, see the device's documentation. The following list contains general guidelines that are not applicable in all cases:
  - Most printers do not recognize 802.1Q tags.
  - If a computer must be a tagged member of a VLAN, you must configure a VLAN ID on the network interface controller (NIC) of the computer. All other computers must be untagged.
  - Most network attached storage (NAS) devices either support 802.1Q tagging, support multiple NICs with multiple Ethernet ports (which can be added to different VLANs), or both.
  - Most Voice over Internet Protocol (VoIP) phones recognize 802.1Q tags.
  - Most WiFi access points recognize 802.1Q tags.
  - Unmanaged switches and some switches with limited management functions do not recognize 802.1Q tags.
  - Most business routers recognize 802.1Q tags. Most home routers do not.

Management VLAN concepts

A management virtual local area network (VLAN) is a much smaller network that is contained within your regular network. The primary benefit of using a management VLAN is improved network security. When all management traffic is on a separate VLAN, it is much harder for unauthorized users to make changes to your network or monitor network traffic.

Another potential benefit is that a management VLAN can help you minimize the impact of a broadcast storm on other VLANs by giving you a separate path to access your network.

On NETGEAR devices that support management VLANs, the management VLAN, VLAN 1, is also the native or default VLAN. By default, all ports are members of the default VLAN. For the management VLAN to be secure, it must be used only for controlling and managing your network devices. You must restrict access to the management VLAN and configure other VLANs to carry all regular network traffic.

If you decide to restrict access to the management VLAN, especially with an access control list

(ACL), make sure that you make your computer or device a member of the VLAN and add its MAC address to the ACL (if applicable). Otherwise, you must log in from an allowed device or lose access to the management functions of the switch. If you are unable to log in on an allowed device, you must reset the switch to factory default settings to regain management access.

How a VLAN works on an Insight-managed switch

A smart switch treats incoming packets in the following way:

- If an untagged packet enters a port, it is automatically tagged with the port's default VLAN ID. Each port is assigned a default VLAN ID that you can configure. The default setting is 1.
- If a tagged packet enters a port, the tag for that packet is unaffected by the default VLAN ID. The packet proceeds to the VLAN that is specified by the VLAN ID in the packet.
- If a packet enters through a port that is a member of the VLAN that is specified by the VLAN ID in the packet, the packet can be sent to other ports with the same VLAN ID.
- If a packet enters through a port that is not a member of the VLAN that is specified by the VLAN ID in the packet, the packet is dropped.
- Packets that leave the switch are either tagged (T) or untagged (U), depending on the VLAN to which the port belongs.

Create a custom VLAN

Create and configure a custom VLAN in the portal, to segment traffic, enhance security, and optimize network performance, including VLAN ID, ports, IP, security, and advanced settings.

To create a custom VLAN and configure default port VLAN IDs using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of the location where you want to add a VLAN.
   The location dashboard displays.
4. Click **Wired > VLAN/Network Setup**.
   The VLAN/Network Setup page displays.
5. Click the **Add** button.
   The Create New VLAN pop-up window displays.

6. In the **VLAN ID** field, enter a VLAN ID.
   VLAN IDs can be any number from 1 to 4093 except the IDs that are already reserved.

   The following VLAN IDs are reserved:
   - 1 (management VLAN)
   - 4088 (Auto-VoIP VLAN)
   - 4089 (Auto-Video VLAN)

7. In the **VLAN Name** field, enter a VLAN name.
8. From the **Network Type** menu, select one of the following options:
   - Data Network
   - Voice VLAN
   - Video LAN
9. Click the **Next** button.
   Graphics of the devices in the location display.

CAUTION: Configuring a VLAN on the network uplink port (the port that connects your device to the Internet) might disconnect your device from the Insight cloud. We recommend that you do not configure a VLAN on the cloud uplink port until you configure an alternate network cloud uplink.

NOTE: If you tag a port that is a member of a link aggregation group (LAG), all member ports of that LAG are automatically tagged.

10. Do one of the following:
    - Click the **Skip** button.
    - Configure the following settings:
        a. For routers only: Select the **Inter-VLAN Routing** toggle.
        b. For routers only: Select the **Device Management** toggle.
        c. Select the device ports that must be members of the VLAN by using the following options:
            - **Individual port**: Click an individual port to select it. Click a selected port again to clear the port. A selected port is indicated by a green check mark.
            - **Select All**: Click the **Select All** button under a switch to select all ports on that switch. After you click the **Select All** button, the button changes to a **Deselect All** button, allowing you to clear the selection of the ports on that switch.
            - **Delete**: Click the **Delete** button under a switch to clear all selected ports on that switch.
            - **Access Port**: Click an individual port and then click the **Access Port** button under the switch to make the selected port an access port on that switch. An access port is a port on which traffic is untagged. An access port is intended for a connection between the switch and an end device, for which the port does not need to process tagged frames because all traffic belongs to the same VLAN.
            - **Trunk Port**: Click an individual port and then click the **Trunk Port** button under the switch to make the selected port a trunk port on that switch. A trunk port is a port on which traffic can be tagged. A trunk port is intended for a connection between two switches (or a switch and a WiFi access point), for which the port must be capable of processing tagged frames to segment the traffic for different VLANs.
        d. To configure further VLAN settings, select the **Advanced Settings** tab, and do the following:
            i. Select the **Video Optimization (IGMP Snooping)** toggle.
            ii. From the **Traffic Classification** menu, select a priority from 0 to 7 to specify how important the traffic on this VLAN is. The highest priority is 7.
            iii. For switches only: Select the **RADIUS Configuration** toggle. If RADIUS Configuration is enabled, the toggle is positioned to the right.
            iv. For switches only: To configure MAC ACL, do the following:
                i. Select the **MAC ACL** toggle. If MAC ACL is enabled, the toggle is positioned to the right.
                ii. From the **Policy** menu, select one of the following modes:
                    - **Allow**: No devices are allowed to connect, except for devices for which you specify the IP addresses and that are then allowed access. To enable this mode, you must add at least one allowed device.
                    - **Deny**: All devices are allowed to connect, except for devices for which

you specify the IP addresses and that are then denied access. To enable this mode, you must add at least one denied device.

iii. To enable IP address filtering for the new VLAN, do the following:

i. Select the **Enable IP Filtering** toggle. If IP filtering is enabled, the toggle is positioned to the right.

ii. From the **Policy** menu, select one of the following modes:

- **Allow**: No devices are allowed to connect, except for devices for which you specify the IP addresses and that are then allowed access. To enable this mode, you must add at least one allowed device.
- **Deny**: All devices are allowed to connect, except for devices for which you specify the IP addresses and that are then denied access. To enable this mode, you must add at least one denied device.

iii. Click the **Add Devices** button. In the **Access Management** pop-up window, which shows automatically detected devices, select the check boxes with the IP addresses for the devices that you want to add, and click the **Add** button.

iv. To add a device manually, click the **Manual** button. In the **Manual Access Management** pop-up window, enter a device name or a range name in the **Device Name** field, and add a single IP address or a range of IP addresses by doing one of the following:

- **Single IP address**: In the **IP Address** field, enter an IP address, and click the **Add** button.
- **Range of IP addresses**: Select the **Add range of devices** check box, enter a network mask address in the **IP Mask** field, and click the **Add** button.

NOTE: If you set up a list with allowed devices, no matter how you add devices, be sure that you add the MAC address of the device that you are using to configure the VLAN, or you will lose access to the VLAN after you enable MAC authentication.

11. Click the **Next** button.
    The page adjusts.

12. Do one of the following:
    - To add a new SSID, click the **Add** button. For more information, see Add an SSID on page 75.
    - Click the **Skip** button.

    The page adjusts.
13. **(Optional)** For switches only: In the **IP Configuration** section, select one of the following:
    - **DHCP**: The routable DHCP mode IP settings that you can configure to the VLAN interface of the switch.
    - **Static**: The routable static mode IP settings that you can configure to the VLAN interface of the switch.
    - **None**: No routable IP settings can be configured to network interface of the switch, and you can configure the switch IP settings mode from the switch dashboard (**Dashboard > IP Settings**).
14. Click the **Next** button.
    The page adjusts.
15. **(Optional)** Routers only: Configure the DHCP Server settings:
    a. From the **Router Device** menu, select a router.
    b. In the **Gateway IP Address** field, enter a gateway IP address.
    c. Select the **DHCP Server** toggle, and configure the DHCP server settings.
    d. Click the **Next** button.
    The page adjusts and displays all the devices in the VLAN.
16. Click the **Next** button.
    The Creating VLAN pages displays and confirms that the VLAN is successfully created.
17. **(Optional)** To view the new VLAN in your location, click the **View Networks** button.
    The VLAN/Network Setup displays to show all VLANs in your network.
18. Configure port VLAN IDs. For more information, see Configure port VLAN IDs for switch ports on page 72.

Configure port VLAN IDs for switch ports

You can configure port VLAN IDs (PVIDs) on switch ports to control how untagged traffic is assigned to VLANs, improving network segmentation, security, and traffic separation for connected devices. When a device sends untagged frames, the switch tags those frames with the port's PVID and forwards them within that VLAN, ensuring proper segmentation and isolation of traffic for connected devices.

To configure port VLAN IDs for one or more ports on the same switch:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click the **Devices** tab.
   The Devices dashboard displays.
5. Click the name of a device.
   The dashboard of the device displays.
6. In the graphic, click the port that you want to configure.
   The Port Summary page displays.
7. Click the **Configure Port** button.
   The Port Configuration tab displays.
8. **(Optional)** Select a single port.
   The selected port displays green.
9. **(Optional)** To select multiple ports:
   a. Select the **Batch Port** radio button.
      A warning pop-up window opens to advise that this action overwrites any existing settings on the selected ports. It also advises that the values that are shown on the wizard page for the ports are the default values and not the currently configured values.
   b. In the graphic, select the ports that you want to configure.
      The selected ports display green.
10. From the **Default VLAN (PVID)** menu, select a VLAN.
    By default, the management VLAN with ID 1 is assigned to a port, so the PVID for the port is 1. When you select a VLAN, the PVID changes.
11. Click the **Save** button.
    Your settings are saved.

Configure multiple VLANs

You can configure multiple VLANs on individual switch ports in the interface using the Port Configuration options to tag, untag, or clear VLAN assignments, allowing you to segment network traffic, isolate devices, and apply different security or quality-of-service policies on the same physical port.

To configure multiple VLANs on switch ports using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click the **Devices** tab.
   The Devices dashboard displays.
5. Click the name of a device.
   The dashboard of the device displays.
6. In the graphic, click the port that you want to configure.
   The Port Summary page displays.
7. Click the **Configure Port** button.
   The Port Configuration tab displays.
8. Navigate to VLAN ID and select the VLANs you want to assign to the port.
9. Click the **TAG** or **UNTAG** buttons for each VLAN.
   When you tag a VLAN, a "T" displays in the top right corner of the VLAN check box.
   When you untag a VLAN, a "U" displays in the top right corner of the VLAN check box.
10. **(Optional)** To clear the configuration on a VLAN, click the **Clear** button.
11. Click the **Save** button.
    The Port Configuration pop-up window displays to advise you to ensure all VLAN settings are configured correctly.
12. Click the **Yes** button.
    Your settings are saved.

Manage the WiFi network and SSIDs for a location

You can configure and manage an Insight-managed WiFi network for a specific location or organization, including creating and updating SSIDs (WiFi network names) and controlling which features apply to the entire network or to individual SSIDs. A WiFi network consists of one or more SSIDs at a single network location. Some features apply to the entire WiFi network (all SSIDs at the location), while other features apply only to individual SSIDs. An SSID that you create on one access point at a location is deployed on all access points at that location.

This chapter contains the following sections:

- Add an SSID to a location or organization
- Configure customized SSID settings
- Add a captive portal for an SSID
- Add an SSID schedule
- Add a MAC ACL for an existing SSID
- Configure a management VLAN for access points in a location
- Configure rate limits for an existing SSID
- Configure Instant WiFi for a location
- Configure WiFi scheduling
- Configure the 802.1Q VLAN and management VLAN to tagged VLAN
- Manage load balancing
- Manage RF Profiles
- View WiFi Customer Profiles

Add an SSID to a location or organization

A WiFi network name is referred to as an SSID, which stands for service set identifier. Each Insight-managed access point can support multiple SSIDs.

You can add an SSID at either the organization level or the location level:

- **Organization-level SSIDs** are intended for use across all locations within an organization. Use this option to define a standard wireless network once and make it available across all sites, allowing centralized management of shared SSIDs such as a corporate network. Any changes to an organization-level SSID, such as a password update or security policy change, can be made in one place rather than repeated at each location.
- **Location-level SSIDs** are specific to a single network location and are suited to deployments where wireless networks differ between sites, for example a guest network with location-specific settings, a different VLAN assignment, or a local captive portal configuration.

To add a new WiFi network (SSID) to a location or organization:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Do one of the following:
   - Select the **Locations** tab, select the name of a location, and select **Wi-Fi > SSID**.
   - Select the **Organizations** tab, select the name of an organization, and select the **SSID** tab.

   The SSID page displays.
3. Click the **Add** button.
   The Add SSID pop-up window displays.
4. In the **SSID Name** field, enter an SSID name.
5. From the **Authentication Type** menu, select one of the following security methods:

Open	Provides an unsecured WiFi network that does not require a password. Data that clients send over the network is not encrypted.
WPA2 Personal	Uses a shared WiFi password (pre-shared key) to control access and encrypt wireless traffic with WPA2 security.
WPA2 Personal Mixed	Supports both older WPA and WPA2 clients using a shared WiFi password, allowing mixed client compatibility while using WPA2 when available.
WPA2 Enterprise	Uses WPA2 security with 802.1X authentication and a RADIUS server to provide per-user credentials and centralized access control.
WPA3 Personal	Uses WPA3 security with enhanced encryption and protection against password guessing, while still using a shared WiFi password.
WPA3 Personal Mixed (WPA2 + WPA3)	Allows both WPA2 and WPA3 capable clients to connect using the same SSID and password, using WPA3 security when supported by the client.
WPA3 Enterprise	Uses WPA3 security with 802.1X authentication and a RADIUS server, providing stronger encryption and per-user credentials for enterprise networks.

6. In the **Password** field, enter a password.

   NOTE: If you configure the authentication type as Open, you do not need to configure a password.

7. To assign a VLAN other than the Management VLAN to the SSID, do one of the following:
   - From the **VLAN ID** menu, select an existing VLAN.
   - Click the **Add VLAN ID** button. For more information, see Create a custom VLAN on page 68.

8. In the **Advanced Settings** section, select one of the following radio buttons:
   - Use Default Settings
   - Customize: Configure the customized SSID settings. For more information, see Configure customized SSID settings on page 77.
9. Click the **Save** button.
   A pop-up window displays to advise the SSID is created successfully.

> **NOTE:** An SSID that you create on one access point at a network location is deployed on and broadcast by all access points at the network location. That is, even though you create the SSID on a single access point, the SSID is not specific to that single access point.

Configure customized SSID settings

Configure advanced SSID options in the Insight Cloud Portal to control how clients connect, secure wireless traffic, and optimize performance for your network. You can configure these settings in both location-level and organization-level SSIDs.

To configure customized SSID settings using the Insight Cloud Portal:

1. To add an SSID, complete the steps outlined in Add an SSID to a location or organization on page 75.
2. In the **Advanced Settings** section, select the **Customize** radio button.
3. In the **Broadcasting Selection** section, select one of the following options.
   a. **Broadcast**: The SSID is visible to client devices and displays in the list of available WiFi networks.
   b. **Don't Broadcast**: The SSID is hidden and does not display in the list of available WiFi networks. Clients must know and enter the SSID name to connect.
4. In the **Band** section, select the check boxes of the radio bands you want the SSID to use.
   a. **2.4GHz**: This WiFi band provides longer range and better wall penetration, but supports lower maximum data rates and is more prone to interference from other devices. It is available for all authentication types.
   b. **5GHz**: This WiFi band provides higher data rates and less interference than 2.4GHz, but with shorter range and less wall penetration. It is available for all authentication types.
   c. **6GHz**: This WiFi band provides the highest potential data rates and the most spectrum, with very low interference, but has the shortest range and requires WiFi 6E or later client devices. It is available only when using WPA3-based authentication types (WPA3 Personal, WPA3 Personal Mixed, or WPA3 Enterprise).

5. To activate customer profiles on the SSID, select the **Customer Profiles** toggle, and from the **Customer Profiles** list, select a customer profile.
   Customer profiles let you apply predefined policies and settings to wireless clients that connect to this SSID. By assigning a customer profile, you can consistently enforce access controls, bandwidth limits, content filtering, and other network policies based on the type of customer or use case, without configuring these options separately for each SSID.

   NOTE: This setting is available only for location-level SSIDs.

6. Select the **Enhanced Open (OWE)** check box to enable encrypted connections on open WiFi networks without requiring a password.
   Enhanced Open (OWE) automatically encrypts each client's wireless traffic, protecting data on an otherwise open SSID while still allowing users to connect without entering a password. This helps improve privacy and security for guest or public networks without adding extra steps to the connection process.

   NOTE: This option is available only for the Open authentication type.

   Depending on the band you selected in step 4, note the following behavior:
   - **Dual-Band (2.4 GHz and 5 GHz)**: Enhanced Open (OWE) applies as described above.
   - **Tri-Band (2.4 GHz, 5 GHz and 6 GHz)**: Enhanced Open / Legacy Open (OWE Transition Mode) is automatically selected and the Enhanced Open option is greyed out.
   - **6 GHz only**: Enhanced Open (OWE) is automatically selected and greyed out.

7. If you select the **Enhanced Open (OWE)** check box, select or deselect the **Allow clients to authenticate using legacy open (OWE Transition Mode)** check box. When you select this check box, the 802.11w setting is automatically set to Optional. To configure Enhanced Open (OWE) as Mandatory, deselect this check box.

8. To configure the 802.11w (PMF) settings, select one of the following radio buttons:
   - **Mandatory**: Requires all client devices to use Protected Management Frames (PMF). This provides the highest level of protection against management frame attacks, but older clients that do not support PMF cannot connect.
   - **Optional**: Enables PMF for clients that support it, while still allowing non-PMF clients to connect. This offers improved protection without blocking legacy devices.
   - **Disable**: Turns off PMF protection for management frames. All clients can connect, including legacy devices, but they are more vulnerable to attacks such as deauthentication or disassociation.

   802.11w (Protected Management Frames) is a WiFi security feature that encrypts and authenticates management frames, which helps prevent attacks such as deauthentication or disassociation that can disconnect clients from the network.

   NOTE: Depending on the what authentication type you select, only certain 802.11w (PMF) settings are configurable.

9. To create and assign a multi pre-shared key, activate the **Multi Pre-Shared Key (MPSK)** toggle.
   a. From the **Select Multi Pre-Shared Key** menu, select a MPSK key.
   b. Click the **Add Pre-Shared Key** button.
      The Add MPSK Key page displays.
   c. Complete the steps outlined in Create a multi pre-shared key for an SSID on page 100.

   NOTE: This setting is available only for location-level SSIDs.

   NOTE: This feature is only applicable to WPA2 Personal and WPA2 Personal Mixed Authentication Types. This option does not display if you select a different authentication type.

10. To add a captive portal to the SSID, click the **Add Captive Portal** button.
    For more information, see Add a captive portal for an SSID on page 82.

11. From the **Addressing and Traffic** menu, select one of the following options.
    a. **Bridge**: Places wireless clients on this SSID directly onto the selected VLAN, using IP addresses from your existing DHCP server. Use this option when clients must be on the same network as wired devices to discover or access local resources such as printers, servers, or controllers.
    b. **NAT**: Assigns IP addresses to wireless clients and performs network address translation on their traffic. Use this option to isolate wireless clients from the wired network, hide client IP addresses, and simplify IP management for guest or high density networks. If you configure this setting, you must assign the management VLAN to the SSID.

12. If you select **NAT** from the **Addressing and Traffic** menu, do the following:
    a. In the **Network Address** field, enter the base IP address of the private subnet. This defines the range from which clients are assigned IP addresses. The default value is 0.0.0.0.
    b. From the **Subnet** drop-down menu, select the subnet mask for your network. This determines the size of the private IP address pool. For example, 255.255.252.0 provides up to 1022 usable client addresses, suitable for larger deployments.
    c. In the **DNS** field, enter the IP address of the DNS server that clients will use to resolve domain names. The default value is 8.8.8.8, but this can be replaced with an internal or preferred DNS server.
    d. In the **Lease Time (Mins)** field, enter the duration in minutes that a client can hold its assigned IP address before it must renew. The default value is 1440 (24 hours). Shorter leases free up addresses faster in high-turnover environments such as guest networks.
    e. If your deployment is a Multi-Dwelling Unit such as an apartment building or shared space, activate the **MDU (Multi-Dwelling Unit)** toggle to isolate independent users sharing the same infrastructure.

13. To activate client isolation on the SSID, click the **Client Isolation** toggle.
    If you enable client isolation, the access points block communication between WiFi clients that are associated with the same SSID or different SSIDs on the access points. This option can be useful for an open network. By default, client isolation is disabled.

14. If you activate the **Client Isolatio**n toggle, select or deselect the **Add wired IP Address or Device Name with Domain to allow access** check box.
    This setting lets you define specific wired resources that wireless clients on this SSID are allowed to reach.

15. To activate multi-link operation (MLO), select the **Multi Link** toggle.
    Multi Link lets compatible WiFi 7 clients use multiple bands at the same time on this SSID to improve performance and reliability. By default, MLO is disabled.

16. To activate band steering, select the **Band Steering** toggle.
    Band steering encourages dual-band and tri-band client devices to connect on higher-capacity bands (such as 5GHz or 6GHz) instead of 2.4GHz. This helps reduce congestion on 2.4GHz, improve overall WiFi performance, and provide better speeds and reliability for supported clients. By default, band steering is disabled.

17. To activate SSID scheduling, select the **SSID Schedule** toggle.
    SSID scheduling lets you automatically turn the SSID on or off based on a defined schedule. You can use this to restrict WiFi availability to specific days or times, such as business hours, to improve security and reduce unnecessary network usage.

    The page adjusts.

    NOTE: The setting is availble only for location-level SSIDs.

18. To set an SSID schedule, from the **Select Schedule** menu, select an existing schedule or click the **Add Schedule** button. For more information, see Add an SSID schedule on page 84.

19. Activate or deactivate the **802.11k/v** toggle.
    This setting helps devices automatically switch to the strongest access point as you move around, reducing lag and dropped connections in multi-AP environments.

20. Activate or deactivate the **Fast Roaming 802.11r** toggle.
    This setting speeds up the security re-authentication when a device roams between access points, reducing interruptions for latency-sensitive traffic such as VoIP and video conferencing.

21. To activate MAC ACL, select the **MAC ACL** toggle.
    MAC ACL (Media Access Control Access Control List) lets you allow or block specific client devices based on their MAC addresses. You can use this to restrict SSID access to known devices, add an extra layer of security for sensitive networks, or prevent unauthorized or misbehaving clients from connecting.

22. From the **Type** menu, select one of the following types of MAC ACL.
    a. **Local ACL**: A MAC address list that you configure and store directly on the device or in this management system. The access point checks client MAC addresses against this local list to allow or block devices.
    b. **Radius ACL**: A MAC address list that is stored on an external RADIUS server. The access point sends the client MAC address to the RADIUS server, which returns an allow or deny decision based on its configured policies.

    For more information on configuring local ACLs and radius ACLs, see Add a MAC ACL for an existing SSID on page 85.

    NOTE: You must disable MLO before you can enable a MAC ACL.

23. Activate or deactivate the **Block Devices by OS** toggle.
    This setting allows the AP to identify the operating system of connecting devices and selectively deny network access based on OS type.

24. To activate rate limiting on the SSID, click the **Rate Limiting** toggle.
    Rate limiting lets you cap the maximum bandwidth that wireless clients can use on this SSID, which helps prevent a few devices from consuming excessive bandwidth and improves overall performance for all users. For more information, see Configure rate limits for an existing SSID on page 87.

25. To activate traffic policies, select the **Traffic Policies** toggle.
    Use traffic policies to control which applications or types of traffic are allowed, prioritized, or blocked on this SSID, so you can enforce security rules, optimize bandwidth for critical apps, and limit non-business or high-bandwidth usage.

26. To activate advanced rate selection, select the **Advanced Rate Selection** toggle, and complete the following steps:
    a. Select an available radio band, depending on what bands are supported by your AP model: **2.4GHz**, **5GHz**, or **6GHz**.
    b. From the **Fixed Multicast Rate** menu, select a speed depending on the radio band you are using.
    This setting configures a specific data rate at which the AP transmits multicast and broadcast traffic to all clients on the network.
    c. Select or deselect the **Set Minimum Rate Controls** check box.
    This setting configures the lowest data rate at which an AP communicates with a client. Any device that can only connect at or below that threshold is refused association or dropped.
    This setting fine tunes which data rates clients can use on this SSID.

27. Click the **Save** button.
    Your settings are saved.

Add a captive portal for an SSID

You can configure a captive portal for an SSID in Insight to control network access by requiring users to log in, accept terms, or authenticate before connecting. This feature helps manage guest access, enhance security, and deliver a customizable onboarding experience for users joining the network.

To add a captive portal for a new or existing SSID:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.

2. Click the **Locations** tab.
   The Locations page displays.

3. Select the name of a location.
   The location dashboard displays.
4. Click **Wi-Fi > SSID**.
   The SSID page displays.
5. Click the **Add** or **Edit** button.
   The Add SSID or Edit SSID pop-up window displays.
6. In the **Advanced Settings** section, select the **Customize** radio button.
7. In the **Captive Portal** section, click the **Add Captive Portal** button.
   The Add Captive Portal pop-up page displays.
8. Click either the **Insight Captive Portal (Free)** or **External Captive Portal** radio button.
9. If you select the **Insight Captive Portal (Free)**, do the following:
   a. In the **Welcome Message** field, add a welcome message that users see when the page opens.
   b. In the **Logo** section, select either the **Show Logo** or **Hide Logo** radio button.
   c. Click the **Upload** button to configure a logo for the page.
   d. In the Background section, select either the **Use Image** or **Use Color** radio button.
   e. If you select the **Use Image** radio button, click the Upload button to configure a background image for the page.
   f. If you select the **Use Color** radio button, to change to another color, click the selected color.

10. If you select the **External Captive Portal**, do the following:
    a. From the **External Captive Portal services** menu, select a service.
    b. In the **Splash Page URL** field, enter the URL for client devices to be redirected to upon first HTTP packet access.
    c. In the **Captive Portal Authentication Type** section, select either the **Web/HTTP** or **RADIUS** radio buttons.
    d. If you select the **Web/HTTP** radio button, enter a Web Authentication URL, Secret and Key.
    The Web authentication URL is provided by the External Captive Portal vendor to authenticate the Splash Page session with the Captive Portal Authentication Servers.
    e. If you select the **RADIUS** radio button, enter a NAS Identifier, and optionally configure the settings in the **Primary Authentication Server** and **Accounting Interval** sections.
    A NAS Identifier is a label that tells the RADIUS server which device is making the authentication request.
    f. In the **Failsafe** section, select the **Enable** or **Disable** radio buttons.
    The Failsafe setting provides short-term Internet access in case the external captive portal is disrupted.
    g. In the **Walled Garden** field, enter a URL and click the Add button.
    The Walled Garden is a list of URLs that client devices can access before External Captive Portal authentication.
11. Click the **Save** button.
    Your settings are saved.

Add an SSID schedule

SSID scheduling lets you automatically turn the SSID on or off based on a defined schedule. You can use this to restrict WiFi availability to specific days or times, such as business hours, to improve security and reduce unnecessary network usage.

To add an SSID schedule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Do one of the following:
   - Select the **Locations** tab, select the name of a location, and select **Wi-Fi > SSID**.
   - Select the **Organizations** tab, select the name of an organization, and select the **SSID** tab.

   The SSID page displays.

3. Click the **Add** button, or select the check box of an SSID and click the **Edit** button.
   The Add SSID or Edit SSID pop-up window displays.
4. In the **Advanced Settings** section, select the **Customize** radio button.
5. To activate SSID scheduling, select the **SSID Schedule** toggle.
6. From the **Select Schedule** menu, select an SSID schedule.
7. Alternatively, click the **Add Schedule** button to create a new schedule.
   The Add Schedule pop-up window displays.
8. In the **Schedule Name** field, enter a name.
9. In the **Days** section, select up to three days, and enter up to three time slots for each day.
10. Click the **Save** button.
    Your settings are saved.

Add a MAC ACL for an existing SSID

For added security, an SSID can support either a RADIUS access control list (ACL), for which you must configure a RADIUS server, or a local ACL that you must compose of individual WiFi devices.

If you set up an ACL with a policy that denies access and you enable the ACL, the ACL functions as follows:

- A WiFi device for which you placed the MAC address in the ACL is denied access to the WiFi network.
- All other WiFi devices are allowed access to the WiFi network.

To set up a MAC ACL for an existing SSID at a location using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wi-Fi > SSID**.
   The SSID page displays.
5. Click the three dots on the right of an SSID and click **Edit**.
   The Edit SSID pop-up page displays.
6. In the **Advanced Settings** section, select the **Customize** radio button.

7. Select the **MAC ACL** toggle
8. Do one of the following:
   - From the **Type** menu, select **Local ACL**.
     a. From the **Policy** menu, select **Allow**.
     b. Do one of the following:
        - Manually add devices to the ACL:
          i. Click the **Manual** button. The Manual Access Management pop-up window displays.
          ii. In the **Device Name** field, enter a device name.
          iii. In the **MAC Address** field, enter a MAC address for a device.
          iv. Click the **Add** button. The page adjusts and the device displays in the Allowed Devices table.
        - Add an existing device to the ACL:
          i. Click the **Add Devices** button. The Manual Access Management (Allow) pop-up window displays.
          ii. Select the check box of the device you want to add to the ACL.
          iii. Click the **Allow** button. The page adjusts and the device displays in the Allowed Devices table.
   - From the **Type** menu, select **Radius ACL**.
9. Click the **Save** button.
   Your settings are saved.

Configure a management VLAN for access points in a location

A management VLAN is a dedicated virtual LAN that carries only management traffic for your access points. In Insight, you assign a management VLAN ID to your APs so that Insight can securely discover, monitor, and configure them over the network while keeping management traffic separate from user data.
Verify that the new management VLAN offers internet access to your APs so that you can manage your device with the new VLAN. Your APs might not apply the new VLAN configuration if no network or Internet access is detected with the new VLAN. In such case, the AP keeps the previous configuration where, by default, VLAN 1 is untagged.

To configure a management VLAN for access points using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.

2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wi-Fi > Management VLAN**.
   The Management VLAN page displays.
5. Do one of the following:
   - From the **Management VLAN** menu, select a management VLAN. By default, the management VLAN is VLAN 1.
   - To add a custom management VLAN, click the **Add Custom VLAN** button. The **Add Custom VLAN ID** pop-up window displays.
     a. In the **VLAN Name** field, enter a VLAN name.
     b. In the **VLAN ID** field, enter a VLAN ID. The ID must be between 1 and 4093.
     c. Click the **Add** button.

   NOTE: Ensure that the new management VLAN offers internet access to your APs so that you can manage your device with the new VLAN.
6. Select the **Untagged VLAN** toggle.
   By default, this setting is enabled.
7. Repeat step 5.
8. Click the **Save** button.
   Your settings are saved.

Configure rate limits for an existing SSID

If you notice WiFi speeds slowing because of heavy use on an SSID, you might want to set upload rate limits, download rate limits, or both. These limits apply only to the SSID that you configure the limits on. The minimum bandwidth rate is 64 Kbps, the maximum bandwidth rate is 1024 Mbps. You can set one rate for the upload bandwidth and another rate for the download bandwidth.

To configure rate limits for an existing SSID using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.

3. Select the name of a location.
   The location dashboard displays.
4. Click **Wi-Fi > SSID**.
   The SSID page displays.
5. Click the three dots on the right of an SSID and click **Edit**.
   The Edit SSID pop-up window displays.
6. In the **Advanced Settings** section, select the **Customize** radio button.
7. Select the **Rate Limiting** toggle.
   The page adjusts.
8. From the **Upload Rate Unit** menu, select either **Kbps** (kilobits per second) or **Mbps** (megabits per second).
9. In the **Upload Limit** section, move the slider to choose an upload bandwidth rate.
10. From the **Download Rate Unit** menu, select either **Kbps** or **Mbps**.
11. In the **Download Limit** section, move the slider to choose a download bandwidth rate.
12. Click the **Save** button.
    Your settings are saved.

Configure Instant WiFi for a location

The Instant WiFi Preferred Channels feature monitors the RF environment of all connected wireless devices (such as access points, routers, and satellites) and continuously changes the wireless settings when needed to sustain performance levels.

NOTE: Instant WiFi is applicable to access points only.

Instant WiFi automatically computes the best channels and transmit powers for its devices. Insight administrators can optimize channels with two options:

- **On Demand (Optimize Now)**
- **Scheduled**

To configure Instant WiFi for a location using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.

3. Select the name of a location.
   The location dashboard displays.
4. Click **Wi-Fi > Instant WiFi**.
   The Instant WiFi page displays.
5. To trigger the Instant WiFi optimization immediately, click **Optimize Now > Confirm**.
   A pop-up warning displays to advise of scheduled optimization.
6. To set a WiFi optimization schedule, from the **Schedule** menu, select one of the following:
   - None
   - Aggressive (Every 6 hours)
   - Moderate (Every 12 hours)
   - Mild (Every 24 hours) The date and time of the next scheduled run displays under the Schedule menu.
7. Enable or disable the **Automatic Transmit Power Allocation** toggle.
8. From the **Default Transmit Power** menu, select one of the following:
   - Full
   - Half
   - Quarter
9. Enable or disable the **Automatic Channel Allocation** toggle.
   The Automatic Channel Allocation feature detects known and unknown neighbors, considers the channel load and other critical parameters, and selects the best channel of operation for each device. By default, this setting is enabled.
10. In the **Preferred Channels** section, select three channels in each band that includes at least one non-DFS channel.
    - To select all the channels, click the **Select All** button.
    - To see the recommended channels, click the **Restore Defaults** button.
11. In the **Radio Settings** section, for each AP on the network, override the channel and power settings, and specify the channel width.
12. Click the **Save** button.
    Your settings are saved.

NOTE: For more information on Instant WiFi configuration, click the **Learn More** link at the top of the page.

Configure WiFi scheduling

You can set a WiFi schedule for each Insight network location. WiFi schedules allow you to limit the hours during which WiFi is available at an Insight location. For example, if your business is only open on certain days of the week, you can turn off WiFi on the days your business is closed.

To configure WiFi scheduling:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wi-Fi > Schedules**.
   The Schedules page displays.
5. Click the **Add** button.
   The Add New Schedule pop-up window displays.
6. In the **Schedule Name** field, enter a name for the WiFi schedule.
7. In the **Description** field, enter a description for the schedule.
   The description must be 1 to 32 characters in length.
8. In the **Days** section, select the days that WiFi is available.
9. For each day, in the **Start Time** field, manually enter a start time or click the clock icon to select a start time.
   You can add a maximum of three time slots for each day.
10. **(Optional)** To delete a time slot, click the red **X** icon.
11. Click the **Save** button.
    Your settings are saved.

Configure the 802.1Q VLAN and management VLAN to tagged VLAN

The 802.1Q VLAN protocol on APs logically separates traffic on the same physical (wired) network, which is essential for keeping management, user, and guest traffic isolated and secure. By using 802.1Q with a tagged VLAN for management traffic, you ensure that critical control traffic is clearly identified, easier to monitor, and less likely to be disrupted by other network segments. You can use Insight to configure this behavior on APs in a network location.

The AP tags all frames that it sends from its Ethernet interface, ensuring that management, user, and guest traffic remain logically separated on the same physical network. Only incoming frames that are tagged with known VLAN IDs are accepted. The management VLAN controls traffic such as Telnet, SNMP, HTTP, and HTTPS traffic sent to and from the AP. Frames that belong to the management VLAN and that are sent over the trunk do not receive an 802.1Q header.

To set the 802.1Q VLAN and management VLAN to tagged VLAN:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wi-Fi > Management VLAN**.
   The Management VLAN page displays.
5. Do one of the following:
   - From the **Management VLAN** menu, select a management VLAN. By default, the management VLAN is VLAN 1.
   - To add a custom management VLAN, click the **Add Custom VLAN** button. The **Add Custom VLAN ID** pop-up window displays.
     a. In the **VLAN Name** field, enter a VLAN name.
     b. In the **VLAN ID** field, enter a VLAN ID. The ID must be between 1 and 4093.
     c. Click the **Add** button.

NOTE: If you change the VLAN ID, make sure that your network supports it. For example, devices on VLAN 60 get DHCP addresses in VLAN 60. The AP tags all frames it sends from its Ethernet interface and accepts only incoming frames tagged with known VLAN IDs. Your LAN switches must support 802.1Q VLAN, and the new VLAN ID must be tagged on the ports of your Insight APs.

6. Deselect the **Untagged VLAN** toggle.
   By default, this setting is enabled.
7. Click the **Save** button.
   Your settings are saved.

Manage load balancing

Load balancing facilitates equal distribution of wireless clients across your Insight-managed access points. When an Insight-managed AP reaches its capacity based on the number of clients or channel utilization, load balancing prevents new clients from connecting. It can prevent far away clients with low received signal strength indication (RSSI) from connecting, and dissociate connected clients so they can connect to a neighboring Insight-managed AP. You can also enable the disassociate sticky client feature to forcibly disconnect clients when their RSSI gets too low so they can reconnect to a stronger AP.

To manage load balancing on an access point using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wi-Fi > Load Balancing**.
   The Load Balancing page displays.
5. Select the **Load Balancing** toggle.
   The page adjusts.
6. Expand the **Maximum number of clients per radio** menu.
7. Move the slider of each radio band to configure the maximum number of clients allowed per radio band.
8. Expand the **Balance on client RSSI** menu.
9. Move the slider of each radio band to set the minimum RSSI value required for clients to stay associated.
   Clients with an RSSI below the configured threshold might be rejected or disassociated, encouraging them to roam to a closer access point. This setting is measured in dBm (decibel-milliwatts), and must be between -94 dBm and -45 dBm.
10. Expand the **Balance on channel utilization** menu.

11. Move the slider of each radio band to set the channel utilization threshold.
    This setting is measured in percentage, and must be between 50 and 90%.
12. Enable or disable the **Disassociate sticky clients** toggle.
    If you enable this toggle, the page adjusts.
13. Move the slider of each radio band to configure the RSSI thresholds at which sticky clients are disassociated.
    This setting is measured in dBm, and must be between -90 dBm and -70 dBm.
14. Click the **Save** button.
    Your settings are saved.

Manage RF Profiles

An RF profile is a named collection of radio frequency settings that controls how access points transmit and receive wireless signals at a location.

Why RF profiles matter

Rather than configuring each access point individually, you define an RF profile once and apply it to any number of access points at a location. When you update the profile, all assigned access points inherit the change automatically.

The radio environment varies significantly between deployment scenarios. RF profiles let you tune radio behavior to match the physical environment and the density of connected devices, without reconfiguring access points one by one.

Radio bands

Each RF profile can include settings for up to four radio bands. Access points apply only the bands their hardware supports.

> **2.4 GHz**
>
> Longest range and best wall penetration, but the most congested band due to overlap with other devices (microwaves, Bluetooth, neighboring networks). Best suited for legacy clients and coverage-critical areas.

> **5 GHz (Low and High)**
>
> Higher throughput and less interference than 2.4 GHz, with shorter effective range. Splitting the band into low and high sub-bands gives the access point more flexibility in channel selection and client steering.

> **6 GHz**
>
> Available on Wi-Fi 6E and Wi-Fi 7 (802.11ax/be) hardware. Provides the most available channels and the least interference, delivering the highest potential throughput for capable

clients.

Key settings

Radio Mode

The 802.11 standard the radio advertises, such as 11be (Wi-Fi 7). Higher modes support faster speeds but require compatible client devices.

Channel Width

The amount of spectrum allocated to a single channel (20, 40, or 80 MHz). Wider channels increase throughput but reduce the number of non-overlapping channels available, which can increase interference in dense environments.

Preferred Channel

One or more channels the access point should prefer during startup and after a radar event. Pinning channels reduces roaming disruption in multi-AP deployments.

DTIM Interval

Delivery Traffic Indication Message interval. Controls how many beacon periods pass before the access point wakes sleeping clients to deliver buffered broadcast and multicast traffic. Lower values reduce latency for mobile devices; higher values extend battery life.

Output Power

The transmit power of the radio (for example, Half or Full). Higher power extends range but can cause co-channel interference in dense deployments. Lower power tightens cell size and improves client roaming behavior.

Beacon Interval

How frequently the access point broadcasts beacon frames, in milliseconds. The default value of 100 ms suits most deployments. Increasing the interval reduces overhead; decreasing it helps clients discover the network faster.

Common profile types

Two profiles cover most enterprise and hospitality deployments:

Open Office

Optimized for indoor, high-density environments with many clients in close proximity. Typically uses conservative channel widths on 2.4 GHz and moderate output power to reduce co-channel interference.

> **Outdoor**
> Designed for exterior access points that need longer range. May use higher output power and wider channels on 5 GHz where spectrum is less contested. Note that 6 GHz is not included in outdoor profiles in many regulatory domains.

You can create additional profiles to match specific environments, such as warehouses, auditoriums, or high-security areas with restricted channel sets.

Add an RF profile to an access point in a location

To add an RF profile to an access point:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click the **Devices** tab.
   The Devices page displays.
5. In the **Devices** table, click the three dots to the right of a device, and click **Assign RF Profile**.
   The Assign RF Profile pop up displays.
6. From the RF Profile menu, select a profile to assign to the device.
7. **Alternatively, click the Create New Profile button to create a new RF profile.**
8. Click the **Assign** button.
   Your settings are saved.

Edit an RF profile

You can update the configuration of an RF profile at any time.

To edit an RF profile:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.

3. Select the name of a location.
   The location dashboard displays.
4. Click **Wi-Fi > Profiles**.
   The RF Profiles page displays.
5. In the table, select the check box of the RF profile you want to edit, and click the **Edit** button.
   The Edit RF Profile pop-up page displays.
6. Update the existing configuration, and click the **Save** button.
   Your settings are saved.

Delete an RF profile

When you no longer require an RF profile, you can delete it at any time.

To delete an RF profile:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wi-Fi > Profiles**.
   The RF Profiles page displays.
5. In the table, select the check box of the RF profile you want to delete, and click the **Delete** button.
   The Delete RF Profile pop-up window displays.
6. Click the **Delete** button.
   The RF profile is deleted and no longer displays in the table.

View WiFi Customer Profiles

A WiFi customer profile is a named set of wireless policies that defines the experience for clients connecting to the network at a location. Where an RF profile controls how the access point radio behaves, such as channel, power, and radio mode, a customer profile controls what connecting clients can do and how they are treated on the network.

Customer profiles are applied for SSID or client group, allowing you to enforce different policies for different types of users or devices at the same location. For example, a retail location might use one customer profile for point-of-sale devices with strict rate limiting, and another for guest WiFi with more permissive settings.

Each customer profile configures the following:

- **802.11k/v**: Enables roaming assistance standards that help clients identify and transition to a stronger access point. 802.11k provides neighbor reports so clients know which APs are nearby; 802.11v allows the network to suggest roaming targets to clients.
- **Rate Limiting**: Sets maximum upload and download speeds for clients using this profile, preventing any one client or group from consuming excessive bandwidth.
- **Multicast Rate**: Defines the transmission rate for multicast traffic on each radio band (2.4 GHz, 5 GHz or 5 GHz Low, 5 GHz High, and 6 GHz). A higher multicast rate reduces airtime used by multicast frames but requires clients to be within good signal range.

The Customer Profiles tab lists all WiFi customer profiles configured for the location.

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wi-Fi > Profiles > Customer Profiles**.
   The Customer Profiles page displays.

   The customer profiles table displays, listing all profiles for the location with the following columns:
   - **Name**: The profile name. Click the arrow to sort alphabetically.
   - **802.11k/v**: Whether 802.11k/v roaming assistance is enabled or disabled for this profile.
   - **Rate Limiting (Upload / Download)**: Whether bandwidth rate limiting is applied to clients using this profile.
   - **Multicast Rate**: The multicast transmission rate configured per radio band: 2.4 GHz, 5 GHz or 5 GHz Low, 5 GHz High, and 6 GHz.
   - **Description**: An optional description of the profile.

5. (Optional) To find a specific profile, enter its name in the **Search** field.

   The customer profiles table displays all profiles configured at the location. The total number of profiles is displayed below the table.

Next steps

To export the profile list, click the download icon at the top right of the table. To add or edit a customer profile, use the **Add** and **Edit** buttons that appear when a profile is selected.

Manage advanced wireless features and optimization

Use advanced wireless features to fine-tune your network's performance, enhance security, and optimize coverage for all connected devices in your home or office.

This chapter contains the following sections:

- Configure fast roaming for a WiFi network
- Create a multi pre-shared key for an SSID
- Configure mesh settings for access points in a location
- Configure a multi pre-shared key group for an SSID
- Configure WiFi traffic policies
- Configure wireless band settings
- Configure WiFi network settings
- Configure URL filtering for all WiFi clients in a location

Configure fast roaming for a WiFi network

You can configure fast roaming on a WiFi network so client devices can switch between access points with minimal interruption.

To configure fast roaming for a WiFi network:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The location dashboard displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wi-Fi > Advanced > Fast Roaming**.
   The Fast Roaming page displays.
5. Activate the **Enable** toggle so that the button is positioned to the right.
   Your settings are saved.

NOTE: When this setting is activated, you can also activate Fast Roaming (802.11r) for an SSID in **Wi-Fi > Advanced > Wireless Roaming**.

Create a multi pre-shared key for an SSID

A Multi Pre-Shared Key (MPSK) allows wireless devices to connect to a single SSID using one of several passwords. The wireless device becomes a member of the VLAN associated with the password.

MPSK has the following limitations:

- The SSID must use WPA2 Personal or WPA2 Personal Mixed encryption mode. The 6 GHz band does not support these encryption modes and cannot be used with MPSK.
- WAC5XX series wireless access points do not support MPSK.
- MPSK cannot be used with the captive portal, client isolation, or NAT mode features.

To create a multi pre-shared key for an SSID using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select **Wi-Fi > Advanced > Multi PSK Settings**.
   The Multi PSK Settings page displays.
5. Click the **Add** button.
   The Add MPSK Key page displays.
6. Select the **Single MPSK** radio button.
7. In the **Key Identifier** field, enter a unique name that identifies this pre-shared key.
8. In the **Password** field, enter a password.
   Wireless devices use this password to connect to the SSID and join the associated VLAN.
9. In the **VLAN ID** field, enter the VLAN ID that you want to associate with this pre-shared key.
10. In the **Max Clients** field, configure the maximum number of clients that can use this pre-shared key simultaneously.
    The Max Clients range is between 1-50. If you enter an amount outside of this range, an input validation error displays.
11. In the **MAC Address** field, enter the MAC address of the client device that you want to associate with this pre-shared key.

12. From the **Expiry** menu, select one of the following:
    - **Never**
    - **Date and Time**
      a. In the **Date** field, click the calendar icon to specify a date.
      b. In the **Time** field, click the clock icon to specify a time.
13. Click the **Add** button.
    Your settings are saved.

> **NOTE:** You can also create and assign a multi pre-shared key when you create an SSID. For more information, see Configure customized SSID settings on page 77.

Configure mesh settings for access points in a location

Mesh settings in Insight let you create and manage wireless mesh networks for your access points, improving coverage, reliability, and roaming performance across a location without requiring wired connections for every device. For more information about mesh WiFi networks, see What's a mesh WiFi system?

To configure mesh settings for access points in a location using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The location dashboard displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wi-Fi > Advanced > Mesh Settings**.
   The Mesh Settings page displays.
5. To configure mesh mode on access points in the location, select one of the following options:
   - **Auto**: AP automatically detects whether it should act as a Root or Extender.
   - **Disable Mesh**: The AP does not mesh with other APs. This is the default setting.

   The Confirmation pop-up window displays.
6. Click the **Yes** button.
   Your settings are saved.
7. Connect your APs to a wired network with internet access using Ethernet cables.
8. Add your Insight Instant Mesh network APs to your Insight account. For more information, see Add a device on page 13.

9. Wait until the onboarding process finishes for all of the Insight Instant Mesh APs.When the onboarding process is finished, the device status LEDs in the Insight Cloud Portal are green and the power LEDs on the APs are solid blue.
10. Disconnect the Ethernet cables from the extender APs.
    The extender APs enter root AP discovery mode.
11. Wait 5 to 15 minutes for each extender AP to complete the root AP discovery process.
    An extender AP's power LED blinks in multiple colors during the root AP discovery process.
    The root AP discovery process finishes. Each extender AP's power LED is solid blue.

Configure a multi pre-shared key group for an SSID

Multi pre-shared key (MPSK) groups in Insight let you assign and manage multiple unique WiFi passwords under a single SSID, so different users or devices can have their own key with tailored access and security policies.

To configure a multi pre-shared key group for an SSID using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wi-Fi > Advanced > Multi PSK Settings > MPSK Group**.
   The MPSK Group page displays.
5. Click the **Add** button.
   The Add MPSK Group pop-up page displays.
6. In the **Group Name** field, enter a name for the MPSK group.
7. In the **VLAN ID** field, enter the VLAN ID to associate with this MPSK group.
8. In the **Number of Keys** field, enter the number of pre-shared keys to generate for this MPSK group.

9. In the **Create Bulk Key Identifier and Passwords** section, do one of the following:
   - Select the **Auto** radio button and do the following:
     a. From the **Key Identifier** menu, select one of the following:
        - **None**
        - **Prefix**: In the **Prefix** field, enter the text to add before each key identifier.
        - **Suffix**: In the **Suffix** field, enter the text to add after each key identifier.
        - **Prefix and Suffix**: In the **Prefix** and **Suffix** fields, enter the text to add before and after each key identifier.
     b. From the **Password Pattern** menu, select one of the following:
        - **None**
        - **Prefix**: In the **Prefix** field, enter the text to add before each password.
        - **Suffix**: In the **Suffix** field, enter the text to add after each password.
        - **Prefix and Suffix**: In the **Prefix** and **Suffix** fields, enter the text to add before and after each password.
     c. In the **Complexity** section, select one or more of the following check boxes:
        - **Use lowercase letters (eg. a to z)**
        - **Use UPPERCASE letters (eg. A to Z)**
        - **Use Numbers (eg. 1234567890)**
        - **Use Symbols (eg. !@#$%^&*)**
        - **Exclude**: In the **Enter Exclude Value** field, enter any characters that you do not want to include in the generated passwords.
   - Select the **Manually** radio button and in the **Device List** file section, upload a CSV file.

     Click the upload link or drag and drop the CSV files.
10. In the **Max Clients** field, configure the maximum number of clients that can use each pre-shared key simultaneously.
11. From the **Expiry** menu, select one of the following:
    - **Never**
    - **Date and Time**:
      a. In the **Date and Time** field, click the calendar icon to specify a date.
      b. In the **Time** field, click the clock icon to specify a time.
12. Click the **Add** button.
    Your settings are saved.

Configure WiFi traffic policies

WiFi traffic policies let you control which devices and types of network traffic are allowed on your WiFi network. You can configure MAC/IP traffic filters to block or permit specific devices or addresses, and allow broadcast or multicast traffic so that shared services, such as streaming or

device discovery, continue to work correctly.

This chapter contains the following sections:

- Configure wireless noise filter settings for access points in a location
- Configure global traffic filter settings for access points in a location
- Configure custom traffic filter settings
- Configure a traffic policy rule for access points in a location

Configure wireless noise filter settings for access points in a location

The Wireless Noise Filter identifies and blocks unwanted or disruptive wireless traffic to access points by applying prioritized ACL rules to a location. When the Wireless Noise Filter setting is enabled, it has higher precedence over the Global Traffic Filter setting.

To configure wireless noise filter settings for access points using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wi-Fi > Advanced > Traffic Policies**.
   The Traffic Policies page displays.
5. Click the **MAC/IP Traffic Filters** tab.
6. Select the **Wireless Noise Filter** tab, and activate the **Enable Wireless Noise Filter** toggle.
   By default, this toggle is disabled.
7. From the table, select the **Enable** toggle of one or more rules.
   When you enable a rule, it is assigned a priority.
8. To edit a rule, select a check box of an ACL rule and click the **Edit** button.
   The Edit ACL Rule pop-up window displays.
9. **(Optional)** From the list of preconfigured ACL rules, edit the configuration of the ACL rule, and click the **Save** button.
10. Click the **Apply** button.
    Your settings are saved.

Configure global traffic filter settings for access points in a location

Global traffic filter settings let you centrally define which types of client traffic are allowed or blocked for all access points in a location, so you can enforce consistent security and usage policies across the network.

To configure global traffic filter settings for access points in a location:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wi-Fi > Advanced > Traffic Policies**.
   The Traffic Policies page displays.
5. Click the **MAC/IP Traffic Filters** tab.
6. Select the **Global Traffic Filter** tab, and activate the **Enable Global Traffic Filter** toggle.
   By default, this toggle is disabled.
7. **(Optional)** To add an ACL rule, click the **Add** button. For more information, see Configure a traffic policy rule for access points in a location on page 106.
8. Click the **Apply** button.
   Your settings are saved.

Configure custom traffic filter settings

The custom traffic filter setting lets you create and manage custom ACL rules to control network traffic for a location.

To configure custom traffic filter settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.

4. Click **Wi-Fi > Advanced > Traffic Policies**.
   The Traffic Policies page displays.
5. Click the **MAC/IP Traffic Filters** tab.
6. Select the **Custom Traffic Filter** tab, and activate the **Enable Custom ACL** toggle.
   By default, this toggle is disabled.
7. From the table, select the **Enable** toggle of one or more rules.
   When you enable a rule, it is assigned a priority.
8. To add a new ACL rule, click the **Add** button.
   For more information, see Configure a traffic policy rule for access points in a location on page 106.
9. Click the **Apply** button.
   Your settings are saved.

Configure a traffic policy rule for access points in a location

Create and apply a MAC or IP traffic policy rule to control network access for access points in a location.

To add a traffic policy rule for access points in a location:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wi-Fi > Advanced > Traffic Policies**.
   The Traffic Policies page displays.
5. Click the **MAC/IP Traffic Filters** tab.
6. Do one of the following:
   - Select the **Global Traffic Filter** tab, and activate the **Enable Global Traffic Filter** toggle.
   - Select the **Custom Traffic Filter** tab, and activate the **Enable Custom ACL** toggle.
7. Click the **Add** button.
   The Add ACL Rule pop-up window displays.
8. In the **Rule Name** field, enter a name for the ACL rule.

9. Select the **Enable** toggle to activate the ACL rule.
10. In the **Action** section, select the **Allow** or **Deny** radio button.
    This setting determines whether traffic that matches this ACL rule is permitted or blocked. Select **Allow** to permit matching traffic, or **Deny** to block matching traffic.
11. In the **Direction** section, select the **In**, **Out** or **Both** radio button.
    This setting specifies how the ACL rule is applied to traffic flow. Select **In** to apply the rule to incoming traffic, **Out** to apply it to outgoing traffic, or **Both** to apply it to traffic in both directions.
12. In the **Network Stack** section, select the **MAC Address** or **IP** radio button.
    This setting determines whether the ACL rule filters traffic based on Layer 2 MAC addresses or Layer 3 IP information. Select **MAC Address** to apply the rule using device MAC addresses, or **IP** to apply the rule using IP addresses.
13. From the **Protocol** menu, select one of the following protocols:

Protocol	Description
Any	Matches traffic that uses any protocol.
ARP	Matches Address Resolution Protocol (ARP) traffic used to map IP addresses to MAC addresses on the local network.
TCP	Matches Transmission Control Protocol (TCP) traffic, such as web browsing, email, and other connection-oriented applications.
UDP	Matches User Datagram Protocol (UDP) traffic, such as DNS, VoIP, and other connectionless applications.
ICMP	Matches Internet Control Message Protocol (ICMP) traffic, such as ping and other network diagnostic messages.
IGMP	Matches Internet Group Management Protocol (IGMP) traffic used for managing multicast group memberships.

14. In the **Source** section, from the **Type** menu, select **ANY** or **MAC Address**.
    This setting specifies how the ACL rule identifies the source of the traffic. Select **ANY** to match traffic from any source, or **MAC Address** to match traffic only from devices with specific source MAC addresses.
15. In the **Port** field, enter the port number to which this ACL rule applies.
16. Click the **Apply** button.
    Your settings are saved.

Configure wireless band settings

You can configure how your network operates across different frequency bands. By adjusting options for the 2.4 GHz, 5 GHz, and 6 GHz radios, you can control coverage, optimize performance, and reduce interference. These settings help you ensure reliable connectivity for various client devices and are particularly useful when managing environments with different device types, densities, and network usage requirements.

To configure wireless band settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wi-Fi > Advanced > Wireless Settings**.
   The Wireless Settings page displays.
5. From the **Radio Mode** menu, select one of the following radio band tabs:

Radio band	Description
2.4GHz	Provides wider coverage and better wall penetration, but typically supports lower data rates and is more prone to interference from other devices (for example, microwaves and Bluetooth).
5GHz or 5GHz Low	Offers higher throughput and less interference than 2.4GHz, with moderate coverage. The lower 5GHz channels are commonly used and supported by most modern Wi-Fi clients.
5GHz High	Uses higher 5GHz channels that can provide additional capacity and performance in dense environments, but may have slightly reduced range compared to lower 5GHz channels.
6GHz	Delivers the highest potential throughput and lowest interference by using new Wi-Fi 6E/7 spectrum, but requires compatible client devices and generally has shorter range than 2.4GHz and 5GHz.

6. Activate the **Enable Radio** toggle.

7. From the **Radio Mode** menu, select one of the following options:

Radio mode	Description and applicable bands
11b	Operates only on the 2.4 GHz band, providing low data rates and primarily used for compatibility with old client devices.
11bg	Mixed 802.11b/g mode that operates on the 2.4 GHz band, allowing both older 11b clients and newer 11g clients to connect on the same radio.
11ng	802.11n mode on the 2.4 GHz band, supporting higher data rates and improved reliability while remaining compatible with 11b/g clients.
11ax	Wi-Fi 6 (802.11ax) mode that can operate on the 2.4 GHz and 5 GHz bands, improving efficiency, capacity, and performance in dense client environments.
11be	Wi-Fi 7 (802.11be) mode that operates on the 2.4 GHz, 5 GHz, and 6 GHz bands (where supported), delivering very high throughput and low latency for next-generation clients.

8. From the **Channel Width** menu, select one of the following options:

Channel width	Description and applicable bands
20 MHz	This channel that provides the widest compatibility and is supported on 2.4 GHz, 5 GHz, and 6 GHz bands.
40 MHz	Two bonded 20 MHz channels that increase throughput. Supported on 2.4 GHz (where allowed), 5 GHz, and 6 GHz bands.
80 MHz	Four bonded 20 MHz channels that provide higher throughput, typically used on 5 GHz and 6 GHz bands.
160 MHz	Eight bonded 20 MHz channels that deliver very high throughput on supported 5 GHz and 6 GHz bands.
320 MHz	Sixteen bonded 20 MHz channels that provide maximum throughput on supported 6 GHz Wi-Fi 7 radios.
Dynamic 20/40/80/160/320 MHz	Allows the radio to dynamically select and adjust the channel width (20, 40, 80, 160, or 320 MHz) based on band support, regulatory limits, and current RF conditions.

9. In the **Beacon Interval (ms) (100 to 300)** field, enter the desired beacon interval in milliseconds (100 to 300).
   The beacon interval is how often the access point sends management frames to announce the WiFi network to nearby client devices.

10. **(Optional)** Activate the **Broadcast/Multicast Rate Limiting (PPS)** toggle.
    Broadcast/Multicast Rate Limiting (PPS) restricts the number of broadcast and multicast packets per second that an access point sends, helping prevent excessive management or discovery traffic from degrading WiFi performance.
11. Move the slider to set the maximum value to define the Broadcast/Multicast Rate limit (0-512).
12. Move the slider to set the DTIM Interval (0-255).
    The DTIM (Delivery Traffic Indication Message) interval controls how often access points send buffered broadcast and multicast traffic to clients.
13. Click the **Save** button.
    Your settings are saved.

Configure WiFi network settings

You can configure advanced WiFi network settings, including multicast optimization, power saving, and logging options for a selected location.

To configure WiFi network settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The location dashboard displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wi-Fi > Advanced > Network Settings**.
   The Network Settings page displays.

5. Activate or deactivate the following toggles:

Setting	Description	Default setting
IGMP Snooping	Optimizes multicast traffic delivery, reduces unnecessary broadcast traffic, and improves performance for multicast applications such as streaming or real-time services.	Disabled
Broadcast to Unicast	Converts broadcast traffic, such as ARP and DHCP, into unicast traffic for associated clients to improve WiFi performance and reliability in dense environments.	Disabled
Energy Efficiency Mode	Reduces power consumption when there is no client activity by operating the radio in 1x1 mode until a client connects.	Disabled
ARP Proxy	Reduces ARP broadcast traffic by allowing the access point to respond to ARP requests on behalf of clients, which can improve network efficiency and performance in busy environments.	Enabled
Hardware Assisted Datapath	Improves packet forwarding performance by offloading data path processing to specialized hardware on the access point, which reduces CPU load and helps maintain higher throughput and lower latency for client traffic. Do not disable this setting unless inter-VLAN routing is necessary.	Enabled
Log Probing Clients to Syslog	Logs information about clients that are probing for the network to an external syslog server, which helps with troubleshooting connectivity issues, monitoring client behavior, and analyzing potential security events.	Disabled

6. Click the **Save** button.
   Your settings are saved.

Configure URL filtering for all WiFi clients in a location

URL filtering is a security feature that monitors and controls access to websites, allowing or blocking them to ensure safe browsing. If enabled, the blocklist applies to all WiFi clients at a location.

To set up URL filtering for all WiFi clients at a location:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wi-Fi > Advanced > URL Filtering**.
   The URL Filtering page displays.
5. Select the **Enable** toggle.
   The page adjusts.
6. In the **Add Domain** field, add a URL or domain name, and click **Enter**.
   When you block a URL, the domain and all URLs in the domain are blocked.
7. To add more URLs or domain names, click the **+** icon.
   A pop-up displays to advise the URL is successfully added.

Manage the wired network for a location

You can manage features that are specific to Insight-managed wired networks at a location. A wired network and VLAN that you create for one location can be used by multiple devices at that location. For information about VLANs, see Manage VLANs and VLAN-based features for a location on page 65. For information about features that are specific to a single switch at a location, see Manage individual switches on page 203.

This chapter contains the following sections:

- Configure link aggregation between two devices in a location
- Configure Spanning Tree for a location
- PoE downtime schedules
- Configure RADIUS authentication for a switch port

Configure link aggregation between two devices in a location

Link aggregation lets you combine multiple Ethernet links into a single logical link between two network devices. The most common combinations involve connecting a switch to another switch, a server, a network attached storage (NAS) device, or a multiport WiFi access point. Network devices treat the link aggregation group (LAG) as a single link, which increases throughput, fault tolerance, or both between the two devices.

Insight-managed switches support both static LAGs and Link Aggregation Control Protocol (LACP) for dynamic LAGs. If a physical link in a dynamic LAG goes down, other physical links in the same dynamic LAG continue to dynamically and transparently pass traffic. NETGEAR Insight assigns all VLANs at the network location to the LAG.

When you set up a LAG between two devices, the following requirements apply:

- The devices must be capable of supporting LAGs. For a dynamic LAG, both devices must be capable of supporting LACP
- You must configure the LAG on each device. However, if both switches are Insight-managed switches, you can set up the LAG on one switch and specify the partner switch.
- On each device, specify at least two ports as members of the LAG.
- After you configure the LAG, connect the member ports with Ethernet cables. (If you do it before, you might create a network loop.)

To set up link aggregation between two devices using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.

2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wired > LAG**.
   The LAG page displays.
5. Click the **Add LAG** button.
   The Create New LAG pop-up window displays.
6. Select the check box for each device that you want to add to the LAG.
   You can select up to two devices to be part of a LAG.
7. Click the **Next** button.
8. In the **LAG Name** field, enter a name for the LAG.
   The LAG name must be 1-15 characters and should not contain special characters.
9. If you do not want to enable the LAG immediately, deactivate the **Enable** toggle.
   By default, the Enable toggle is activated.
10. If you are setting up a dynamic LAG on switches that both support IEEE 802.3ad Link Aggregation Control Protocol (LACP), deactivate the **Static LAG** toggle.
    By default, the Static LAG toggle is activated.

    NOTE: Insight-managed switches support LACP so that you can set up a dynamic LAG between them
11. From the **Default VLAN (PVID)** menu, select the VLAN that you want to assign as the native VLAN for this LAG.
    Any untagged traffic that enters the LAG is automatically associated with this VLAN.
12. Click the **Save and Continue** button.
13. Depending on the type of devices that you are using for the LAG, select the member ports by doing one of the following:
    - **LAG between two Insight-managed switches:** For each Insight-managed switch, select at least two ports as members of the LAG.
    - **LAG between one Insight-managed switch and another type of device that supports LAGs:** For the Insight-managed switch, select at least two ports as members of the LAG. For the other device, configure matching LAG settings.
14. Click the **Save** button.
    Your settings are saved and the LAG page displays again, showing the configured LAG.

15. If you are setting up a LAG for an Insight-managed switch and another type of device, you must manually configure the LAG on other type of the device.

> **NOTE:** If you set up a static LAG, be sure that the switch ports that you are making members of the static LAG are using the same port speed, duplex mode, and flow control settings as on the Insight-managed switch.

16. Use Ethernet cables to connect the members ports of the LAG on each device.
    Unless you configured the LAG to be disabled (see step 9), the LAG becomes active immediately.

Configure Spanning Tree for a location

The Spanning Tree Protocol (STP) provides a tree topology for any arrangement of switches and bridges in a network. STP also provides one path between end stations on a network, avoiding and eliminating loops and preventing network congestion. Insight supports STP and Rapid STP (RSTP).

RSTP can recognize full-duplex connectivity and ports that are connected to end stations, allowing for rapid transitioning of such ports to the forwarding state. By default, both STP and RSTP are disabled. You can enable either STP or RSTP. You cannot configure any specific STP or RSTP settings.

To configure Spanning Tree for a location:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wired > Spanning Tree**.
   The Spanning Tree page displays.
5. Activate the **Enable** toggle.
   The Spanning Tree settings display.
6. From the **Spanning Tree** Mode menu, select STP or RSTP.
7. For each switch for which you want to configure Spanning Tree, in the graphic of the switch, select the ports, LAGs, or both to enable Spanning Tree for them.
8. **(Optional)** For each switch, click the **Select All** button to select all ports on that switch.
   After you click the **Select All** button, the name of the button changes to **Deselect All**.

9. **(Optional)** Click the **Deselect All** button for a switch to clear all selected ports on that switch.
10. Click the **Save** button.
    Your settings are saved.

PoE downtime schedules

When a PoE Downtime Schedule is active, PoE power is disabled on the PoE ports to which you assign the schedule, that is, the ports do not deliver PoE power. When the PoE Downtime Schedule is not active, PoE power is enabled on those ports, that is, the ports do deliver PoE power. You can use PoE Downtime Schedules to reduce energy consumption, extend the life of connected PoE devices, and improve security by ensuring that powered devices, such as wireless access points or IP cameras, are turned off automatically during nonbusiness hours or other times when they are not needed.

Create a PoE downtime schedule

Create a schedule that disables PoE power on selected ports during specified days and times.

To create a PoE downtime schedule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wired > PoE Downtime Schedules**.
   The PoE Downtime Schedules page displays.
5. Click the **Add Schedule** button.
   The Create New PoE Downtime Schedule pop-up window displays.
6. Using the controls on the page, give the schedule a name, set the days and time, if applicable, set the recurrence, and set the start date and end date for the schedule. You are setting up a schedule for a period during which PoE power is disabled. That is, the ports to which you assign this schedule do not deliver PoE power while the schedule is active.

7. Do one of the following:
   - To save the schedule, click the **Save** button. Your settings are saved. The PoE Schedules page displays, showing the new schedule. By default, the schedule is enabled.
   - To save the schedule and immediately assign it to PoE ports, click the **Save and Choose Ports** button. The Edit Ports pop-up window opens and shows graphics of the switches at the network location Do the following:
     a. Select the PoE ports to which you want to assign the new schedule. Selected ports display green.
     b. Click the **Save** button. Your settings are saved. The PoE Schedules page displays, showing the new schedule. By default, the schedule is enabled.

Edit a PoE downtime schedule

You can edit an existing PoE downtime schedule for a selected location to control when PoE power is active or not.

To edit a PoE Downtime Schedule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wired > PoE Downtime Schedules**.
   The PoE Downtime Schedules page displays.
5. Click the **Edit** button for a schedule.
   The Edit PoE Schedule pop-up window displays.
6. Edit the configuration, and click the **Save** button.
   Your settings are saved.
7. To edit the configured ports in the PoE Downtime Schedule, click the **Edit Ports** button.
   The Apply Schedule pop-up window displays.
8. Edit the port configuration, and click the **Save** button.
   Your settings are saved.

Delete a PoE downtime schedule

You can remove a PoE downtime schedule that is no longer needed so it does not affect device power availability.

To delete a PoE downtime schedule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wired > PoE Downtime Schedules**.
   The PoE Downtime Schedules page displays.
5. Click the **Delete** button for a schedule.
   The Delete Schedule pop-up window displays.
6. Click the **Delete** button.
   A pop-up window displays to advise the schedule is successfully deleted.

Configure RADIUS authentication for a switch port

RADIUS Configuration lets you control access at the switch port level so that only authorized devices can connect to your network.
By setting each port's authentication mode, Auto, Authorized, or Unauthorized, you can determine whether connected devices must authenticate through a RADIUS server, receive immediate access, or are blocked. This feature strengthens network security at the edge by enforcing identity-based access control for devices connecting at your physical network location.

To configure RADIUS authentication for a switch port:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Wired > RADIUS Configuration**.
   The RADIUS Configuration page displays.

5. Select one or more switch ports, or to select all ports, click the **Select All** button.
6. Select the **Port Authentication Mode** button.
   The Select Authentication Mode pop-up window displays.
7. Select one of the following radio buttons:

Authentication mode	Description
Auto	Requires authentications for all client connections.
Authorized	Any client connected to the port is automatically authenticated.
Unauthorized	Connections are not allowed.

8. Click the **Apply** button.
9. Click the **Save** button.
   Your settings are saved.
10. **(Optional)** If you want to change the automatic authentication mode for the individual switch, click the **Device Configuration** button.
11. **(Optional)** To reset the RADIUS Configuration on all switches in the location, click the **Reset** button.

Manage alarms

You can view, configure, and respond to system alarms using the Insight Cloud Portal.

This chapter contains the following sections:

- View alarms
- Mark alarms as read
- Filter alarms
- Refresh alarms
- Download alarms CSV file
- Configure alarm settings
- Search alarms
- Share alarms

View alarms

To view all alarms in your organization using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Select the **Alarms** tab.
   The Alarms page displays.
   All alarms display in the table.

Mark alarms as read

To mark alarms as read using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Alarms** tab.
   The Alarms page displays.
3. Click the **Mark All As Read** button.
   All alarms are marked as read.

Filter alarms

To filter an alarm:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Alarms** tab.
   The Alarms page displays.
   All alarms display in the Alarms table.
3. In the top-right corner, click the filter icon.
   The Filter pop-up page displays.
4. In the **Type** section, select an alarm type:
   - **All** (default)
   - **Critical**
   - **Warning**
   - **Info**
5. In the **Time Range** section, select a time range:
   - **Last Month**
   - **Last Week**
   - **Last 24Hrs**
   - **Last 8Hrs**
6. In the **Devices** section, select one or more devices:
   - **AP**
   - **SW**
   - **ORBI**
   - **BR**
   - **MHS**
   - **PR**
   - **AV**
7. In the **Organizations** section, select one or more organizations.
8. Select a status:
   - **Read**
   - **Unread**
9. Click the **Apply** button.
   Your settings are saved.
10. **(Optional)** To reset all selections, click the **Clear All** button. To exit without saving changes, click the **Cancel** button.

Refresh alarms

To refresh alarms:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Select the **Alarms** tab.
   The Alarms page displays.
   All alarms display in the table.
3. In the top-right corner, click the refresh icon.
   The Alarms table refreshes displaying all alarms.

Download alarms CSV file

To download alarms CSV file:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Select the **Alarms** tab.
   The Alarms page displays.
   All alarms display in the table.
3. In the top-right corner, click the **Download CSV** icon.
   The alarms CSV file downloads.

Configure alarm settings

To configure alarm settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Select the **Alarms** tab.
   The Alarms page displays.
   All alarms display in the table.
3. Click the **Column Settings** tab.

4. Under **Show Column**, select the check box next to each column you want to display.
   - **Type**
   - **Time**
   - **Location**
   - **Organization**
   - **Log Message**
5. Under **Left Pin**, select the check box next to each column you want to pin to the left of the table.
   - **Type**
   - **Time**
   - **Location**
   - **Organization**
   - **Log Message**
6. Click the **Apply** button.
   Your settings are saved.
7. **(Optional)** To restore default settings, click the **Reset** button.

Search alarms

To search for specific alarms:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Select the **Alarms** tab.
   The Alarms page displays.
   All alarms display in the table.
3. In the top-right corner, in the search box, enter a specific description of an alarm, such as the organization, location or type.
   The specific alarm or alarms displays in the table.

Share alarms

To share an alarm or alarms using the Insight Cloud Portal:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.

2. Click the **Alarms** tab.
   The Alarms page displays.

   All alarms display in the Alarms table.

3. Select the check box next to one or more alarms.

4. Click the **Share** button.
   The Share Notifications pop-up window displays.

5. Enter the email address to which you want to share the alarm or alarms.

   NOTE: To share an alarm or alarms to multiple emails, separate each email address with a comma.

6. Click the **Send** button.
   A pop-up notification displays to advise the alarm or alarms are shared successfully.

   The email contains a `.xls` detailing the alarm details.

Manage reports

You can create, manage, and distribute tailored reports across network deployments in the Insight Cloud Portal. You can configure reports with specific scopes, time periods, and widget selections to surface the data most relevant to each organization from location health trends and inventory to troubleshooting insights. Once created, reports can run instantly or scheduled for recurring delivery, and shared directly with administrators, business owners, managers, or any email recipient.

This chapter contains the following sections:

- Add a report
- Edit a report
- Delete a report

Add a report

To add a report:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Administration > Reports**.
   The Reports page displays.
   All reports display in the table.
3. Click the **Add** button.
   The Add Report page displays.
4. In the **Report Name** field, enter a unique report name.
   The name must be alphanumeric and may include hyphens, underscores, and spaces, up to 128 characters.
5. **(Optional)** In the **Description** field, enter a description.
   The description can be a maximum of 255 characters.
6. **(Optional)** In the **Other Email** section, enter additional email addresses, separated by commas.
   If you enter more than one email, separate the emails by commas.
7. In the **Report Type** section, select one of the following options:
   - **Custom Report**
   - **Troubleshooting**

8. If you select the **Custom Report** option, choose one of the following run types and configure the schedule as needed:
   - **Instant**: Generates a one-time report for the selected time period: 24 hours, 7 days, or 30 days.
   - **Scheduled**: Configures a recurring report that is delivered weekly or monthly.
9. In the **Branding Logo** section, select one of the following options:
   - **Use VAR Logo**
   - **Use Organization Logo**
10. Click the **Next** button.
    The Filter & Scope displays.
11. Select an existing filter from the list, or click **Create New** and enter a unique filter name (required, up to 128 characters).
    You must select exactly one filter from the list before proceeding.
12. Select a location, with or without a specific device. You can search for organizations or locations and select all entries from a parent organization.
13. Click the **Next** button.
14. Select one or more widgets to include in the report.
    Available widget categories depend on the selected report type.
15. **(Optional)** In the **Rearrange Widgets** section, drag widgets to reorder.
16. In the **Review** section, review the report configuration.
    The Review section displays a summary of all selections.
17. Click the **Save** button.
    Your settings are saved.

Edit a report

To edit a custom report:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Administration > Reports**.
   The Reports page displays.

   All reports display in the table.
3. Click the check box of a report, and click the **Edit** button.
   The Edit Report page displays.

4. Update the report configuration, and click the **Update** button.
   Your settings are saved.

Delete a report

To delete a custom report:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Administration > Reports**.
   The Reports page displays.

   All reports display in the table.
3. Click the check box of a report, and click the **Delete** button.
   The Delete Report Template pop-up window displays.
4. Click the **Delete** button.
   The report is deleted and does not display in the table.

Manage individual routers

Using the Insight Cloud Portal, you can remotely manage all aspects of an individual router's network configuration, security, and connectivity. The portal provides a centralized interface for controlling how the router connects to the internet, routes traffic, and protects the network.

This chapter contains the following sections:

- Monitor connected clients
- Activate or deactivate Link Layer Discovery Protocol
- Manage Simple Network Management Protocol
- Configure certificate authorities
- Configure server certificates
- Manage WAN settings
- Manage the QoS Settings
- Manage Static Routes
- Manage VPN settings
- Manage security settings

Monitor connected clients

Track and monitor all clients connected to a router managed through Insight, providing visibility into device activity and network usage.

To monitor clients connected to an Insight-managed router:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. From the **Devices** table, select the router.
   The router dashboard displays.
5. Click the **Connected Clients** tab.
   A list of all clients currently connected to the router displays.
6. To see more details about a client, next to its name, click **+** icon.

Activate or deactivate Link Layer Discovery Protocol

Link Layer Discovery Protocol (LLDP) is a layer 2 protocol allowing devices to advertise link and endpoint information that can be used to discover link neighbors and their capabilities. You can enable or disable LLDP on an Insight-managed router at a global level (that is, for all ports) and at a port level. For example, you can enable LLDP globally, but then disable it on individual ports. By default LLDP is enabled globally and on all ports, except, for security reasons, on the WAN1 port.

To activate or deactivate LLDP:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > General > LLDP**.
6. Click the **LLDP (Link Layer Discovery Protocol)** toggle to activate or deactivate LLDP globally:
   - The toggle is positioned to the right: LLDP is globally activated. This is the default setting.
   - The toggle is positioned to the left: LLDP is globally deactivated.
7. Click the **Summary** tab, and select a port on the router.
   The router pop-up page displays.
8. Click the **Port Settings** tab, and click the **LLDP** toggle to activate or deactivate LLDP on the port.
9. Click the **Save** button.
   Your settings are saved.

Manage Simple Network Management Protocol

You can configure SNMP settings for SNMPv1, SNMPv2, and SNMPv3 on an Insight-managed router. The router supports the configuration of SNMP communities and users that can monitor the router over SNMP, receive trap messages, or do both. The router uses both standard public MIBs for standard functionality and private MIBs that support additional router functionality.

Configure the SNMP system name, contact and location

You can configure the SNMP system name, system contact, and system location.

To add the SNMP system name, contact and location:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > General > SNMP > Configuration**.
6. In the **System Name** field, enter a system name.
7. In the **System Contact** field, enter a system contact.
8. In the **System Location** field, enter a system location.
9. Click the **Save** button.
   Your settings are saved.

Configure the SNMPv1 and SNMPv2 client

You can configure the SNMPv1 and SNMPv2 client so that SNMP community members can view the configuration of the Insight-managed router as read-only through SNMP. To access the router over SNMP, you need to use the client IP address of the router.

To configure the SNMPv1 and SNMPv2 client:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.

5. Click **Settings > General > SNMP > Configuration**.
6. Activate the **SNMPv1/v2** toggle so that it is positioned to the right.
   By default, the SNMPv1/v2 toggle is activated.
7. In the **Community Name** field, type a community private string.
8. In the **Client IP Address** field, type the IP address of the client.
9. Click the **Save** button.
   Your settings are saved.

Configure an SNMPv3 user account

Any Insight user can connect to the Insight-managed router using the SNMPv3 protocol, but for authentication and encryption, the router supports only one user, which is the admin user. Therefore, you can configure only one SNMPv3 profile. To access the router over SNMPv3, you need to use the SNMPv3 user name.

To configure the SNMPv3 user account:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > General > SNMP > Configuration**.
6. Activate the **SNMPv3** toggle so that it displays blue and is positioned to the right.
   By default, the SNMPv3 toggle is activated.
7. Click the **Add** button.
   The Add SNMP v3 Configuration pop-up window displays.
8. In the **User Name** field, type the SNMP v3 user name.
   This defines who is allowed to authenticate and access the SNMP data.

9. From the **Authentication Algorithm** menu, select one of the following options as the algorithm to use for authentication:
   - **MD5**: Message Digest 5(MD5) is the authentication protocol.
   - **SHA**: Secure Hash Algorithm (SHA) is the authentication protocol. SHA provides stronger security than MD5.
   - **SHA-224**: A cryptographic hash algorithm that is reliable and safe in a variety of applications.
   - **SHA-256**: A secure hash algorithm that is superior to SHA-224.
   - **SHA-384**: A secure hash algorithm that is superior to SHA-256.
   - **SHA-512**: A secure hash algorithm that is superior to SHA-384.
10. In the **Authentication Password** field, enter a password.
    The password must be between 8 and 64 characters, and cannot contain spaces. With any selection of authentication protocol, the admin login password for the device UI is used as the SNMPv3 authentication password.
11. To enable encryption, do the following:
    a. From the **Encryption Algorithm** menu, select one of the following options as the algorithm to use for encryption:
       - **AES**
       - **AES-128**
       - **AES-192**
       - **AES-256**
    b. In the **Encryption Password** field, type an encryption code of eight or more alphanumeric characters.
       The password must be between 8 and 64 characters and cannot contain spaces.
12. Click the **Add** button.
    Your settings are saved.

Edit an SNMPv3 user account

You can update the settings for an existing SNMPv3 user account on an Insight-managed router.

To edit an SNMPv3 user account:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.

4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > General > SNMP > Configuration**.
6. Activate the **SNMPv3** toggle so that it displays blue and is positioned to the right.
   By default, the SNMPv3 toggle is activated.
7. Select the check box of the user account you want to edit, and click the **Edit** button.
   The Edit SNMP v3 Configuration pop-up window displays.
8. Update the SNMPv3 user account settings, and click the **Update** button.
   Your settings are saved.

Delete an SNMPv3 user account

You can remove an existing SNMPv3 user account from an Insight-managed router.

To delete an SNMPv3 user account:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > General > SNMP > Configuration**.
6. Activate the **SNMPv3** toggle so that it displays blue and is positioned to the right.
   By default, the SNMPv3 toggle is activated.
7. Select the check box of the user account you want to edit, and click the **Delete** button.
   Your settings are saved.

Manage the SNMPv1, SNMPv2, and SNMPv3 trap recipients

You can configure SNMPv1 and SNMPv2 communities, and SNMPv3 users that must be able to receive traps.

Add an SNMPv1 or SNMPv2 community or an SNMPv3 user that must receive traps

You can add an SNMPv1 or SNMPv2 community or an SNMPv3 user that must receive traps.

To add an SNMPv1 or SNMPv2 community or an SNMPv3 user that must receive traps:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > General > SNMP > Trap**.
6. Depending on the community or user you want to add, do one of the following:
   - **Add an SNMPv1 or SNMPv2 community as a trap recipient:**
     a. In the **SNMP v1/v2 Trap** section, click the **Add** icon.
        The Add v1/v2 Trap pop-up window displays.
     b. In the **Community Name** field, type the name of the SNMP community that includes the host.
     c. In the **Trap Receiver IP** field, type the IP address of the SNMP community.
     d. In the **Port** field, type the port number on the device you want to use.
     e. From the **Version** menu, select the SNMP version that is used by the host:
        - **SNMP v1**: The Insight uses SNMPv1 to send traps to the receiver. This is the default setting.
        - **SNMP v2**: The Insight uses SNMPv2 to send traps to the receiver.
     f. Click the **Add** button. Your settings are saved and the trap configuration is added.
   - **Add an SNMPv3 user as a trap recipient:**
     a. In the **SNMP v3 Trap** section, click the **Add** icon.
        The Add v3 Trap pop-up window displays.
     b. From the **SNMP v3 Users** menu, select the SNMP v3 user. This name defines who can authenticate and access the SNMP data.
     c. In the **Trap Receiver IP** field, type the IP address of the client.
     d. In the **Port** field, type the port number on the router you want to use.
     e. From the **Security Level** menu, select the security level for the SNMPv3 user.
     f. Click the **Add** button. Your settings are saved.

Change an SNMPv1 or SNMPv2 community or an SNMPv3 user that must receive traps

You can change an existing SNMPv1 or SNMPv2 community or an SNMPv3 user that receives traps.

To change an SNMPv1 or SNMPv2 community or an SNMPv3 user that receives traps:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > General > SNMP > Trap**.
6. Depending on the community or user you want to edit, do one of the following:
   - **Edit an SNMPv1 or SNMPv2 community that must receive traps:**
     a. Select the check box of a community, and click the **Edit** button. The Edit v1/v2 Trap pop-up window displays.
     b. Update the community configuration, and click the **Update** button. Your settings are saved.
   - **Edit an SNMPv3 user that must receive traps:**
     a. Select the check box of a SNMPv3 user account, and click the **Edit** button. The Edit v3 Trap pop-up window displays.
     b. Update the user configuration, and click the **Update** button. Your settings are saved.

Delete an SNMPv1 or SNMPv2 community or an SNMPv3 user that must receive traps

You can delete SNMPv1 or SNMPv2 community or an SNMPv3 user that receives traps.

To delete an SNMPv1 or SNMPv2 community or an SNMPv3 user that receives traps:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.

3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > General > SNMP > Trap**.
6. Depending on the community or user you want to delete, do one of the following:
   - **Delete an SNMPv1 or SNMPv2 community that must receive traps:**
     a. In the **SNMP v1/v2 Trap** section, select the check box of a community.
     b. Click the **Delete** button. Your settings are saved.
   - **Delete an SNMPv3 user that must receive traps:**
     a. In the **SNMP v3 Trap** section, select the check box of a user account.
     b. Click the **Delete** button. Your settings are saved.

Configure trap flags

You can enable or disable specific traps. When the condition that is identified by an active trap occurs on the Insight-managed router, a trap message is sent to any enabled SNMP trap receivers (also referred to as hosts), and a message is written to the trap log.

To configure the trap flags:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.

5. Click **Settings > General > SNMP > Trap**.
   The Trap Flags page displays.

Trap Flag	Description
Authentication Failure	Generates a trap when an SNMP request uses invalid credentials.
Ethernet Port Link Up/Down	Generates a trap when an Ethernet port link changes state between up and down.
IPSec Site-to-Site Tunnel Up/Down	Generates a trap when an IPSec site-to-site VPN tunnel goes up or down.
IPSec Client-to-Site Tunnel Up/Down	Generates a trap when an IPSec client-to-site VPN tunnel goes up or down.
OpenVPN Tunnel Up/Down	Generates a trap when an OpenVPN tunnel goes up or down.

6. To activate a trap flag, click the associated toggle.
   When a trap is activated, the toggle is positioned to the right.
7. Click the **Save** button.
   Your settings are saved.

Display the supported MIBs

To display the MIBs that are supported by the router:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > General > SNMP > MIBs**.
   The Supported MIBs page displays.

   The Name field displays the RFC number, if applicable, and the name of the MIB.

   The Description field displays the RFC title or MIB description.

6. To download a list of the supported MIBs, click the **Download** button, and enter your email address and click the **Send** button.
   The supported MIB list is sent to your email address.

Configure certificate authorities

The Certificate Authority (CA) feature lets you create and manage certificates for VPN tunnels (IPSec IKEv2 and OpenVPN) on Insight-managed routers. The router must have a CA before you can create any server certificates or configure VPN connections.

Add a certificate authority

To add a certificate authority:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > General > Certificate Authority > Add**.
   The Add Certificate Authority page displays.
6. In the **Certificate Authority Name** field, enter a name for the certificate authority.
7. In the **Type** section, select one of the following radio buttons:
   - **Intermediate Certificate Authority**: Select this option if you are importing a certificate that was issued by an external root CA and will be used to sign server certificates for this router.
   - **Internal/Private**: Select this option to create a self-signed certificate authority on the router that will be used to sign server certificates for internal VPN connections.
8. From the **Key Type** menu, select the **RSA** algorithm.

9. From the **Key Length (Bits)** menu, select one of the following options:
   - **1024**: Select this option for a lower security level that may be suitable only for testing or legacy environments.
   - **2048**: Select this option for a recommended balance of security and performance for most production deployments.
   - **4096**: Select this option for a higher security level when stronger encryption is required and additional processing overhead is acceptable.
10. From the **Digest Algorithm** menu, select one of the following options:
    A digest algorithm is a cryptographic hash function that creates a fixed-length value from input data to help verify data integrity and detect tampering.
    - **SHA1**: Select this option only for legacy compatibility when stronger algorithms are not supported. SHA1 is considered weak and is not recommended for new deployments.
    - **SHA256**: Select this option for a recommended balance of security and performance for most production environments.
    - **SHA384**: Select this option when you require stronger security than SHA256 and can accommodate the additional processing overhead.
    - **SHA512**: Select this option for the highest level of security when maximum integrity protection is required and performance impact is acceptable.
11. In the **Lifetime (Days)** field, enter the number of days that the certificate authority remains valid.
12. In the **Common Name** field, enter a descriptive name that identifies this certificate authority, such as the organization or site name.
13. From the **Country Code** menu, select the country code that corresponds to the location of your organization.
14. From the **State/Province** menu, select the state or province where your organization is located.
15. In the **City** field, enter the city where your organization is located.
16. In the **Organization** field, enter the legal name of your organization.
17. In the **Organization Unit** field, enter the department or business unit that is responsible for managing this certificate authority.
18. Click the **Add** button.
    Your settings are saved.

Import an existing certificate authority

To import an existing certificate authority:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > General > Certificate Authority > Import**.
   The Import Certificate Authority page displays.
6. In the **Certificate Name** field, enter a descriptive name for the certificate authority.
7. In the **Select Certificate** section, click **Open file** to upload a certificate authority, or drag and drop the file.
8. In the **Select Private Key** section, click **Open file** to upload a select private key, or drag and drop the file.
9. In the **Password** field, enter the password for the certificate authority.
10. Click the **Import** button.
    Your settings are saved.

Delete a certificate authority

To delete a certificate authority:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > General > Certificate Authority**.

6. Select the check box of the certificate authority you want to delete, and click the **Delete** button.
   Your settings are saved.

Configure server certificates

You can create a server certificate that is based on a certificate authority (CA) that you imported or a CA that you created. You can use a certificate to set up a client-to-site VPN configuration.

NOTE: After you save the certificate, you cannot change it. However, you can delete it and create a new certificate.

Add a server certificate

To add a server certificate:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > General > Certificate > Add**.
   The Add Certificate Authority page displays.
6. In the **Descriptive Name** field, type a name.
   The name is for identification purposes.
7. In the **Internal Certificate** section, specify the settings as described in the following table.
8. In the **Certificate Attributes** section, specify the settings as described in the following table.
9. Click the **Apply** button.
   Your settings are saved. The CA is added to the Authorities table.

Import an existing server certificate

You can import an existing server certificate that you can use to set up a client-to-site VPN configuration on your Insight-managed router. Typically, you import an existing certificate file and associated private key file from a computer or a storage device that is connected to your computer or directly to the Insight-managed router.

To import an existing server certificate:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > General > Certificate > Import**.
   The Import Certificate pop-up page displays.
6. In the **Certificate Name** field, enter a name.
7. In the **Encoding** section, select one of the following radio buttons:
   - **PEM**: Privacy Enhanced Mail (PEM) encoding requires both a certificate file and a private key file.

     Do the following:
     a. In the **Select Certificate** section, click **Open file**, or drag and drop the file.
     b. In the **Select Private Key** section, click **Open file**, or drag and drop the file.
   - **PKCS#12**: This type of encoding includes both a certificate and a private key in a single file. In the Select PKCS#12 (P12 File) section, click **Open file**, or drag and drop the file.
8. To require a password to later export the P12 file or private key file, in the **Password** field, enter a password.
9. Click the **Import** button.
   Your settings are saved.

Delete a server certificate

To delete a server certificate:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > General > Certificate > Delete**.
   Your settings are saved.

Manage WAN settings

The WAN settings let you configure how your router connects to the internet, including IP addressing, DNS, and advanced options such as MTU size and connection type. Use these settings to optimize performance, ensure reliable connectivity, and meet the requirements of your internet service provider.

This chapter contains the following sections:

- Manage the Internet Settings for the WAN1 port
- Configure a Dual WAN Connection
- Manage dual WAN traffic rules

Manage the Internet Settings for the WAN1 port

When you manage a router through Insight, you can view or manually configure the Internet settings for the WAN1 port.

This chapter contains the following sections:

- Configure a dynamic Internet connection for the WAN1 port
- Configure a static Internet connection for the WAN1 port
- Configure a PPPoE Internet connection for the WAN1 port

Configure a dynamic Internet connection for the WAN1 port

You can manually configure an Internet connection with dynamic IP address settings for the WAN1 port on an Insight-managed router. Use this setup if your ISP assigns the WAN IP address dynamically, or another router in your existing network does. Before you start the configuration procedure, ensure that you have the dynamic IP address information that your ISP gave you.

To manually configure a dynamic Internet connection for the WAN1 port:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > WAN > WAN1**.
6. Activate or deactivate the **VLAN Tagging** toggle.
   When enabled, the router adds an 802.1Q VLAN tag with the configured VLAN ID to all traffic on the WAN1 port. When disabled, WAN1 traffic is sent untagged.
7. In the **VLAN ID** field, enter the VLAN ID.
   A VLAN ID is a number from 1 to 4094 that identifies a specific virtual LAN (VLAN) on your network. Use the VLAN ID that your ISP or network administrator provides for the WAN1 connection.
8. From the **Connection Type** menu, select **DHCP**.
   The page adjusts.
9. To add an additional WAN IP address, click the **Additional WAN IP Address** toggle so that it is positioned to the right.
   The page adjusts.
10. Depending on how you want to specify additional IP addresses, do one of the following:
    - **Manually enter an additional IP address**:
      a. Select the **Add IP Address** radio button.
      b. In the **IP Address** field, enter an IP address.
      c. Click the **Add** button.
    - **Set an IP address range**:
      a. Select the **Add IP Range** radio button.
      b. Enter a start and end IP address.
      c. Click the **Add** button.
11. In the **Router MAC Address** section, select one of the following radio buttons:
    - **Use Default MAC Address**: Use the default router MAC address that displays on the router dashboard page and the router label. This is the default setting.
    - **Configure MAC Address**: In the MAC address field, enter a specific MAC address.

12. In the **MTU** section, select one of the following radio buttons:
    - **Auto**: Your ISP assigns the MTU size automatically. In most situations, the default size is 1500 bytes.
    - **Manual**: Your ISP requires you to use a specific MTU size. In the MTU size field, enter the size.
13. In the **DHCP Options** section, in the **Vendor Class** field, enter the vendor class identifier provided by your ISP.
14. In the **Client Identifier** field, enter the client identifier provided by your ISP.
15. Click the **Save** button.
    Your settings are saved.

Configure a static Internet connection for the WAN1 port

You can manually configure an Internet connection with static (fixed) IP address settings for the WAN1 port. Before you start the configuration procedure, ensure that you have the static IP address information that your ISP gave you.

To configure a static Internet connection for the WAN1 port:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > WAN > WAN1**.
6. Activate or deactivate the **VLAN Tagging** toggle.
   When enabled, the router adds an 802.1Q VLAN tag with the configured VLAN ID to all traffic on the WAN1 port. When disabled, WAN1 traffic is sent untagged.
7. In the **VLAN ID** field, enter the VLAN ID.
   A VLAN ID is a number from 1 to 4094 that identifies a specific virtual LAN (VLAN) on your network. Use the VLAN ID that your ISP or network administrator provides for the WAN1 connection.
8. From the **Connection Type** menu, select **Static**.
   The page adjusts.

9. In the **External IP Address**, **Subnet Mask**, and **Gateway IP Address** fields, type the static IP address, subnet mask, and gateway IP address that your ISP gave you.
10. To add an additional WAN IP address, click the **Additional WAN IP Address** toggle so that it is positioned to the right.
    The page adjusts.
11. Depending on how you want to specify additional IP addresses, do one of the following:
    - **Manually enter an additional IP address**:
      a. Select the **Add IP Address** radio button.
      b. In the **IP Address** field, enter an IP address.
      c. Click the **Add** button.
    - **Set an IP address range**:
      a. Select the **Add IP Range** radio button.
      b. Enter a start and end IP address.
      c. Click the **Add** button.
12. In the **Router MAC Address** section, select one of the following radio buttons:
    - **Use Default MAC Address**: Use the default router MAC address that displays on the router dashboard page and the router label. This is the default setting.
    - **Configure MAC Address**: In the MAC address field, enter a specific MAC address.
13. In the **MTU** section, select one of the following radio buttons:
    - **Auto**: Your ISP assigns the MTU size automatically. In most situations, the default size is 1500 bytes.
    - **Manual**: Your ISP requires you to use a specific MTU size. In the MTU size field, enter the size.
14. Click the **Save** button.
    Your settings are saved.

Configure a PPPoE Internet connection for the WAN1 port

You can manually configure a PPPoE Internet connection for the WAN1 or WAN2 port. A PPPoE (Point-to-Point Protocol over Ethernet) connection is a type of Internet connection in which your Insight-managed router uses a user name and password to authenticate with your ISP over Ethernet. Before you start, be sure that you have the PPPoE information that your ISP gave you.

To configure a PPPoE Internet connection for the WAN1 port:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.

3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > WAN > WAN1**.
6. Activate or deactivate the **VLAN Tagging** toggle.
   When enabled, the router adds an 802.1Q VLAN tag with the configured VLAN ID to all traffic on the WAN1 port. When disabled, WAN1 traffic is sent untagged.
7. In the **VLAN ID** field, enter the VLAN ID.
   A VLAN ID is a number from 1 to 4094 that identifies a specific virtual LAN (VLAN) on your network. Use the VLAN ID that your ISP or network administrator provides for the WAN1 connection.
8. From the **Connection Type** menu, select **PPPoE**.
   The page adjusts.
9. In the **External IP Address**, **Subnet Mask**, and **Gateway IP Address** fields, type the static IP address, subnet mask, and gateway IP address that your ISP gave you.
10. To add an additional WAN1 IP address, click the **Additional WAN IP Address** toggle so that it is positioned to the right.
    The page adjusts.
11. Depending on how you want to specify additional IP addresses, do one of the following:
    - **Manually enter an additional IP address**:
      a. Select the **Add IP Address** radio button.
      b. In the **IP Address** field, enter an IP address.
      c. Click the **Add** button.
    - **Set an IP address range**:
      a. Select the **Add IP Range** radio button.
      b. Enter a start and end IP address.
      c. Click the **Add** button.
12. In the **Internet Information** section, enter a user name, password, and service name.
13. In the **Connection Mode** section, select one of the following radio buttons:
    - **Always On**
    - **Manually Connected**: If you choose this option, click the **Connect** button to establish the connection.

    Your settings are saved.

Configure a Dual WAN Connection

Use a dual WAN connection on an Insight-managed router to increase network reliability and optimize bandwidth by connecting the router to two separate Internet service providers. In Insight, you can configure the router to balance traffic across both connections or designate one as a backup, helping ensure continuous Internet access for critical applications and services managed through the Insight Cloud Portal.

This chapter contains the following sections:

- Dual WAN and WAN failover
- Configure dual WAN with a dynamic Internet connection for the WAN2 port
- Configure dual WAN with a static Internet connection for the WAN2 port
- Configure dual WAN with a PPPoE Internet connection for the WAN2 port
- Configure dual WAN failover detection

Dual WAN and WAN failover

Dual WAN is a configuration in which two WAN ports on your Insight-managed router are connected to the Internet. At any time, traffic passes through one WAN port only.

One WAN port functions as the primary interface and the other as the secondary (backup) interface. If the primary interface goes down for example, because your ISP has an outage, the router automatically switches (fails over) to the secondary WAN interface. When the primary interface comes back up, the router automatically switches back.

You configure and manage dual WAN through the Insight Cloud Portal.

WAN ports

The router supports two WAN ports:

> **WAN1 port**
>
> The default primary WAN port. Supports speeds of up to 2.5 Gbps.

> **LAN5 WAN2 port**
>
> Functions as a LAN port by default. You can reconfigure it as the WAN2 port in the Insight Cloud Portal, but it cannot function as both simultaneously. When configured as WAN2, it supports speeds of up to 10 Gbps.

Primary and secondary interfaces

In a dual WAN configuration, WAN1 is the primary interface and WAN2 is the secondary interface by default. You can change this in Insight so that WAN2 is primary and WAN1 is secondary.

Ideally, each WAN port connects to a different ISP. For example, you could connect WAN1 over a cable modem to one ISP and WAN2 over a mobile router with a 5G connection to another ISP. This ensures that a single ISP outage does not interrupt connectivity.

NOTE: A separate Internet connection is required for each WAN port. You must have two active ISP connections to use dual WAN with failover.

Configure dual WAN with a dynamic Internet connection for the WAN2 port

By default, the LAN5 WAN2 port operates as a LAN port. To configure dual WAN, you must change this port to function as a WAN port (the WAN2 port) and then set up a dynamic Internet connection on the WAN2 port.

Before you start, be sure that you have the dynamic IP address information that your ISP gave you. This information is different from the information for the WAN1 port.

To configure dual WAN with a dynamic Internet connection for the WAN2 port:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > WAN > WAN2**.
   The page displays the options for the WAN2 port.
6. Activate or deactivate the **VLAN Tagging** toggle.
   When enabled, the router adds an 802.1Q VLAN tag with the configured VLAN ID to all traffic on the WAN2 port. When disabled, WAN2 traffic is sent untagged.
7. In the **VLAN ID** field, enter the VLAN ID.
   A VLAN ID is a number from 1 to 4094 that identifies a specific virtual LAN (VLAN) on your network. Use the VLAN ID that your ISP or network administrator provides for the WAN1 connection.
8. From the **Connection Type** menu, select **DHCP**.
   The page adjusts.
9. In the **External IP Address**, **Subnet Mask**, and **Gateway IP Address** fields, type the static IP address, subnet mask, and gateway IP address that your ISP gave you.

10. To add an additional WAN IP address, click the **Additional WAN IP Address** toggle so that it is positioned to the right.
    The page adjusts.
11. Depending on how you want to specify additional IP addresses, do one of the following:
    - **Manually enter an additional IP address**:
      a. Select the **Add IP Address** radio button.
      b. In the **IP Address** field, enter an IP address.
      c. Click the **Add** button.
    - **Set an IP address range**:
      a. Select the **Add IP Range** radio button.
      b. Enter a start and end IP address.
      c. Click the **Add** button.
12. In the **Router MAC Address** section, select one of the following radio buttons:
    - **Use Default MAC Address**: Use the default router MAC address that displays on the router dashboard page and the router label. This is the default setting.
    - **Configure MAC Address**: In the MAC address field, enter a specific MAC address.
13. In the **MTU** section, select one of the following radio buttons:
    - **Auto**: Your ISP assigns the MTU size automatically. In most situations, the default size is 1500 bytes.
    - **Manual**: Your ISP requires you to use a specific MTU size. In the MTU size field, enter the size.
14. In the **DHCP Options** section, in the **Vendor Class** field, enter the vendor class identifier provided by your ISP.
15. In the **Client Identifier** field, enter the client identifier provided by your ISP.
16. Click the **Save** button.
    Your settings are saved.

Configure dual WAN with a static Internet connection for the WAN2 port

By default, the LAN5 WAN2 port functions as a LAN port. You can change the function of the port to a WAN port (the WAN2 port) and set up a static ISP connection for the WAN2 port.

Before you start, be sure that you have the static IP address information that your ISP gave you. This information is different from the information for the WAN1 port.

To configure dual WAN with an ISP connection that uses a static IP address for the WAN2 port:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.

2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > WAN > WAN2**.
6. Activate or deactivate the **VLAN Tagging** toggle.
   When enabled, the router adds an 802.1Q VLAN tag with the configured VLAN ID to all traffic on the WAN2 port. When disabled, WAN2 traffic is sent untagged.
7. In the **VLAN ID** field, enter the VLAN ID.
   A VLAN ID is a number from 1 to 4094 that identifies a specific virtual LAN (VLAN) on your network. Use the VLAN ID that your ISP or network administrator provides for the WAN2 connection.
8. From the **Connection Type** menu, select **Static**.
   The page adjusts.
9. In the **External IP Address**, **Subnet Mask**, and **Gateway IP Address** fields, type the static IP address, subnet mask, and gateway IP address that your ISP gave you.
10. To add an additional WAN IP address, click the **Additional WAN IP Address** toggle so that it is positioned to the right.
    The page adjusts.
11. Depending on how you want to specify additional IP addresses, do one of the following:
    - **Manually enter an additional IP address**:
      a. Select the **Add IP Address** radio button.
      b. In the **IP Address** field, enter an IP address.
      c. Click the **Add** button.
    - **Set an IP address range**:
      a. Select the **Add IP Range** radio button.
      b. Enter a start and end IP address.
      c. Click the **Add** button.
12. In the **Router MAC Address** section, select one of the following radio buttons:
    - **Use Default MAC Address**: Use the default router MAC address that displays on the router dashboard page and the router label. This is the default setting.
    - **Configure MAC Address**: In the MAC address field, enter a specific MAC address.

13. In the **MTU** section, select one of the following radio buttons:
    - **Auto**: Your ISP assigns the MTU size automatically. In most situations, the default size is 1500 bytes.
    - **Manual**: Your ISP requires you to use a specific MTU size. In the MTU size field, enter the size.
14. Click the **Save** button.
    Your settings are saved.

Configure dual WAN with a PPPoE Internet connection for the WAN2 port

By default, the LAN5 WAN2 port functions as a LAN port. You can change the function of the port to a WAN port (the WAN2 port) and set up a PPPoE ISP connection for the WAN2 port.

Before you start, be sure that you have the PPPoE information that your ISP gave you. This information is different from the information for the WAN1 port.

To configure dual WAN with an ISP connection that uses a PPPoE IP address for the WAN2 port:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > WAN > WAN2**.
6. Activate or deactivate the **VLAN Tagging** toggle.
   When enabled, the router adds an 802.1Q VLAN tag with the configured VLAN ID to all traffic on the WAN2 port. When disabled, WAN2 traffic is sent untagged.
7. In the **VLAN ID** field, enter the VLAN ID.
   A VLAN ID is a number from 1 to 4094 that identifies a specific virtual LAN (VLAN) on your network. Use the VLAN ID that your ISP or network administrator provides for the WAN1 connection.
8. From the **Connection Type** menu, select **PPPoE**.
   The page adjusts.

9. In the **External IP Address**, **Subnet Mask**, and **Gateway IP Address** fields, type the static IP address, subnet mask, and gateway IP address that your ISP gave you.
10. To add an additional WAN1 IP address, click the **Additional WAN IP Address** toggle so that it is positioned to the right.
    The page adjusts.
11. Depending on how you want to specify additional IP addresses, do one of the following:
    - **Manually enter an additional IP address**:
      a. Select the **Add IP Address** radio button.
      b. In the **IP Address** field, enter an IP address.
      c. Click the **Add** button.
    - **Set an IP address range**:
      a. Select the **Add IP Range** radio button.
      b. Enter a start and end IP address.
      c. Click the **Add** button.
12. In the **Internet Information** section, enter a user name, password, and service name.
13. In the **Connection Mode** section, select one of the following radio buttons:
    - **Always On**
    - **Manually Connected**: If you choose this option, click the **Connect** button to establish the connection.

    Your settings are saved.

Configure dual WAN failover detection

Failover detection is the mechanism that lets the Insight detect if the primary WAN interface is up. If the interface is down, the Insight initiates a failover to the secondary WAN interface. The Insight then monitors both WAN interfaces and when the primary WAN interface comes back up, failover detection lets the Insight switch back to the primary WAN interface.

NOTE: A dual WAN traffic rule takes priority over a WAN failover. Therefore, after a WAN failover, traffic that is subject to a traffic rule for a specific WAN interface might stop if that WAN interface is down. That is, the traffic might not failover to the other WAN interface. For more information, see Manage dual WAN traffic rules on page 156.

By default the WAN1 port is set as the primary interface and the WAN2 port is set as the secondary interface. However, you can change the setting so that the WAN2 port functions as the primary interface and the WAN1 port as the secondary interface.

The Insight supports the following types of failover detection:

- **Ping the WAN DNS server**: The Insight sends pings to the DNS servers that are already configured for the primary and secondary interfaces.
- **Ping the WAN gateway**: The Insight sends pings to the gateways that are already configured for the primary and secondary interfaces.
- **Ping custom IP addresses**: The Insight sends pings to custom gateways that you must configure for the primary and secondary interfaces.
- **Query the WAN DNS server**: The Insight sends DNS queries to the DNS servers that are already configured for the primary and secondary interfaces.
- **Query custom IP addresses**: The Insight sends DNS queries to custom DNS servers that you must configure for the primary and secondary interfaces.

In a WAN failover configuration with two WAN interfaces, if a query or ping does not yield a response after the specified number of retries (the default is five retries), the Insight automatically switches to the active interface. This process is referred to as a *failover*.

For example, if four queries to the WAN1 port fail, the Insight automatically switches to the WAN2 port. After the failover, the Insight continues to sends queries or pings to the WAN1 port. If the WAN1 port comes back up and responds to a query or ping, the Insight switches back to the WAN1 port.

To configure failure detection:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.

3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > WAN > Dual WAN**.
6. Select **Configuration**.
   The page displays the failure detection options.
7. From the **Primary WAN** menu, select **WAN1** or **WAN2**.
   The selected interface functions as the primary WAN interface in the dual WAN configuration. The selection from the Secondary WAN menu is automatically adjusted because there are only two WAN interfaces.
8. From the **Failure Detection Method** menu, select the failure detection method and select where the Insight must send the ping or query:
   - **ping**: The Insight sends pings to the device that you select:
     - **Ping WAN DNS**: The Insight sends pings to the DNS server IP addresses that you already configured for the WAN1 and WAN2 ports.
     - **Ping WAN Gateway:** The Insight sends pings to the gateway IP addresses that you already configured for the WAN1 and WAN2 ports.
     - **Ping WAN Custom**: The Insight sends pings to custom IP addresses that you must enter in the IP address fields in the WAN1 Internet Connection Test and WAN2 Internet Connection Test sections.
   - **DNS lookup**:
     - **Lookup WAN DNS**: The Insight sends queries to the DNS server IP addresses that you already configured for the WAN1 and WAN2 ports.
     - **Lookup WAN Custom**: The Insight sends queries to custom IP addresses that you must enter in the IP address fields in the WAN1 Internet Connection Test and WAN2 Internet Connection Test sections.
9. If you select the **Lookup WAN Custom** or **Ping WAN Custom** radio button in the previous step, type the IP addresses in the **DNS 1** and **DNS 2** fields in both the **WAN 1 Configuration** and **WAN 2 Configuration** sections.
10. In the **Number of retries** field, enter the number of times that the Insight resends a query or ping after it did not receive a response from the WAN1 or WAN2 ports. The default number of retries is 5. That means that after five failed responses, the Insight fails over to the active interface. For example, if four queries to the WAN1 port fail, the Insight automatically fails over to the WAN2 port. The range is from 1 to 10 retries.

11. In the **Retry Interval** field, type the period in seconds after which the Insight sends a query or ping to the WAN1 and WAN2 ports to determine their status (up or down).
    By default, the period is 5 seconds. The range is from 1 to 3600 seconds.
12. Click the **Save** button.
    Your settings are saved.

Manage dual WAN traffic rules

The dual WAN traffic rule allows you to configure the WAN interface through which traffic leaves the router.

A dual WAN traffic rule takes priority over a WAN failover. As an example, assume that you set up two Dual WAN traffic rules:

- Rule 1: Traffic of VLAN 1 is sent to the WAN1 interface.
- Rule 2: Traffic of VLAN 2 is sent to the WAN2 interface.

Traffic of VLAN 3 is not subject to a dual WAN traffic rule. If the WAN1 interface goes down and traffic fails over to the WAN2 interface, traffic from VLAN2 and VLAN3 remains flowing, but traffic from VLAN 1 stops because rule 1 requires traffic to pass through the WAN 1 interface.

A dual WAN traffic rule on the Insight defines the following components:

- **Name**: The name of the dual WAN rule.
- **Service**: The rule can apply to all traffic or only to traffic for a specific predefined service or protocol.
- **Source interface**: The source interface is the interface from which the traffic originates. The rule can apply to all source interfaces or only to a specific WAN interface or VLAN interface.
- **Source address**: The source address is the IP address from which the traffic originates. The rule can apply to all source addresses or only to one IP address, a range of IP addresses, or a subnet of IP addresses.
- **Destination address**: The destination address is the IP address to which the traffic is sent. The rule can apply to all destination addresses, to a single IP address, or a subnet of IP addresses.
- **WAN Interface**: The WAN interface to which the rule applies.

⚠ **CAUTION:** You need some networking knowledge to set up traffic rules, because incorrectly configured traffic rules might block communication on the Insight.

Add a dual WAN traffic rule

You can add dual WAN traffic rules to prevent or allow traffic based on its protocol, source, destination, and other criteria.

To add a dual WAN traffic rule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > WAN > Dual WAN Rules**.
6. Click the **Add** button.
   The Add dual WAN Rule pop-up window displays.
7. In the **Rule Name** field, enter a name for the rule.
   The name is for identification purposes.
8. Choose whether to activate the new rule immediately at creation.
   - To enable the rule immediately, keep the **Enable Rule Status** toggle positioned to the right. This is the default setting.
   - To not enable at creation, click the **Enable Rule Status** toggle so that it is positioned to the left.
9. From the **Service** menu, select the predefined service.
10. From the **Source Interface** menu, select the VLAN or WAN interface from which the traffic originates. By default, the traffic can originate from any interface.
11. From the **Source Address** menu, select the IP address, subnet, or address range from which the traffic can originate:

Any	The traffic can originate from any IP address. This is the default setting.
Single	The traffic can originate from a single address. Enter the IP address.
Subnet	The traffic can originate from a subnet. Enter the IP address and subnet.

12. From the **Destination Address** menu, select the IP address, subnet, or address range to which the traffic can be sent:

Any	The traffic can be sent to any IP address. This is the default setting.
Single	The traffic can be sent to a single address. Enter the IP address.
Subnet	The traffic can be sent to a subnet. Enter the IP address and subnet.

13. From the **Policy** menu, select a WAN interface:
    - **WAN1 Only**
    - **WAN2 Only**
14. Click the **Add** button.
    If you enabled the new dual WAN rule (that is, the **Enable Rule Status** toggle is blue and positioned to the right), it goes into effect immediately. The dual WAN traffic rule is assigned the lowest priority in relation to existing rules.
    The new dual WAN rule is added to the table on the Rules page.
15. To change the priority for a rule, select the Move icon, and drag to your desired position in the table.

Change a dual WAN traffic rule or its priority, or enable or disable the rule

You can change an existing dual WAN traffic rule or its priority, or enable or disable the rule.

To change a dual WAN traffic rule or its priority, or enable or disable the rule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > WAN > Dual WAN Rules**.
6. To change the settings for a rule or enable or disable the rule, do the following:
   a. In the table, select the check box for the rule.
   b. Click the **Edit** icon. The Edit Dual WAN Rule pop-up window displays.
   c. Change the settings for the rule, or enable or disable the rule: For more information about the settings, see Add a dual WAN traffic rule on page 156.
   d. To activate or deactivate the rule, click the **Enable Rule Status** toggle:
      - **The toggle is positioned to the right**: The rule is activated.
      - **The toggle is gray and positioned to the left**: The rule is deactivated.
   e. Click the **Apply** button. Your settings are saved. The modified rule displays in the table on the Rules page.

7. To change the priority for a rule, select the Move icon, and drag to your desired position in the table.

Delete a dual WAN traffic rule

If you no longer need a dual WAN traffic rule, you can remove it.

To remove a dual WAN traffic rule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > WAN > Dual WAN Rules**.
6. In the table, select the check box for the rule.
7. Click the **Delete** icon.
   A pop-up window display.
8. Click the **OK** button.
   Your settings are saved. The rule is removed from the table.

Manage the QoS Settings

You can manage the quality of service (QoS) settings in the form of Smart Queue Management (SQM) for WAN interfaces. SQM keeps latency as low as possible on all types of traffic.You can activate and configure SQM for each WAN interface.

- Use a speed test to automatically configure SQM for a WAN port
- Manually configure SQM for a WAN port

Use a speed test to automatically configure SQM for a WAN port

On Insight-managed routers, you can run a speed test from the Insight Cloud Portal to automatically configure SQM.

To use a speed test to automatically configure Smart QueueManagement(SQM) for a WAN port:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > QoS**.
   The QoS page displays. By default the WAN1 tab is selected.
6. To configure the WAN2 port instead of the WAN1 port, click the **WAN2** tab.
7. Activate the **Enable Smart Queue Management** toggle so that it is positioned to the right.
   By default, this setting is disabled.
8. In the **Internet Bandwidth** section, select the **Let speed test detect my Internet bandwidth** radio button.
9. Click the **Speed Test** button.
   A pop-up window displays.
10. Click the **Speed Test** button.
    The Privacy Policy pop-up window displays.
11. Click the **Agree** button.
    After a short delay, the pagedisplays the measuredlatency(delay) in ms,download speed in Mbps, and upload speed in Mbps.
12. **(Optional)** To stop the speed test, click the **Stop Test** button.

    NOTE: If you do not stop the speed test, the test stops by itself after about 40 seconds, and the name of the Test Speed button changes to Done.
13. Click the **Done** button.
    The detected speeds are automatically entered in the Download and Upload fields.
14. To change the speeds in the fields, manually overwrite them.
15. Click the **Apply** button.
    Your settings are saved.

Manually configure SQM for a WAN port

For Insight-managed routers, you can manually specify upload and download speed limits for a WAN port to ensure SQM operates with values that reflect your actual network conditions.

To manually configure SQM for a WAN port:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > QoS**.
   The QoS page displays. By default the WAN1 tab is selected.
6. To configure the WAN2 port instead of the WAN1 port, click the **WAN2** tab.
7. Activate the **Enable Smart Queue Management** toggle so that it is positioned to the right.
   By default, this setting is disabled.
8. In the **Internet Bandwidth** section, select the **I want to define Internet bandwidth** radio button.
   The Internet Bandwidth section displays.
9. In the **Download** field, type the maximum download bandwidth that must be applied to the WAN port, and set the unit as **Kbps**, **Mbps**, or **Gbps**.
   The range is from 300 Kbps to 5 Gbps.
10. In the **Upload** field, type the maximum upload bandwidth that must be applied to the WAN port, and set the unit as **Kbps**, **Mbps**, or **Gbps**.
    The range is from 300 Kbps to 5 Gbps.
11. Click the **Save** button.
    Your settings are saved.

Manage Static Routes

For almost all Internet traffic, routes are automatically and dynamically selected. You can also set up a fixed, static IPv4 route on an Insight-managed router. Typically, you only need to add static routes when you have more than one router or multiple IP subnets on your network.

This chapter contains the following sections:

- Add a static route
- Edit a static route
- Delete a static route

Add a static route

You can add an IPv4 static route to a destination IP address and specify the subnet mask, next hop IP address, and metric.

To add a static route:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Routing > Static Route > Add**.
   The Add Static Route page displays.
6. In the **Route Name** field, enter a name for the static route.
7. In the **Network** field, enter the IPaddress for the final destination of the route.
8. In the **Subnet Mask** field, enter theIPsubnet maskfor the final destination of the route.
   If the destination is a single host, enter 255.255.255.255.
9. In the **Gateway** field, enter the IP address of the gateway for the route.
   This is the gateway or next router in the path from your network to the final destination of the route.
10. In the **Metric** field, enter a value between 1 and 254 to set the route priority.
    The metric value represents the number of routers between your network and the final destination of the route.
11. From the **Interface** menu, select a VLAN profile (in this context referred to as a VLAN interface) or a WAN interface.
12. Click the **Status** toggle to activate or deactivate the static route.

13. Click the **Save** button.
    Your settings are saved.

Edit a static route

You can change an existing static route.

To edit a static route:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Routing > Static Route**.
   The Add Static Route page displays.
6. In the table, select the check box for the route.
7. Click the **Edit** button.
   The Edit Static Route pop-up page displays.
8. Change the settings for the route.
9. Click the **Save** button.
   Your settings are saved.

Delete a static route

If you no longer need a static route, you can remove it.

To delete a static route:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.

4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Routing > Static Route**.
   The Add Static Route page displays.
6. In the table, select the check box for the route.
7. Click the **Delete** button.
   Your settings are saved.

Manage VPN settings

You can use the VPN settings to securely connect remote devices to your network, configure VPN services, and manage access for users and clients. You can manage VPN features, set up VPN profiles, and adjust advanced options to meet your security and connectivity requirements.

This chapter contains the following sections:

- Configure OpenVPN settings
- Manage site-to-site VPN connections
- Manage custom IPSec VPN profiles
- Manage client-to-site VPN connections
- Manage VPN users
- Manage WireGuard settings and clients

Configure OpenVPN settings

You can configure OpenVPN on your Insight-managed router to allow remote client devices to connect securely to your network over the internet. OpenVPN is an open-source VPN protocol that provides a flexible and reliable solution for remote access, supporting a wide range of client devices including Windows, macOS, iOS, and Android. Configuring OpenVPN on the router establishes it as the VPN server, enabling remote workers to access network resources such as file servers and printers securely from any location.

To configure OpenVPN settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.

4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > VPN > OpenVPN**.
6. To enable the OpenVPN connection, click the **Enable Connection** so it is positioned to the right.
7. In the **Mode Services** section, configure the following settings:
   a. From the **Service Mode Type** menu, select **TUN** or **TAP** as the tunnel mode.
      TUN is a routed IP tunnel mode that operates at Layer 3 (network layer) and is typically used for routing IP traffic between networks. TAP is a bridged Ethernet tunnel mode that operates at Layer 2 (data link layer) and is typically used to extend a LAN over the VPN, allowing Ethernet frames, broadcasts, and non-IP traffic to pass through the tunnel.
   b. From the **Protocol** menu, select UDP or TCP as the protocol for the VPN connection.
      UDP is generally faster and preferred for most VPN connections because it has lower overhead and does not require connection-oriented handshakes. TCP provides reliable, connection-oriented delivery and might be preferred in environments where packet loss is common or where strict reliability is required, but it can introduce additional latency.
   c. In the **Port** field, enter the port number for the OpenVPN server.
      The default port is 1194.
8. In the **Client Address Pool** section, configure the IPv4 virtual network for VPN clients:

   NOTE: The IP address range for VPN clients must not overlap with any WAN, VLAN, or VPN subnet. Express the range in network address and subnet mask format, for example 10.0.8.0/255.255.255.0. The OpenVPN server is allocated the first usable address within the network; the remaining usable addresses are allocated to clients.

   a. In the **IP Address** field, enter the network address for the client address pool.
   b. In the **Netmask** field, enter the subnet mask that defines the size of the client address pool.
9. In the **DNS Server** section, select one of the following options:

Auto	The router assigns the DNS server automatically.
Custom	Enter a primary DNS server IP address in the **Primary DNS Server** field. Optionally, enter a secondary DNS server IP address in the **Secondary DNS Server** field.

10. To configure advanced settings, click **Advanced Settings** and configure the following options as required:
    a. To allow multiple simultaneous connections using the same client credentials, click the **Duplicate Connection** toggle so it is positioned to the right.
    b. To prevent VPN clients from communicating with each other, click the **Client Isolation** toggle so it is positioned to the right.
    c. In the **Domain Name** field, enter the domain name to assign to VPN clients.
    d. To route only traffic destined for configured subnets through the VPN tunnel, click the **Split Tunnel** toggle so it is positioned to the right.
    The Add Split Tunnel pop-up window displays.
    e. In the **IP Address** field, enter the network address for the client address pool.
    f. In the **Netmask** field, enter the subnet mask that defines the size of the client address pool.
    g. Click the **Add** button.
11. Click **Save**.
    Your settings are saved.
12. **(Optional)** To export the client configuration, do the following:
    a. In the **Export Client Configuration** section, from the **Server Certificate** menu, select the server certificate to use for the VPN connection.
    b. In the **Connection Name** field, enter a name for the VPN connection.
    c. In the **Server Address** field, enter the IP address or domain name of the OpenVPN server.
    d. Click the **Export** button.

Manage site-to-site VPN connections

Your router can also connect to another router that supports IPSec. IPsec is a secure network protocol that authenticates and encrypts data sent through a VPN tunnel. IPSec uses either version one or version two of the Internet Key Exchange protocol (IKEv1 or IKEv2) to establish a secure connection.

Add a site-to-site IPSec VPN connection

To set up an IPSec VPN tunnel:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.

3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > VPN > IPSec > Site-to-Site**.
6. Click the **Add** button.
   The Add New Connection pop-up page displays.
7. Click the **Enable Connection** toggle to activate a VPN connection.
   The router attempts to establish a VPN tunnel with the remote VPN router immediately after you click the **Add** button. To prevent this, deactivate the **Enable Connection** toggle so that it is positioned to the left.
8. In the **Connection Name** field, enter a name for the connection.
   The name is for identification purposes only.
9. From the **IPSec Profile** menu, select an existing profile.
10. In the **Remote Endpoint** field, enter the WAN IP address or domain name of the remote VPN router.
11. In the **Pre-Shared Key** field, enter the password used for IKE authentication between the router and the remote VPN router.
    - The same password must be configured on the remote VPN router.
    - To display the password, click the **eye** icon.
12. In the **Local Networks** section, click the **Add** button to create a local network.
    a. Select a **VLAN** or **Custom** from the menu.
    b. In the **IP Address** field, enter an IP address.
    c. In the **Subnet Mask** field, enter a subnet mask.
    d. Click **Add**.
       The settings are saved and the local network is created.
    - To modify the local network, click the **Edit** icon and update the IP address or subnet mask.
    - To delete the local network, select the associated check box and click the **Delete** icon.
13. In the **Remote Networks** section, click the **Add** icon to create a remote network.
    a. In the **IP Address** field, enter an IP address.
    b. In the **Subnet Mask** field, enter a subnet mask.
    c. Click **Add**.
       The settings are saved and the remote network is created.

14. In the **Local Identifier** section, specify one of the following settings:

IP Address	The IP address that you want to use for the router.
Local WAN IP	The WAN IP address of the router. When you select this option, the Local Identifier field automatically displays the WAN IP address of the router.
Local FQDN	The domain name for the router.

These settings specify the LAN IP addresses that the VPN tunnel can access on the router.

15. In the **Remote Identifier** section, specify one of the following settings:

Remote WAN IP	The WAN IP address of the remote VPN router.
Remote FQDN	The domain name for the remote VPN router.

These settings specify the LAN IP addresses that the VPN tunnel can access on the remote VPN router.



CAUTION: The remote LAN IP addresses and the local LAN IP addresses cannot be in the same subnets. For example, if the local subnet specified in the previous step is 192.168.1.x, the remote subnet must be a different subnet such as 192.168.2.x, and cannot be 192.168.1.x.

16. Click **Add**.
Your settings are saved. The VPN connection is added to the table on the **Site-to-Site** page.
If you activated the connection in step 7, the router immediately attempts to establish the VPN tunnel with the remote VPN router. If you did not enable the connection, you can do so later.

Configure DPD options for a site-to-site IPSec VPN connection

Dead Peer Detection (DPD) is a mechanism on an Insight-managed router that can prevent a VPN tunnel from being disconnected when traffic is idle. In some cases, you might not want a VPN tunnel to be disconnected when traffic is idle, for example, when client-server applications over the tunnel cannot tolerate the tunnel establishment time. If you require a VPN tunnel to remain connected, you can use DPD to prevent the tunnel from being disconnected and to force a reconnection if the tunnel disconnects for any reason. For DPD to function, the peer VPN device on the other end of the tunnel also needs to support DPD.

To configure DPD options for a site-to-site IPSec VPN connection:

1. Access the Insight Cloud Portal.
For more information, see Log in to the Insight Cloud Portal on page 9.
The Insight Dashboard displays.

2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > VPN > IPSec > Site-to-Site**.
6. In the table, select the check box for the connection.
7. Click the **Edit** button.
   The Edit Connection pop-up page displays.
8. To configure Dead Peer Detection (DPD), in the **DPD Options** section, do the following:
   a. Activate the **Enable DPD Options** toggle.
      By default, the toggle is positioned to the right, which means DPD is activated. If the toggle is positioned to the left, DPD is disabled.
   b. In the **DPD Delay** field, enter the interval in seconds between consecutive DPD messages.
      The default is 10 seconds. The valid range is 10–300 seconds.
   c. In the **Detection Time** field, enter the period in seconds during which the client must receive a DPD response from the remote VPN router.
      If this period is exceeded, the VPN tunnel is torn down and the client attempts to reestablish it. The default is 30 seconds. The valid range is 30–1800 seconds.
   d. From the **DPD Action** menu, select the action to take if no timely DPD response is received from the remote VPN router:

DPD action	Description	Result
None	No action is taken.	The VPN tunnel remains in its current state.
Restart	The router tears down the VPN tunnel and attempts to reestablish it.	The VPN tunnel is re-created if the remote peer is reachable.
Clear	The router tears down the VPN tunnel but does not attempt to reestablish it.	The VPN tunnel remains down until it is manually reestablished.

If the router detects a connection failure with the remote VPN router, it tears down the VPN tunnel and attempts to reestablish it.

9. Click **Add**.
   Your settings are saved.

Edit a site-to-site IPSec VPN connection

To edit a site-to-site IPSec VPN connection:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > VPN > IPSec > Site-to-Site**.
6. In the table, select the check box for the connection.
7. Click the **Edit** button.
   The Edit Connection pop-up page displays.
8. Change the settings for the connection.
9. Click the **Apply** button.
   Your settings are saved.

Delete a site-to-site VPN connection

To delete a site-to-site VPN connection:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > VPN > IPSec > Site-to-Site**.

6. In the table, select the check box for the connection you want to delete.
7. Click the **Delete** button.
   Your settings are saved.

Manage custom IPSec VPN profiles

An IPSec VPN profile is a set of protocols and encryption, authentication, and integrity-check algorithms that form a building block for a VPN tunnel configuration on an Insight-managed router. You must first set up an IPSec profile (or use a predefined IPSec profile), and then define the VPN tunnel settings.

An IPSec VPN profile defines the protocols and algorithms, also referred to as security association (SA) settings, that secure a VPN tunnel on an Insight-managed router.

- **IKE version**: The Internet Key Exchange (IKE) version 1 (IKEv1) or 2 (IKEv2) protocol. IKEv2 is more advanced than IKEv1, but there might be situations in which you need to use the older IKEv1. The version that you must select depends on your network requirements.
- **Phase I options**: The IKE Phase I settings define the authentication and negotiation exchange between the two VPN partners before the VPN tunnel is established. You must specify the encryption and authentication algorithms, as well as the Diffie-Hellman group algorithm for the verification and exchange of keys. Both VPN partners must use the same algorithms so that the communication between the routers can be authenticated and is secure *before* the VPN tunnel is established.
- **Phase II options**: The IKE Phase II settings define how the VPN tunnel is set up and encapsulated between the two VPN partners, and how the VPN tunnel traffic is kept secure *after* the tunnel is established. You must specify the encapsulation protocol, encryption algorithm, and an integrity check algorithm that allows the VPN partner to guard against modification of the tunnel and the traffic that is transported through the tunnel. Here too, both VPN partners must use the same algorithms.

Supported IPSec Algorithm Options

The following table lists the encryption, authentication, and Diffie-Hellman (DH) Group options available when configuring an IPSec VPN profile. The same options apply to both Phase I and Phase II settings, with the exception of the GCM-based encryption algorithms (AES128-GCM-16 and AES256-GCM-16), which are available in Phase II only. In ascending order, each algorithm provides more security.

Table 3.IPSec Algorithm Options

Setting	Option	Description	Availability
Encryption	AES-128	Advanced Encryption Standard (AES) with	Phase I and Phase II

Setting	Option	Description	Availability
In ascending order, each algorithm provides more security.		a 128-bit key size. This is the default setting.	
	AES-192	AES with a 192-bit key size.	Phase I and Phase II
	AES-256	AES with a 256-bit key size.	Phase I and Phase II
	3DES	Triple Data Encryption Standard (3DES).	Phase I and Phase II
	AES128-GCM-16	AES with a 128-bit key size and Galois/ Counter Mode (GCM) with an Integrity Check Value (ICV) of 16 bytes.	Phase II only
	AES256-GCM-16	AES with a 256-bit key size and GCM with an ICV of 16 bytes.	Phase II only
Authentication In ascending order, each algorithm provides more security.	MD5	Hash algorithm that produces a 128-bit digest.	Phase I and Phase II
	SHA1	Hash algorithm that produces a 160-bit digest. This is the default setting.	Phase I and Phase II
	SHA2-256	Hash algorithm that produces a 256-bit digest.	Phase I and Phase II
DH Group	—	The Diffie-Hellman (DH) group sets the strength of the key exchange algorithm in bits. In ascending order, each group provides more security but may require more computing power, which can slow down Phase I traffic. Select one of the available DH groups from the menu.	Phase I and Phase II
SA Lifetime	Phase I default	28800 seconds (8 hours). Valid range: 3600–86400 seconds.	Phase I
	Phase II default	3600 seconds (1 hour). Valid range: 3600–28800 seconds.	Phase II

Add a custom IPSec VPN profile

Before you begin

Before you can set up a VPN tunnel between two VPN routers at different sites or between a client and a VPN router, you must define the IPSec profile that secures the VPN tunnel.

Internet Key Exchange (IKE) is the protocol used to set up security associations (SAs) between VPN routers or between a client and a VPN router to ensure the following:

- The VPN tunnel is established between the correct partners.
- The VPN tunnel cannot be altered while traffic is passing through.
- Traffic passing through the VPN tunnel is secured.

The strength of the algorithms you select for an IPSec VPN profile depends on the sensitivity and speed of the traffic that must travel through the VPN tunnel for which the profile will be used.

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > VPN > IPSec Profiles > Add**.
   The Add/Edit IPSec Profile pop-up window displays.
6. In the **Profile Name** field, enter a name for the profile.
   The name is for identification purposes only.
7. Select an IKE Version radio button:
   - **IKEv1**: The profile uses IKEv1. IKEv1 is superseded by IKEv2 but some VPN routers do not support IKEv2. If you do not know if the remote VPN router supports IKEv2, we recommend that you use IKEv1.
   - **IKEv2**: The profile uses IKEv2. If the remote VPN router also supports IKEv2, we recommend that you use IKEv2.

   NOTE: The only selection from the **Protocol Selection** menu is ESP (Encapsulating Security Payload), which authenticates, encrypts, and guards against data changes during transmission. This is the fixed setting.
8. In the **Phase I Options** section, specify the algorithm settings.
   For available options, see Supported IPSec Algorithm Options on page 171.
9. To add another algorithm for the same Phase I, click **Add Algorithm** and repeat the previous step.
   Adding more algorithms improves the chances of devices being able to find a match, because not all devices might support the same types of algorithms.

10. In the **SA Lifetime** field for Phase I, enter the period in seconds during which the IKE security association (SA) is valid.
    If the period times out, the next rekeying occurs. The default is 28800 seconds (8 hours). The period can be between 3600 and 86400 seconds.
11. In the **Phase II Options** section, specify the algorithm settings.
    For available options, see Supported IPSec Algorithm Options on page 171.
12. To add another algorithm for the same Phase II, click **Add** and repeat the previous step.
    Adding more algorithms improves the chances of devices being able to find a match, because not all devices might support the same types of algorithms.
13. In the **SA Lifetime** field for Phase II, enter the period in seconds during which the IKE security association is valid.
    If the period times out, the next rekeying occurs. The default is 3600 seconds (1 hour). The period can be between 3600 and 28800 seconds.
14. Click the **Add** button.
    Your settings are saved.

Edit an IPSec VPN profile

NOTE: It is not possible to edit or delete a default VPN profile.

To edit an IPSec VPN profile:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > VPN > IPSec Profiles**.
6. In the table, select the check box for the profile you want to edit.
7. Click the **Edit** button.
   The Edit IPSec Profile pop-up page displays.

8. Edit the profile configuration.
   For more information on the profile settings, see Add a custom IPSec VPN profile on page 172.
9. Click the **Update** button.
   Your settings are saved.

Delete an IPSec VPN profile

To delete an IPSec VPN profile:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > VPN > IPSec Profiles**.
6. In the table, select the check box for the profile you want to delete.
7. Click the **Delete** button.
   The Delete Confirmation pop-up window displays.
8. Click the **Yes** button.
   Your settings are saved.

Manage client-to-site VPN connections

Using the Insight Cloud Portal, you can configure client-to-site VPN connections on your Insight-managed router. A client-to-site VPN connection is typically used to give remote workers secure access to the resources on your local network, such as file servers and printers, from any location.

Each client-to-site VPN connection uses an IPSec profile to define the encryption and authentication algorithms that secure the tunnel. Before you can add a client-to-site VPN connection, you must configure an IPSec profile. For more information, see Manage custom IPSec VPN profiles on page 171.

Add a client-to-site VPN connection

Before you begin Before you add a client-to-site VPN connection, you must configure an IPSec

profile. For more information, see Add a custom IPSec VPN profile on page 172.

Add a client-to-site IPSec VPN connection on your Insight-managed router to allow remote client devices to connect securely to your network.

To add a client-to-site VPN connections:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > VPN > IPSec > Client-to-Site**.
6. Click the **Add** button.
   The Add Client-to-Site Connection pop-up page displays.
7. To enable the VPN connection, click the **Enable Connection** toggle so it is positioned to the right.
8. In the **Connection Name** field, enter a name for the VPN connection.
9. From the **IPSec Profile** menu, select the IPSec profile to use for the VPN connection.
10. From the **Interface** menu, select the WAN interface for the VPN connection.
11. From the **Local Identifier Type** menu, select the type of local identifier for the VPN connection.
12. In the **Local Identifier** field, enter the local identifier for the VPN connection.
13. From the **Remote Identifier Type** menu, select the type of remote identifier for the VPN connection.
14. In the **Remote Identifier** field, enter the remote identifier for the VPN connection.
15. From the **Authentication Method** menu, select the authentication method for the VPN connection.
16. If you selected **Pre-Shared Key (PSK)**, enter the pre-shared key in the **Pre-Shared Key** field.
17. In the **VPN Clients Pool Range** section, configure the IP address pool for VPN clients:
    a. In the **Start IP** field, enter the first IP address in the client address pool.
    b. In the **End IP** field, enter the last IP address in the client address pool.

18. In the **DNS Server** section, select one of the following options:

Auto	The router assigns the DNS server automatically.
Custom	Enter a primary DNS server IP address in the **Primary DNS Server** field. Optionally, enter a secondary DNS server IP address in the **Secondary DNS Server** field.

19. To configure advanced settings, click **Advanced Settings** and configure the following options as required:
    a. To prevent VPN clients from communicating with each other, click the **Client Isolation** toggle so that it is positioned to the right.
    b. To route only traffic destined for configured subnets through the VPN tunnel, click the **Split Tunnel** toggle so that it is positioned to the right.
    c. To activate Dead Peer Detection (DPD), click the **Enable DPD Options** toggle so that it is positioned to the right, then configure the following settings:

    NOTE: DPD allows the router to detect when a VPN peer is no longer reachable and take appropriate action.

    - In the **DPD Delay** field, enter the interval in seconds between DPD checks. The default value is 10 seconds.
    - In the **Detection Time** field, enter the time in seconds after which an unresponsive peer is considered dead. The default value is 30 seconds.

20. Click the **Add** button.
    Your settings are saved.

Edit a client-to-site VPN connection

To edit a client-to-site VPN connection:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > VPN > IPSec > Client-to-Site**.
6. From the table, select the check box of the connection you want to edit.

7. Click the **Edit** button.
   The Edit Client-to-Site Connection pop-up page displays.
8. Edit the connection configuration.
9. Click the **Update** button.
   Your settings are saved.

Delete a client-to-site VPN connection

To delete a client-to-site VPN connection:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > VPN > IPSec > Client-to-Site**.
6. From the table, select the check box of the connection you want to delete.
7. Click the **Delete** button.
   The Delete Configuration pop-up window displays.
8. Click the **Yes** button.
   Your settings are saved.

Manage VPN users

You can add, edit, and remove VPN user accounts, configure authentication settings, and control remote access to your network resources.

This chapter contains the following sections:

Add a VPN user

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.

2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > VPN > VPN User > Add**.
   The Add VPN User pop-up page displays.
6. To activate the VPN user account, click the **Enable** toggle.
7. In the **Username** field, enter a user name for the VPN user.
8. In the **Password** field, enter a password for the VPN user.
   The password must be 8–64 characters.
9. **(Optional)** To restrict the VPN user's access to specific VLANs, activate the **Enable Access Control** toggle, then configure the access control entries:
   a. From the **VLAN Name** menu, select the VLAN to which the user requires access.
   b. In the **IP Address** field, enter the IP address for the access control entry.
   c. In the **Subnet Mask** field, enter the subnet mask for the access control entry.
   d. To add more access control entries, click the add (**+**) icon.
10. Click the **Add** button.
    Your settings are saved.

Edit a VPN user

To edit a VPN user:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > VPN > VPN User**.
   The Add/Edit IPSec Profile pop-up window displays.

6. From the table, select the check box of the VPN user you want to edit.
7. Click the **Edit** button.
8. Update the VPN user configuration.
9. Click the **Update** button.
   Your settings are saved.

Delete a VPN user

To delete a VPN user:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > VPN > VPN User**.
   The Add/Edit IPSec Profile pop-up window displays.
6. From the table, select the check box of the VPN user you want to delete.
7. Click the **Delete** button.
   The Delete Configuration pop-up window displays.
8. Click the **Yes** button.
   Your settings are saved.

Manage WireGuard settings and clients

Using the Insight Cloud Portal, you can configure WireGuard on your Insight-managed router. WireGuard uses state-of-the-art cryptography to provide a fast, secure, and simple VPN solution. You configure WireGuard in two stages: first set up the VPN server settings on the router, then add the client devices that will connect to it.

The WireGuard VPN server settings define how the router operates as a VPN server, including the server address, port, encryption keys, and the IP address pool from which client devices are assigned addresses. The router supports up to 30 concurrent connected client devices through a single WireGuard VPN connection.

Each device that requires VPN access must be added as a separate WireGuard client entry on the router. When a client connects, traffic is routed in one of the following modes:

- **Full tunnel**: All traffic from the client device, including internet traffic, is routed through the VPN tunnel and encrypted. This mode provides the highest level of security.
- **Split tunnel**: Only traffic destined for configured subnets is routed through the VPN tunnel. Other traffic is sent directly to the internet unencrypted. This mode can improve performance for applications that do not require VPN protection.

This chapter contains the following sections:

- Configure WireGuard settings
- Add a WireGuard client
- Edit WireGuard client settings
- Delete a WireGuard client

Configure WireGuard settings

Configure the WireGuard VPN server settings on your Insight-managed router, including the server connection, IP address range for VPN clients, and DNS server settings.

To configure WireGuard settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > VPN > WireGuard > WireGuard Settings**.
6. To activate the WireGuard VPN connection, click the **Enable Connection** toggle so it is positioned to the right.
7. In the **Server Address** field, enter the IP address or domain name of the WireGuard VPN server.
8. In the **Port** field, enter the port number for the WireGuard VPN server.
   The default port is 51820.
9. To generate a key pair, click the **Generate Key** button.
   The router populates the **Private Key** and **Public Key** fields automatically.

10. In the **IP Address Range for VPN Clients** section, configure the IPv4 virtual network for VPN clients:

> **NOTE:** The IP address range for VPN clients must not overlap with any WAN, VLAN, or VPN subnet. Express the range in network address and subnet mask format, for example 10.0.8.0/255.255.255.0. The WireGuard server is allocated the first usable address within the network; remaining usable addresses are allocated to clients.

    a. In the **IP Address** field, enter the network address for the VPN client IP address range.
    b. In the **Netmask** field, enter the subnet mask for the VPN client IP address range.

11. In the **DNS Server** section, select one of the following radio buttons:

Auto	The router assigns the DNS server automatically.
Custom	Enter a custom DNS server address. If you select this option, enter a custom DNS.

12. To prevent VPN clients from communicating with each other, click the **Client Isolation** toggle so it is positioned to the right.

13. In the **MTU** field, enter the maximum transmission unit size in bytes.
    The default MTU value is 1420.

14. Click the **Save** button.
    Your settings are saved.

Add a WireGuard client

Before you begin Before you can add a WireGuard client, you must enable the WireGuard VPN connection. See Configure WireGuard settings on page 181.

A WireGuard client is a device that connects to the WireGuard VPN server on your Insight-managed router. Each device that requires VPN access must be added as a separate client entry, identified by its own public key and assigned a dedicated IP address within the VPN client address pool. The router supports up to 30 concurrent connected client devices.

When a client connects, you can choose how its traffic is routed. With Split Tunnel disabled, all traffic including internet traffic is routed through the VPN tunnel. With Split Tunnel enabled, only traffic destined for configured subnets is routed through the tunnel, which can improve performance for applications that do not require VPN protection.

To add a WireGuard client:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.

2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > VPN > WireGuard > WireGuard Clients**.
6. Click the **Add** button.
   The Add WireGuard Client page displays.
7. To activate the WireGuard client, click the **Enable** toggle so that it is positioned to the right.
8. In the **Name** field, enter a name for the client.
9. In the **IP Address** field, enter the IP address to assign to the client.
10. In the **Keep Alive** field, enter the keepalive interval in seconds.
    The default value is 25 seconds. The valid range is 0–600 seconds.
    The router populates the **Public Key** and **Pre-Shared Key** fields automatically.
11. **(Optional)** To generate a new pre-shared key at any time, click **Regenerate**.
12. To allow the client to access only specific VLANs through the VPN tunnel, click the **Split Tunnel** toggle so that it is positioned to the right, and configure the split tunnel entries:

    NOTE: When **Split Tunnel** is disabled, the client operates in full tunnel mode and all traffic, including internet traffic, is routed through the VPN tunnel. When enabled, only traffic for the configured subnets is routed through the tunnel.

    a. From the **VLAN Name** menu, select the VLAN to route through the VPN tunnel.
    b. In the **IP Address** field, enter the IP address for the split tunnel entry.
    c. In the **Netmask** field, enter the subnet mask for the split tunnel entry.
    d. Click the add (**+**) button to add a split tunnel entry.
13. Click the **Add** button.
    Your settings are saved.

Edit WireGuard client settings

To edit WireGuard client settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.

2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > VPN > WireGuard > WireGuard Clients**.
6. From the table, select the check box next to the WireGuard Client you want to edit.
7. Click the **Edit** button.
   The Edit WireGuard Client page displays.
8. Edit the WireGuard client settings.
9. Click the **Update** button.
   Your settings are saved.

Delete a WireGuard client

To delete a WireGuard client:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > VPN > WireGuard > WireGuard Clients**.
6. From the table, select the check box next to the WireGuard Client you want to delete.
7. Click the **Delete** button.
   The Delete Configuration pop-up window displays.
8. Click the **Yes** button.
   Your settings are saved.

Manage security settings

The security settings in the Insight Cloud Portal gives you control over the traffic passing through your router. You can configure SASE security, firewall rules and schedules, NAT, traffic rules, port triggering and forwarding, and Dynamic DNS profiles. These settings protect your network and manage how traffic flows in and out.

This chapter contains the following sections:

- Configure Secure Access Service Edge security on a router
- Configure basic firewall settings
- Manage Firewall Schedules
- Manage Outbound NAT rules
- Manage Traffic Rules
- Manage Port Triggering Rules
- Manage Port Forwarding Rules
- Service Management
- Manage Dynamic DNS Profiles

Configure Secure Access Service Edge security on a router

Before you begin to configure SASE security on a router, you must activate it at the organization-level first. For more information, see Configure Secure Access Service Edge security for an organization on page 47.

You can configure NETGEAR Exium Secure Access Service Edge (SASE) security on your router using the Insight Cloud Portal and Exium Security Cloud to achieve a unified approach to network protection and management. This protects your devices with advanced safeguards, including secure remote access, traffic inspection, and policy-based control for users and applications across all connected locations. By centralizing configuration and monitoring in the cloud, you can consistently enforce security policies, gain visibility into network activity, and quickly respond to emerging threats.

To configure SASE security on an Insight-managed router:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.

5. Click **Settings > Security > Security Center**.
   The Gateway Security page displays.
6. Click the **Enable SASE Security** button.
   Your settings are saved.
7. **(Optional)** Click the **Security Configurations** button to open the Exium Security page.

Configure basic firewall settings

Use the basic firewall settings on an Insight-managed router to control how external traffic reaches your network, protect against port scans and denial-of-service (DoS) attacks, and manage how long inactive sessions remain open. You can also configure a DMZ server for a specific device, decide whether the router responds to internet ping requests, optimize VoIP traffic with SIP ALG, and set limits on concurrent connections. In addition, VPN passthrough options let you allow IPSec, PPTP, and L2TP VPN traffic so remote users can securely connect through the router.

To configure basic firewall settings for an Insight-managed router:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Security > Firewall > Basic Settings**.
6. From the table, select the check box of a service, and click the **Edit** button.
   The Edit Service pop-up page displays.
7. Activate or deactivate the Enable Port Scan and DoS (Denial of Service) Protection toggle to detect and block suspicious port scans and potential denial-of-service attacks.
8. Activate or deactivate the **Enable DMZ Server** toggle to expose a specified internal device to the internet while keeping other devices protected behind the firewall. If you activate this setting, enter the IP address of the device.
9. Activate or deactivate the Enable Respond to Ping on Internet Port toggle to control whether the router responds to ping (ICMP) requests on the internet port.

10. Activate or deactivate the **SIP ALG** toggle to enable or disable SIP (Session Initiation Protocol) Application Layer Gateway processing for VoIP traffic.
11. In the **TCP Session Timeout** field, enter the duration in seconds that an inactive TCP session remains open before the router automatically terminates it.
12. In the **UDP Session Timeout** field, enter the duration in seconds that an inactive UDP session remains open before the router automatically terminates it.
13. In the **ICMP Session Timeout** field, enter the duration in seconds that an inactive ICMP session remains open before the router automatically terminates it.
14. In the **Maximum Concurrent Connections** field, enter the maximum number of simultaneous connections that the router allows before it starts dropping new connection requests.
15. In the **Current Connections** field, view the current number of active connections on the router.
16. In the **VPN Passthrough** section, activate or deactivate the following toggles:
    - **Enable IPSec Passthrough**: Allow IPSec VPN traffic to pass through the router so devices on the local network can establish IPSec VPN connections through this router:
    - **Enable PPTP Passthrough**: Allow PPTP VPN traffic to pass through the router so devices on the local network can establish PPTP VPN connections through this router.
    - **Enable L2TP Passthrough**: Allow L2TP VPN traffic to pass through the router so devices on the local network can establish L2TP VPN connections through this router.
17. Click the **Save** button.
    Your settings are saved.

Manage Firewall Schedules

You can centrally configure firewall schedules that control when specific devices or applications can access the internet or your network on Insight-managed routers. By defining time-based rules in Insight, you can automatically allow or block traffic during certain hours or days, helping you enforce usage policies and improve network security consistently across all your supported locations.

Add a firewall schedule

You can add firewall schedules to control when specific traffic rules are active on your network.

To add a firewall schedule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.

2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Security > Firewall > Schedule > Add**.
   The Add Schedule pop-up page displays.
6. In the **Rule Name** field, enter a name for the schedule.
7. Enter a start time and end time for the schedule.
8. From the **Select Days** menu, select the frequency of the schedule from the following options:
   - **Daily**
   - **Weekdays**
   - **Weekends**
   - **Custom**: If you select this option, you can then select the specific days of the week
9. Click the **Add** button.
   Your settings are saved.

Edit a firewall schedule

You can edit the configuration of an existing firewall schedule at any time.

To edit a firewall schedule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Security > Firewall > Schedule**.
   The Edit Schedule pop-up page displays.
6. Select the check box of the schedule you want to edit, and click the **Edit** button.
   The Edit Schedule pop-up window displays.

7. Edit the configuration, and click the **Update** button.
   Your settings are saved.

Delete a firewall schedule

You can permanently remove a firewall schedule from an Insight-managed router, for example to retire an outdated policy, eliminate conflicting rules, or simplify the firewall configuration in the Locations dashboard.

To delete a firewall schedule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Security > Firewall > Schedule**.
6. Select the check box of the schedule you want to delete, and click the **Delete** button.
   The Delete Confirmation pop-up window displays.
7. Click the **Yes** button.
   Your settings are saved.

Manage Outbound NAT rules

Configure outbound NAT rules on an Insight-managed router to control how internal devices access external networks, enforce consistent IP addressing, and apply security or traffic policies. Properly defined rules help ensure reliable connectivity, protect internal resources, and support advanced use cases such as port forwarding, VPNs, and segmented network traffic.

Add an outbound NAT rule

Create an outbound NAT rule to translate internal IP addresses to a public IP address for outbound traffic from a selected location.

To add an outbound NAT rule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Security > Firewall > Outbound NAT > Add**.
   The Add Outbound NAT pop-up page displays.
6. In the **Rule Name** field, add a name for the rule.
7. Activate the **Enable Rule Status** toggle to display the status of the rule.
8. Configure the following settings:
   - **Service**: Select the predefined application or service (for example, HTTP or HTTPS) that this rule applies to.
   - **Source Address**: Select whether the source is Any, or a single IPv4 address.
   - **Destination Address**: Select whether the destination is Any, or a single IPv4 address.
   - **Outbound Interface**: Select the WAN interface through which translated outbound traffic is sent.
   - **Translate Source IP Address**: Select the IPv4 address that replaces the original source address in outbound traffic.
9. Click the **Add** button.
   Your settings are saved.

Edit an outbound NAT rule

You can modify an existing outbound NAT rule for a selected router location.

To edit an outbound NAT rule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.

4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Security > Firewall > Outbound NAT**.
6. Select the check box of the rule you want to edit, and click the **Edit** button.
   The Edit Outbound NAT page displays.
7. Edit the configuration of the outbound NAT rule, and click the **Update** button.
   Your settings are saved.

Delete an outbound NAT rule

You can delete an existing outbound NAT rule from a router in your network.

To delete an outbound NAT rule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Security > Firewall > Outbound NAT**.
6. Select the check box of the rule you want to delete, and click the **Delete** button.
   The Delete Confirmation pop-up window displays.
7. Click the **Yes** button.
   Your settings are saved.

Manage Traffic Rules

In Insight-managed routers, traffic rules are configured and enforced centrally through the Insight cloud, ensuring that policies are consistently applied across all managed sites and devices, and that changes can be deployed quickly without requiring direct access to each router. Traffic rules allow you to define how network traffic is prioritized, filtered, or redirected so that critical applications receive the bandwidth and protection they need. By creating specific rules based on criteria such as source or destination IP address, port, protocol, or application type, you can control which devices or services receive higher priority and which are limited or blocked. Traffic rules can also be used to enforce security policies, prevent unauthorized access, and optimize

overall network performance by reducing congestion and ensuring that essential services remain available.

Add a traffic rule

You can create a traffic rule on an Insight-managed router so you can control how specific types of network traffic are handled.

To add a traffic rule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Security > Firewall > Traffic Rules > Add**.
   The Add Traffic Rule pop-up page displays.
6. In the **Rule Name** field, enter a name for the traffic rule.
7. Activate the **Enable Rule Status** toggle to display the status of the rule.
8. Configure the following settings:
   - **Service**: Select the predefined application or service (for example, HTTP or HTTPS) that this traffic rule applies to.
   - **Source Interface**: Select the interface from which the traffic originates (for example, LAN, WAN, or a specific VLAN) that this rule will apply to.
   - **Source Address**: Select whether the source is a single IP address, a subnet, or an IP range.
   - **Destination Interface**: Select the interface where the traffic is going (for example, LAN, WAN, or a specific VLAN) that this rule will apply to.
   - **Destination Address**: Select whether the destination is a single IP address, a subnet, or an IP range.
   - **Schedule Name**: Select the schedule that defines when this traffic rule is active.
9. Click the **Add** button.
   Your settings are saved.

Edit a traffic rule

You can edit a traffic rule on an Insight-managed router at any time.

To edit a traffic rule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Security > Firewall > Traffic Rules**.
   The Traffic Rules tab displays.
6. Select the check box of the traffic rule you want to edit, and click the **Edit** button.
   The Edit Traffic Rule pop-up page displays.
7. Edit the traffic rule configuration, and click the **Update** button.
   Your settings are saved.

Delete a traffic rule

You can remove an existing firewall traffic rule from an Insight-managed router when it is no longer needed or when you want to simplify or update your network security configuration.

To delete a traffic rule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Security > Firewall > Traffic Rules**.
   The Traffic Rules tab displays.

6. Select the check box of the traffic rule you want to delete, and click the **Delete** button.
   The Delete Confirmation pop-up window displays.
7. Click the **Yes** button.
   Your settings are saved.

Manage Port Triggering Rules

Use port triggering rules to dynamically open specific ports on your router when an application or service needs them, and close the ports automatically when they are no longer in use. When your router is Insight-managed, you configure and monitor these port triggering rules, ensuring that remote access and application traffic remain secure and centrally managed.

Add a port triggering rule

You can create a port triggering rule to dynamically open specific inbound ports only when an application on your network needs them.

To add a port triggering rule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Security > Port Triggering > Add**.
   The Add Port Triggering pop-up page displays.
6. In the **Name** field, add a name for the rule.
7. Configure the following settings:
   - **External IP Address**: Select the external IPv4 address used for the port triggering rule.
   - **Internal IP Address**: Select the internal IPv4 address used for the port triggering rule.
   - **Trigger Service**: Select the predefined application or service whose outbound traffic triggers the opening of the specified inbound ports.
   - **Incoming Service**: Select the predefined application or service whose inbound traffic is allowed when the trigger condition is met.
8. Activate the **Status** toggle to display status of the rule.

9. Click the **Add** button.
   Your settings are saved.

Edit a port triggering rule

You can modify the configuration of an existing port triggering rule for an Insight-managed router at any time.

To edit a port triggering rule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Security > Port Triggering**.
6. Select the check box of the rule you want to edit, and click the **Edit** button.
   The Edit Port Triggering pop-up page displays.
7. Edit the configuration of the port triggering rule, and click the **Update** button.
   Your settings are saved.

Delete a port triggering rule

You can remove an existing port triggering rule from an Insight-managed router.

To delete a port triggering rule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.

5. Click **Settings > Security > Port Triggering**.
6. Select the check box of the rule you want to delete, and click the **Delete** button.
   The Delete Confirmation pop-up window displays.
7. Click the **Yes** button.
   Your settings are saved.

Manage Port Forwarding Rules

You can configure and manage port forwarding rules on your Insight-managed router to allow secure, controlled access to internal services, such as web servers or remote desktops from the internet. By mapping specific external ports to devices on your local network, you can support remote access and specialized applications while maintaining centralized management and visibility through Insight.

Add a port forwarding rule

Add a port forwarding rule to your Insight-managed router to allow external network traffic to reach specific devices or services on your internal network.

To add a port forwarding rule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Security > Port Forwarding > Add**.
   The Add Port Forwarding pop-up page displays.

6. Configure the following settings:
   - **External Services**: Select a predefined service to automatically apply its standard ports and protocols for port forwarding.
   - **External IP Address**: Select the external IPv4 address used for the port triggering rule.
   - **Internal Services**: Select the internal application or service to which the forwarded traffic is directed.
   - **Internal IP Address**: Select the internal IPv4 address used for the port triggering rule.
   - **WAN Interface**: Select the internet (WAN) connection on which this port forwarding rule will apply. This determines which external interface accepts incoming traffic and forwards it to your internal device.
7. Activate the **Status** toggle to display the status of the rule.
8. Click the **Add** button.
   Your settings are saved.

Edit a port forwarding rule

You can modify the configuration of an existing port forwarding rule for an Insight-managed router at any time.

To edit a port forwarding rule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Security > Port Forwarding**.
6. Select the check box of the rule you want to edit, and click the **Edit** button.
   The Edit Port Forwarding pop-up page displays.
7. Edit the configuration of the port forwarding rule, and click the **Update** button.
   Your settings are saved.

Delete a port forwarding rule

You can remove an existing port forwarding rule from an Insight-managed router.

To delete a port forwarding rule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Security > Port Forwarding**.
6. Select the check box of the rule you want to delete, and click the **Delete** button.
   The Delete Confirmation pop-up window displays.
7. Click the **Yes** button.
   Your settings are saved.

Service Management

Service Management is a centralized way to define, organize, and control the services that run on your network devices. It provides tools to add new services, edit existing ones, and delete services that are no longer needed, helping you keep your network configuration consistent, secure, and aligned with your operational requirements.

Add a service

Your router supports many services without the need to add a custom service.

If the service that you want to set up is not included in the default settings, you can set up custom service through Insight.

To configure a service for an Insight-managed router:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.

3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Setting > Security > Service Management > Add**.
   The Add Service pop-up page displays.
6. In the **Name** field, enter a name for the service.
7. From the **Protocol** menu, select a protocol for the service.
8. Enter the number for the start and end port that the service applies to.
9. Click the **Add** button.
   Your settings are saved and the service displays in the table.

Edit a service

To edit a service on an Insight-managed router:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Security > Service Management**.
6. From the table, select the check box of a service, and click the **Edit** button.
   The Edit Service pop-up page displays.
7. Edit the service configuration, and click the **Update** button.
   Your settings are saved.

Delete a service

You can delete a configured service from an Insight-managed router when it is no longer needed or must be removed for security or policy changes.

To delete a service:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Security > Service Management**.
6. From the table, select the check box of a service, and click the **Delete** button.
   The Delete Confirmation pop-up window displays.
7. Click the **Yes** button.
   Your settings are saved.

Manage Dynamic DNS Profiles

Dynamic DNS (DDNS) profiles let you associate an Insight-managed router that uses a varying WAN (public) IPv4 address with a fully qualified domain name (FQDN), also referred to as an Internet domain name.

If your network has a static (fixed) IP address, you can register an FQDN that public Domain Name Servers (DNS) link directly to your IP address. However, if your Internet account uses a dynamically assigned WAN IP address, you do not know in advance what the router's IP address will be, and the address can change frequently. In this case, you configure a DDNS profile so the router can still be reached by its FQDN.

A DDNS profile uses a DDNS service provider to register an FQDN, allow the router to be located on the Internet through its FQDN, and map DNS requests for that FQDN to the router's frequently changing IP address. The router updates the DDNS service through the profile whenever the WAN IP address changes so that services running on the router's network remain accessible from the Internet.

The router lets you create multiple DDNS profiles by selecting from supported DDNS service providers, but you must set up an account with one or more of these providers before you can configure and use the profiles.

Add a dynamic DNS profile

You can configure a dynamic DNS service so remote users can reliably access your network using a domain name instead of an IP address.

To add a dynamic DNS service:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Security > Dynamic DNS > Add**.
   The Add Dynamic DNS pop-up page displays.
6. Activate the **Dynamic DNS** toggle.
7. In the **Policy name** field, enter a name for the DDNS service.
8. From the **Service provider** menu, select the dynamic DNS provider you want to use.
9. In the **Domain name** field, enter the fully qualified domain name (FQDN) that remote users use to access your network.
10. In the **Username** field, enter the user name for your dynamic DNS account.
11. In the **Password** field, enter the password for your dynamic DNS account.
12. In the **Periodical update** field, enter how often the router should send updates to the dynamic DNS provider.
13. Click the **Add** button.
    Your settings are saved.

Edit a dynamic DNS profile

You can update the configuration of an existing dynamic DNS profile for an Insight-managed router at any time.

To edit a dynamic DNS profile:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Security > Dynamic DNS**.
6. Select the check box of a dynamic DNS service, and click the **Edit** button.
7. Edit the dynamic DNS profile configuration, and click the **Save** button.
   Your settings are saved.

Delete a dynamic DNS profile

If you no longer need a dynamic DNS profile, you can remove it.

To delete a dynamic DNS profile:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Click **Settings > Security > Dynamic DNS**.
6. Select the check box of a dynamic DNS profile, and click the **Delete** button.
   The Delete Dynamic DNS pop-up window displays.
7. Click the **Delete** button.
   Your settings are saved.

Manage individual switches

You can use the Insight Cloud Portal to view, configure, and monitor individual Insight-managed switches in your network so that you can quickly identify issues, apply changes, and keep each device operating as expected.

This chapter contains the following sections:

- Configure port settings
- Configure IP settings
- Configure port mirroring
- Configure static routes
- View connected clients

Configure port settings

You can use the Insight Cloud Portal to configure switch ports through the following options:

- You can configure an individual port.
- You can configure a batch of ports on the same switch.
- You can configure a group of ports on different switches in the same network.

NOTE: In this section, each switch port feature is described in a separate subsection. However, if you are familiar with these features, which are common to many switches, you can simultaneously configure multiple features on multiple ports.

Activate or deactivate one or more ports

You can activate or deactivate a single port or multiple ports at the same time using the Single Port or Batch Port configuration type in the Insight Cloud Portal.

To activate or deactivate one or more ports:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the switch.
   The switch dashboard displays.
5. Click the **Configuration > Port Configuration**.

6. In **Configuration Type** section, select either the **Single Port** or **Batch Port** radio button.
7. Click the port or ports you want to activate or deactivate on the switch diagram.
8. If you select the Single Port radio button, enter a port description.
9. **(Optional)** Click the **Copy Port Configuration** button to apply the individual port configuration to another port on the switch.
10. Click the **Enable Port** toggle so that it is positioned to the right.
11. Click the **Always Active Port** toggle so that it is positioned to the right.
    If activated, the Always Active Port setting keeps sending the notification if the connection or port is down.
12. Click the **Save** button.
    Your settings are saved.

Set the port configuration mode for switch ports

The port configuration mode determines the authorization state of a switch port. This setting is typically used in conjunction with 802.1X network access control.

To set the port configuration mode for switch ports:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the switch.
   The switch dashboard displays.
5. Click the **Configuration > Port Configuration**.
   The Port Configuration page displays.
6. In **Configuration Type** section, select either the **Single Port** or **Batch Port** radio button.
7. On the switch graphic, select the port or ports for which you want to set the configuration mode.

8. From the **Port Configuration Mode** menu, select one of the following options:
   - **Auto**: The port is authorized or unauthorized based on the result of 802.1X authentication.
   - **Authorized**: The port is always authorized and does not require authentication.
   - **Unauthorized**: The port is always unauthorized and denies access to all connected devices.
9. Click the **Save** button.
   Your settings are saved.

Configure custom PoE settings for one or more ports

Configure custom PoE settings for one or more ports to control the power each port delivers to connected devices. You can set the PoE standard, detection type, priority, power limit, and schedule for each port.

To manage custom PoE settings for one or more ports:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the switch.
   The switch dashboard displays.
5. Click the **Configuration > Port Configuration**.
6. Select the **Power Management (PoE Port Only)** tab.
7. Activate the **Enable PoE** toggle so that it is positioned to the right.
   The port displays the current power usage in watts.
8. From the **PoE Standard** menu, select the PoE standard for the port or ports.
9. From the **Detection Type** menu, select the detection type for the port or ports.
10. From the **Priority** menu, select the priority level for the port or ports.
11. From the **Power Limit Type** menu, select how the power limit is determined for the port or ports.

12. In the **Power Limit (Watts)** field, enter the maximum power available to the selected PoE port or ports.
    Enter a value between 3 W (minimum) and 30 W (maximum).
13. From the **PoE Schedule** menu, select a schedule to control when the port or ports supply PoE power, or select **None** to supply power at all times.
14. Click the **Apply** button.
    Your settings are saved.

Configure power-cycling on PoE ports

Power-cycling a PoE port temporarily cuts power to the connected device and then restores it. You can power-cycle one or more PoE ports on an Insight-managed switch from the Insight Cloud Portal, which can help resolve connectivity issues with PoE-powered devices.

To power-cycle PoE ports:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the switch.
   The switch dashboard displays.
5. Click the **Operations > PoE Management**.
   The PoE Management page displays.
6. On the switch graphic, select the ports you want to power cycle, and click the **Start Power Cycle** button.
7. **(Optional)** Click the **PoE Downtime Schedules** link to set up a schedule that automatically power-cycles PoE ports at specified times.

Set the storm rate limit for incoming traffic for one or more ports

You can configure the storm rate on an Insight-managed switch to control the maximum available bandwidth for incoming broadcast, multicast, and unknown unicast packets. A broadcast storm occurs when an excessive number of broadcast packets transmit simultaneously across a network through a single port, which can overload network resources. When incoming traffic exceeds the storm rate that you specify, the switch discards the excess packets.

To set the storm rate limit for incoming traffic for one or more ports:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the switch.
   The switch dashboard displays.
5. Click the **Configuration > Port Configuration > Rate Limiting**.
   The Rate Limiting tab displays.
6. In the **Egress Rate (%)** field, enter the maximum percentage of the port bandwidth that you want to allow for outgoing traffic.
7. Click the **Storm Control Rate (%)** toggle so that it is positioned to the right, and enter the maximum percentage of the port bandwidth that you want to allow for incoming broadcast, multicast, and unknown unicast traffic.
8. Click the **Save** button.
   Your settings are saved.

Configure the duplex mode for switch ports

The duplex mode determines how a switch port sends and receives data. In full-duplex mode, the port can send and receive data simultaneously. In half-duplex mode, the port can only send or receive data at one time. Using the Insight Cloud Portal, you can set the duplex mode for individual ports on a managed switch.

To configure the duplex mode for switch ports:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.

2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the switch.
   The switch dashboard displays.
5. Click the **Configuration > Port Configuration**.
   The Port Configuration page displays.
6. From the **Duplex Mode** menu, select one of the following options:
   - **Auto**: The duplex mode is set by the autonegotiation process. This is the default setting.
   - **Full**: The port transmits between the devices in both directions simultaneously.
   - **Half**: The port transmits between the devices in only one direction at a time.
7. Click the **Save** button.
   Your settings are saved.

Configure port VLAN IDs for switch ports

By default, all switch ports are members of VLAN 1 and are assigned a port VLAN ID (PVID) of 1. If you set up other VLANs, you can assign a different PVID to a port.

To configure port VLAN IDs for switch ports:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the switch.
   The switch dashboard displays.
5. Click the **Configuration > Port Configuration**.
   The Port Configuration page displays.
6. From the **Default VLAN (PVID)** menu, select a default VLAN.
   By default, the Management VLAN is selected. By default, the management VLAN with ID 1 is assigned to a port, so the PVID for the port is 1. Selecting a VLAN changes the PVID.

7. In the **VLAN ID** section, select the check box for each VLAN you want to assign to the port or ports.
8. **(Optional)** To restore the default VLAN assignments, click the **Restore Default** button.
9. **(Optional)** To select all VLANs, click the **Select All** button.
10. For each selected VLAN, click **Tag** or **Untag** to set the frame tagging mode.
11. **(Optional)** To clear your selections, click the **Clear** button.
12. Click the **Save** button.
    Your settings are saved.

Configure the port speed for switch ports

By default, the speed for all ports is set to Auto, allowing the ports to detect the speed of the connection between the port and the attached device. You can also manually set the port speed to 10, 100, 1000 Mbps, or 10 Gbps depending on what speeds are supported on your switch.

To configure the port speed for switch ports:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the switch.
   The switch dashboard displays.
5. Click the **Configuration > Port Configuration**.
   The Port Configuration page displays.

6. From the **Port Speed** menu, select one of the following options:
   - **Auto**: Automatically detects and sets the port speed based on the connected device.
   - **10 Mbps**: Sets the port speed to 10 Mbps for legacy or low-bandwidth devices.
   - **100 Mbps**: Sets the port speed to 100 Mbps for Fast Ethernet connections.
   - **1000 Mbps**: Sets the port speed to 1000 Mbps (1 Gbps) for Gigabit Ethernet connections.
   - **10 Gbps**: Sets the port speed to 10 Gbps for high-bandwidth connections on supported ports.

   NOTE: The above options can vary depending on what speed your switch supports.
7. Click the **Save** button.
   Your settings are saved.

Configure the maximum Ethernet frame size or jumbo frames

In the Insight Cloud Portal, you have granular control over the maximum Ethernet frame size that your Insight-managed switches send and receive. The maximum frame size is adjustable from 1500 bytes to 9198 bytes. Anything larger than 1518 bytes (or 1522 bytes with 802.1Q VLAN tagging) is considered a jumbo frame.

To configure the maximum Ethernet frame size or jumbo frames:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the switch.
   The switch dashboard displays.
5. Click the **Configuration > Port Configuration**.
   The Port Configuration page displays.
6. In **Configuration Type** section, select either the **Single Port** or **Batch Port** radio button.
7. Select the ports that you want to configure.

8. In the **Max Frame Size** field, enter the desired maximum frame size in bytes or select a predefined jumbo frame size.
   The maximum frame size range is between 1500 bytes and 9198 bytes. By default, the maximum frame size is 1522.
9. Click the **Save** button.
   Your settings are saved.

Configure IP settings

Configure the IP address settings for an individual switch, either by activating automatic IP assignment or by entering a static IP address manually.

To configure IP settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the switch.
   The switch dashboard displays.
5. Click **Configuration > IP Settings**.
6. Choose how you want to assign the IP address.

Automatic (DHCP)	Activate the **Assign IP Address Automatically** toggle so it is positioned to the right. The remaining fields become read-only and the switch obtains its address from your DHCP server.
Static IP address	Deactivate the **Assign IP Address Automatically** toggle, so it is positioned to the left, to make the address fields editable, then continue with the steps below.

7. If you selected a static IP address, enter the network details in the following fields.
   a. In the **IP Address** field, enter the static IPv4 address to assign to the switch.
   b. In the **Subnet Mask** field, enter the subnet mask for your network.
   c. In the **Gateway Address** field, enter the IPv4 address of your default gateway.
   d. In the **Primary DNS Server** field, enter the IPv4 address of your primary DNS server.
   e. In the **Secondary DNS Server** field, enter the IPv4 address of a secondary DNS server.

8. Click the **Save** button.
   Your settings are saved.

Configure port mirroring

Port mirroring sends a copy of incoming or outgoing network traffic on a source port to a predefined destination port. You can select multiple switch ports as source ports but only a single switch port as the destination port.

A packet that is mirrored to the destination port is in the same format as the original packet on the wire. If the mirror is copying a received packet, the copied packet is VLAN-tagged or untagged as it was received on the source port. If the mirror is copying a transmitted packet, the copied packet is VLAN-tagged or untagged as it is being transmitted on the source port.

To configure port mirroring:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the switch.
   The switch dashboard displays.
5. Click the **Configuration > Port Mirroring**.
   The Port Mirroring page displays.
6. Activate the **Port Mirroring** toggle so that it is positioned to the right.
7. In the **Select your source port** section, click one or more ports on the switch graphic to mirror traffic from.
8. In the **Select your destination port** section, click the port to which mirrored traffic is sent.
9. Click the **Save** button.
   Your settings are saved.

Configure static routes

A static route is a manually configured path that tells the Insight-managed switch how to forward traffic to a destination network through a specific next-hop IP address. Static routes are used in multi-VLAN or multi-segment Insight networks where the switch must forward traffic to networks that are not directly connected. Unlike dynamic routing, static routes do not change automatically and must be configured manually in the Insight Cloud Portal. You can also configure a default route, which forwards all unmatched traffic to a specified gateway.

To configure static routes:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the switch.
   The switch dashboard displays.
5. Click **Configuration > Static Route**.
   The Static Route page displays.
6. Click the **Add** button.
   The Static Route pop-up page displays.
7. If you want to configure a default route, click the **Default Route** toggle so that it is positioned to the right.
   When activated, the switch forwards all traffic with no matching route to the next-hop IP address you specify.
8. In the **IP Address** field, enter the destination network IP address.
9. In the **IP Subnet Mask** field, enter the subnet mask for the destination network.
10. In the **Nexthop IP Address** field, enter the IP address of the gateway through which the destination network is reachable.
11. Click the **Save** button.
    The static route is added to the table on the Static Route page. The switch forwards traffic to the specified destination network through the configured next-hop IP address.

Next steps

To delete a static route, select the check box next to the route on the Static Route page and click

the **Delete** button.

View connected clients

In the Insight Cloud Portal, you can view the clients connected to the ports of an Insight-managed switch. A graphic of the switch shows which ports have connected clients, and you can view detailed information about each client, including the port it is connected to.

To view connected clients:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the switch.
   The switch dashboard displays.
5. Click the **Connected Clients** tab.
   A graphic of the switch displays highlighting the port or ports that have connected clients.
6. To view the details of the connected clients, in the table, click the port number.
   The Port Summary page displays with the connected client details.

Manage individual access points

You can configure and monitor individual access points independently to meet specific coverage, performance, or security requirements for a location.

This chapter contains the following sections:

- Configure IP settings
- Configure WiFi radio bands, channels, and output power
- Configure LED settings
- Configure hardware assisted datapath
- Configure the PoE input settings
- Configure mesh mode
- Display the device topology
- Locate an access point using the power LED
- View connected clients

Configure IP settings

Configure the IP address settings for an individual access point, either by activating automatic IP assignment or by entering a static IP address manually.

To configure IP settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the access point.
   The access point dashboard displays.
5. Click **Configuration > IP Settings**.

6. Choose how you want to assign the IP address.

Automatic (DHCP)	Activate the **Assign IP Address Automatically** toggle so it is positioned to the right. The remaining fields become read-only and the access point obtains its address from your DHCP server.
Static IP address	Deactivate the **Assign IP Address Automatically** toggle, so it is positioned to the left, to make the address fields editable, then continue with the steps below.

7. If you selected a static IP address, enter the network details in the following fields.
   a. In the **IP Address** field, enter the static IPv4 address to assign to the access point.
   b. In the **Subnet Mask** field, enter the subnet mask for your network.
   c. In the **Gateway Address** field, enter the IPv4 address of your default gateway.
   d. In the **Primary DNS Server** field, enter the IPv4 address of your primary DNS server.
   e. In the **Secondary DNS Server** field, enter the IPv4 address of a secondary DNS server.
8. Click the **Save** button.
   Your settings are saved.

Configure WiFi radio bands, channels, and output power

Each access point has independent radio settings for each WiFi band that an access point supports. Adjusting these settings lets you optimize coverage, reduce interference, and control transmit power for a specific location. By default, all supported radio bands on the access point are activated and channels are selected automatically.

To configure WiFi radio bands and channels:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the access point.
   The access point dashboard displays.
5. Click **Configuration > Radio & Channels**.

6. For each radio band you want to configure, activate the **Enable Radio** toggle so it is positioned to the right.
   When the toggle is activated, the radio is active and the remaining fields in that section are available. When the toggle is deactivated, clients cannot connect on that band.
7. From the **Radio Mode** menu, select the WiFi standard to use for that band.
8. From the **Channel** menu, select a channel.

Auto	The access point selects the least congested channel automatically. This is the recommended setting for most environments.
Fixed channel	The access point uses the channel you specify. Use a fixed channel when your network plan requires channel separation across multiple access points.

9. From the **Channel Width** menu, select the channel width.
   Wider channels provide higher throughput but are more susceptible to interference. Narrower channels offer greater range and stability in dense environments. Common values are **20 MHz** for 2.4 GHz, **40 MHz** or higher for 5 GHz, and **80 MHz** or higher for 6 GHz.
10. From the **Output Power** menu, select the transmit power level.
    Select **Auto(Full)** to let the access point use the maximum permitted power for your region. To reduce coverage area or limit interference between adjacent access points, select a lower fixed power level.
11. Repeat the above steps for the other radio band if necessary.
12. Click **Save**.
    Your settings are saved.

Configure LED settings

The LED settings control how the status LEDs on the access point behave during normal operation.

To configure LED settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.

4. Select the **Devices** tab, and click the name of the access point.
   The access point dashboard displays.
5. Click **Configuration > LED Settings**.
6. From the **LED Settings** menu, select the LED mode.
   The available options vary by access point model. Common options include the following:
   - **Default**: LEDs operate normally, reflecting the current device status.
   - **Off**: All LEDs are deactivated. The access point continues to operate normally but no indicator lights are visible.
   - **Off except Power LED**: All LEDs are deactivated, except for the Power LED.
7. Click **Save**.
   The LED setting is applied to the access point. Changes take effect immediately and do not interrupt WiFi service or connected clients.

Configure hardware assisted datapath

The hardware assisted datapath, which is enabled by default, increases the maximum throughput on the AP. We recommend that you leave the hardware assisted datapath enabled unless you need to enable inter-VLAN routing or support jumbo frames in the network. Disabling the hardware assisted datapath reduces the maximum throughput, but the throughput is still high enough for an enterprise deployment.

NOTE: This option is only available for WiFi 7 access points.

To configure hardware assisted datapath:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the access point.
   The access point dashboard displays.
5. Click **Configuration > Hardware Assisted Datapath**.

6. In the **Hardware Assisted Datapath** section, activate the **Hardware Assisted Datapath** toggle so that it is positioned to the right.
   When the toggle is activated, the access point uses hardware acceleration for packet processing. When the toggle is deactivated, packet processing is handled by the CPU. By default, this setting is activated.

Configure the PoE input settings

NOTE: This feature applies only to access points that support PoE+ and PoE++.

The power setting controls which PoE standard the access point negotiates with the connected switch. Selecting the correct standard ensures the access point receives sufficient power to operate all its radios and features at full capacity. The available standards differ in the maximum wattage they deliver:

- **802.3af** - delivers up to 15.4 W. Suitable for basic single-radio access points.
- **802.3at** (PoE+) - delivers up to 30 W. Suitable for dual-radio access points and models with additional features such as USB ports.
- **802.3bt** (PoE++) - delivers up to 60 W or 90 W depending on type. Required for high-performance tri-radio or WiFi 6E access points.

CAUTION: Selecting a PoE standard that exceeds the capability of your switch port may cause the access point to power cycle or operate in a reduced-power mode. Verify the PoE budget of the connected switch port before changing this setting.

To configure the PoE input settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the access point.
   The access point dashboard displays.
5. Click **Configuration > Power Settings**.

6. From the **Power Settings** menu, select the PoE standard.

Automatic	The access point negotiates the highest PoE standard supported by both the device and the connected switch port. This is the recommended setting for most deployments.
802.3af	Forces the access point to operate within the 802.3af power budget (up to 15.4 W). Use this setting only if your switch port does not support a higher PoE standard.
802.3at	Forces the access point to operate within the 802.3at (PoE+) power budget (up to 30 W). Use when the connected switch port supports 802.3at but not 802.3bt.
802.3bt	Forces the access point to operate within the 802.3bt (PoE++) power budget (up to 90 W). Use for high-performance access points that require maximum power to enable all features.

7. Click the **Save** button.
   Your settings are saved.

Configure mesh mode

Insight Instant Mesh lets you create a wireless mesh network in which access points relay traffic to one another without requiring a wired connection to every device. Each access point in the mesh operates in one of the following roles:

- **Root**: The access point has a wired connection to the network and serves as the entry point for mesh traffic.
- **Extender**: The access point connects wirelessly to a root access point and extends coverage to areas where running a cable is not practical.

You can set the mesh mode at the individual access point level, or leave it on Auto to let Insight assign the role automatically based on the device's wired connectivity.

NOTE: Mesh mode can also be configured at the network location level (see Configure mesh settings for access points in a location on page 101). The individual device setting takes precedence over the location-level setting for that access point only.

To configure mesh mode:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.

3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the access point.
   The access point dashboard displays.
5. Click **Mesh > Mesh Settings**.
6. In the **Mesh Mode** section, select one of the following radio buttons:

Auto	Insight automatically configures the access point as a root or extender based on whether it detects a wired uplink. This is the recommended setting for most deployments.
Extender Only	Forces the access point to operate as a wireless extender and connect to the mesh through a root access point. Use this setting for access points that do not have a wired uplink. **NOTE:** If you plan to set the access point to Extender Only mode, at least one root access point must be active and reachable in the same network location
Root Only	Forces the access point to operate as a root access point regardless of its connectivity state. Use this setting when you want to ensure a specific device always anchors the mesh, even if its wired connection is temporarily unavailable.
Disable Mesh	Turns off mesh functionality for this access point. The device operates as a standalone access point and does not participate in the mesh network. No further configuration is necessary. This is the default setting.

7. In the **Backhaul Settings** section, select a backhaul radio mode.

Shared	The backhaul radio is shared between mesh backhaul traffic and wireless client connections.
Dedicated	The backhaul radio is reserved exclusively for mesh backhaul traffic. Wireless clients cannot connect on this radio.

8. In the **Root AP Selection** section, select a backhaul radio mode.

> **NOTE:** This setting is configurable for Auto or Extender modes only.

Auto	The extender access point selects a root access point automatically based on signal strength and backhaul radio availability.
Manual	Specifies a preferred root access point for this extender to connect to by default. If the preferred root access point is unavailable, the extender falls back to automatic selection.

9. Click the **Save** button.
   Your settings are saved.

> **NOTE:** To verify the access point's active mesh role and connection path after saving, return to the **Mesh** tab and review the mesh topology diagram.

Next steps

If the access point does not connect to the mesh after changing its mode, see How do I troubleshoot my Insight Instant Mesh network?

Display the device topology

The Mesh Topology tab on the access point dashboard provides a visual map of the mesh network, showing how access points are connected to each other as root and satellite nodes. You can use this view to assess the health of backhaul connections between Insight-managed access points in your mesh network.

To access this tab, select the AP in the Devices table, and click **Mesh > Mesh Topology**.

Connection quality between mesh access points is indicated by line color:

- **Excellent:** Bright green line
- **Good:** Dark green line
- **Fair:** Orange line
- **Poor:** Red line

Use the Mesh AP List tab to view a list of access points in the mesh network, and the Mesh Settings tab to configure mesh options. For more information, see Configure mesh mode on page 220.

Locate an access point using the power LED

You can use the Device Locate feature to make the power LED on an Insight-managed device blink blue, helping you physically identify it in your rack.

To locate an access point using the power LED:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the access point.
   The access point dashboard displays.
5. Select the **Locate Device** button.
   The power LED on the device begins blinking blue. A countdown timer displays how long the locate session remains active. This timer typically displays for five minutes.
6. Physically identify the device in your rack by looking for the blinking blue LED.
7. **(Optional)** Click **Stop** to end the locate session before the timer expires.

View connected clients

To display the clients that are connected to the AP over WiFi:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the access point.
   The access point dashboard displays.
5. Click **Clients > Wireless Clients**.
   The connected wireless clients display.
6. From the filter menu, select one of the following options:
   - **All**
   - **Connected**
   - **Disconnected**

Manage individual mobile hotspots

Use NETGEAR Insight to monitor and configure individual mobile hotspots in your network. You can manage mobile profiles, configure a geofence, set up port forwarding rules, and manage access control to determine which devices can connect through the hotspot.

Manage mobile profiles

Mobile Profiles are policy templates that control the behavior and network access of mobile hotspot devices, which often operate outside traditional network boundaries and require consistent, remote management. Because mobile hotspots connect over cellular networks and are frequently used in the field or on the move, mobile profiles ensure that security, usage policies, and connectivity settings remain enforced regardless of location. These profiles include configurations for WiFi settings, cellular data usage, schedule-based access rules, port filtering, domain restrictions, and device control, allowing you to maintain visibility, optimize performance, and enforce security across all deployed mobile hotspots.

This chapter contains the following sections:

- Add a mobile profile
- Edit a mobile profile
- Clone a mobile profile
- Delete a mobile profile

Add a mobile profile

You can create a new WiFi mobile profile for a location by defining the profile name, SSID, password, and device configuration settings.

To configure a mobile profile:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select **Mobile Hotspot > Mobile Profiles**.
   The Mobile Profiles page displays.
5. Click the **Add** button.
   The Add Mobile Profile pop-up window displays.

6. In the **Name** field, enter a name for the mobile profile.
7. Deactivate the **Retain Device Default** toggle.
   By default, this setting is activated. When this setting is activated, each device you onboard to the profile keeps their factory default settings for WiFi. The page adjusts.
8. In the **SSID Name** field, enter a name for the SSID.
9. In the **Password** field, enter a password.
   The password must be between 8 and 63 characters.
10. Click the **Save** button.
    Your settings are saved.

Edit a mobile profile

You can edit the configuration of an existing mobile profile, including its general settings, network policies, device controls, and enforcement rules.

To edit a mobile profile:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select **Mobile Hotspot > Mobile Profiles**.
   The Mobile Profiles page displays.
5. Select the check box of a mobile profile, and click the **Edit** button.
   The Edit Mobile Profile pop-up window displays.

6. Configure the settings in each tab:
   - **General**: Profile name and basic settings. For more information, see Update general mobile profile configuration settings on page 226.
   - **Enforcement Settings**: Policy enforcement options and compliance rules. For more information, see Update enforcement settings for a mobile profile on page 227.
   - **Access Control**: User and device access permissions. For more information, see Update access control settings for a mobile profile on page 228.
   - **Device Control**: Device-specific policies. For more information, see Update device control settings for a mobile profile on page 229.
   - **Schedule**: Time-based access rules. For more information, see Configure a firmware update schedule for a mobile profile on page 231
   - **Port Filtering**: Restricts client access to specific TCP/UDP ports. For more information, see Update port filtering settings for a mobile profile on page 232.
   - **Manage Restriction**: Domain/URL filtering. For more information, see Update the unrestricted device list for a mobile profile on page 234.
   - **Device list**: View and manage associated devices. For more information, see View connected devices on page 235.
7. Click the **Save** button.
   Your settings are saved.

Update general mobile profile configuration settings

You can update the name, description, password, and behavior settings for a mobile profile using the Insight Cloud Portal.

To update general mobile profile configuration settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select **Mobile Hotspot > Mobile Profiles**.
   The Mobile Profiles page displays.
5. Select the check box of the mobile profile you want to edit.
6. Click the **Edit** button.
   The Edit Mobile Profile page displays.

7. Click the **General** tab.
8. In the **Mobile Profile Name** field, enter a name for the profile.
9. **(Optional)** In the **Description** field, enter a description.
10. In the **Device Admin Password** field, enter a password to assign to onboarded devices.
    By default, onboarded devices are assigned the location password. For increased security, you can assign a different password here. Click the eye icon to show or hide the password.
11. Click the **Anonymous Mode** toggle so it is positioned to the right.
    When activated, the client device name is not sent to or stored on Insight.
12. Click the **Auto Rollback** toggle so it is positioned to the right.
    When activated, device settings automatically revert if connectivity is lost.
13. Click the **Apply** button.
    The Confirm pop-up window displays advising that the new configuration is sent to all connected devices and causes those devices to restart.
14. Click the **Yes** button
    The updated general settings are applied to the mobile profile.

Update enforcement settings for a mobile profile

You can control whether settings are admin enforced or allow customization using the Insight Cloud Portal.

To update enforcement settings for a mobile profile:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select **Mobile Hotspot > Mobile Profiles**.
   The Mobile Profiles page displays.
5. Select the check box of the mobile profile you want to edit.
6. Click the **Edit** button.
   The Edit Mobile Profile page displays.
7. Click the **Enforcement Settings** tab.

8. For each setting, select either **Admin Enforced** or **Customization Allowed**.
   The following settings are available:

Category	Settings
General	Device Admin Password
WiFi	General Configuration and SSID Configuration
Cellular	Data Settings and Network Mode; Monthly Usage; Access Point Name
Router	Gateway and DHCP Server; Network Settings; DNS Servers

9. Click the **Apply** button.
   The Confirm pop-up window displays advising that the new configuration is sent to all connected devices and causes those devices to restart.
10. Click the **Yes** button.
    The enforcement settings are applied to the mobile profile.

Update access control settings for a mobile profile

You can manage which devices are permitted to connect to mobile profiles by configuring access control list in the Insight Cloud Portal.

To update access control settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select **Mobile Hotspot > Mobile Profiles**.
   The Mobile Profiles page displays.
5. Select the check box of the mobile profile you want to edit.
6. Click the **Edit** button.
   The Edit Mobile Profile page displays.
7. Click the **Access Control** tab.
8. Activate the **Enable Access Control** toggle so it is positioned to the right.
   The Alert pop-up window displays advising that this action overrides existing device-level configuration.

9. To continue, click the **Yes** button.
10. In the **Filter Type** section, select either the **Allow List** or **Block List** radio button.
    This setting filters what type of access control lists display in the table.
11. **(Optional)** To add a new access control list, do the following:
    a. Click the **Add** button.
       The Add Service in Allow List page displays.
    b. In the **Device Name** field, enter a descriptive name for the device.
    c. In the **MAC Address** field, enter the device MAC address in the format XX:XX:XX:XX:XX:XX.
    d. Click the **Save** button.
       The new access control list displays in the table.
12. To edit an existing access control list, do the following:
    a. Select the check box of an existing access control list.
    b. Click the **Edit** button.
       The Edit Service pop-up page displays.
    c. Update the existing configuration, and click the **Save** button.
13. To delete an access control list:
    a. Select the check box of an existing access control list.
    b. Click the **Delete** button.
       The Confirm Deletion pop-up window displays.
    c. Click the **Yes** button.
14. Click the **Save** button.
    Your settings are saved.

Update device control settings for a mobile profile

You can update a range of pre-configured hardware and connectivity settings for devices assigned to a mobile profile. You can control WiFi behavior, screen settings, LED indicators, data offload, Ethernet port settings, and device discovery settings.

To configure device control settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.

4. Select **Mobile Hotspot > Mobile Profiles**.
   The Mobile Profiles page displays.
5. Select the check box of a mobile profile, and click the **Edit** button.
   The Edit Mobile Profile pop-up window displays.
6. Select the **Device Control** tab.
7. In the **WiFi Control** section, configure the following settings:
   WiFi Control settings define how WiFi behaves on the device, including display visibility, range, standby behavior, and whether WiFi turns off automatically when tethering.
   a. Click the **WiFi Display** toggle to enable or disable WiFi visibility on the device.
   b. From the **WiFi Range** menu, select the WiFi range that best suits your network.
   c. From the **WiFi Standby Timer** menu, select how long the device waits before entering WiFi standby mode.
   d. Activate the **WiFi Off When Tethering** toggle to automatically turn off WiFi when the device is tethering.
8. In the **Device Setting** section, configure the following settings:
   Device Settings allow you to control the tethering mode and whether the physical factory reset button is active, helping to prevent accidental resets.
   a. From the **Tethering Mode** menu, select how the device shares its connection.
   b. Click the **Enable Factory Reset Button** toggle to activate or deactivate the physical factory reset button on the device.
9. In the **Screen Settings** section, configure the following settings:
   Screen settings allow you to control the brightness and standby timer of the device screen to manage power consumption.
   a. Drag the **Screen Brightness** slider to adjust the brightness level of the device screen.
   b. In the **Screen Standby Timer** menu, select how long the screen stays on before turning off.
10. Click the **LED Blinking** toggle to activate or deactivate the LED blinking indicator on the device.
    LED Blinking provides a visual status indicator on MR100 devices. Deactivating this setting turns off the LED indicator.
11. Click the **Data Offload** toggle to activate or deactivate data offload on the device.
    When activated, the Data Offload setting redirects MAVi traffic to wired interfaces instead of the mobile network, reducing mobile data usage.

12. In the **Ethernet Port** section, configure the following settings:
    Ethernet Port settings allow you to enable or disable the Ethernet port on the device and configure how long the Ethernet port display remains active.
    a. Click the **Enable Ethernet Port** toggle to activate or deactivate the Ethernet port on the device.
    b. In the **Ethernet Display Time** field, enter how long the Ethernet port display remains active.
13. In the **Device Discovery Settings** section, configure the following settings:
    Device Discovery Settings allow other devices on the network to automatically discover this device. UPnP enables automatic discovery, UPnP IGD allows internet gateway device functionality, and DLNA enables media sharing between compatible devices.
    a. Click the **UPnP** toggle to activate or deactivate automatic device discovery on the network.
    b. From the **UPnP IGD Version** menu, select the UPnP IGD version.
    c. Click the **DLNA** toggle to activate or deactivate DLNA media sharing between compatible devices.
14. Click the **Save** button.
    The device control settings are saved and applied to all devices assigned to the mobile profile.

Configure a firmware update schedule for a mobile profile

A mobile profile schedule defines when Insight checks for new firmware for the assigned mobile hotspot devices. If a firmware update is available at the scheduled time, the update is triggered automatically. Scheduling firmware updates allows you to control when updates occur, for example, outside business hours to minimize disruption to connected users. You can add a maximum of three time slots per day, and schedules can be configured independently for each day of the week.

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select **Mobile Hotspot > Mobile Profiles**.
   The Mobile Profiles page displays.

5. Select the check box of a mobile profile, and click the **Edit** button.
   The Edit Mobile Profile pop-up window displays.
6. Click the **Schedule** tab.
   The Schedule page displays.
7. Click the **Enable Schedule** toggle so that it is activated and positioned to the right.
8. Click the **Add** button.
   The Add New Schedule pop-up page displays.
9. In the **Schedule Name** field, enter a name for the schedule.
10. (Optional) In the **Description** field, enter a description of up to 32 characters.
11. In the **Days** section, activate the toggle for each day you want Insight to check for firmware updates.
    You can activate any combination of Monday through Sunday. When you activate a day, time slot fields display so you can specify the time at which the firmware check runs. You can add a maximum of three time slots per day.
12. For each activated day, set the time at which the firmware check occurs.
13. Click the **Save** button.
    The new schedule displays in the schedule table on the Schedule tab, showing the schedule name and its details.

    The firmware update schedule is saved to the mobile profile. At each scheduled time, Insight checks whether new firmware is available for the assigned mobile hotspot devices. If an update is found, it is applied automatically.

Next steps

To modify an existing schedule, select it in the table and click the **Edit** button. To remove a schedule, select it and click the **Delete** button. To apply the mobile profile to a device, assign it from the **Device List** tab.

Update port filtering settings for a mobile profile

Port filtering controls which network services and ports are permitted for devices connected to the mobile hotspot. When enabled, you can choose between an Allow List, which permits only specified services and blocks all other traffic, or a Block List, which blocks specified services and permits all other traffic. You can add, edit, and delete services from the list as needed.

NOTE: Port filtering is not supported on MR1100 models.

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select **Mobile Hotspot > Mobile Profiles**.
   The Mobile Profiles page displays.
5. Select the check box of a mobile profile, and click the **Edit** button.
   The Edit Mobile Profile pop-up window displays.
6. Select the **Port Filtering** tab.
7. Click the **Enable** toggle to activate port filtering.
8. Under **Filter Type**, select one of the following options:

Allow List	Only traffic on the specified services and ports is permitted. All other traffic is blocked.
Block List	Traffic on the specified services and ports is blocked. All other traffic is permitted.

9. **(Optional)** To add a service, click the **Add** button and enter the service name and port details.
10. **(Optional)** To edit a service, select the check box next to the service and click the **Edit** button.
11. **(Optional)** To delete a service, select the check box next to the service and click the **Delete** button.
12. Click the **Save** button.
    Your settings are saved.
    Port filtering is configured for the mobile profile and applied to all devices assigned to the profile.

Update the unrestricted device list for a mobile profile

The unrestricted device list allows you to grant specific devices full internet access, bypassing any website filtering rules configured in the mobile profile. This is useful when you need to allow trusted devices unrestricted access while still enforcing website filtering for all other devices connected to the hotspot. You can also configure a filter type to either allow or block specific domains for all other devices. You can add, edit, and delete devices from the list at any time in the Insight Cloud Portal.

To manage the unrestricted device list for a mobile profile:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select **Mobile Hotspot > Mobile Profiles**.
   The Mobile Profiles page displays.
5. Select the check box of a mobile profile, and click the **Edit** button.
   The Edit Mobile Profile pop-up window displays.
6. Select the **Manage Restriction** tab.
   A list of unrestricted devices display.
7. Click the **Enable Restriction** toggle to activate website filtering restrictions for the mobile profile.
8. **(Optional)** In the **Unrestricted Client** section, do the following:
   a. To add a device to the unrestricted device list, click the **Add** button.
      The Add Unrestricted Device pop-up page displays.
   b. In the **Device Name** field, enter a name for the device.
   c. In the **MAC Address** field, enter the MAC address of the device in the format AA:BB:CC:DD:EE:FF.
   d. Click the **Save** button.
      The device displays in the unrestricted client list.
9. **(Optional)** To edit device credentials of an unrestricted device, do the following:
   a. Select the check box next to the device, and click the **Edit** button.
      The Edit Unrestricted Device pop-up page displays.
   b. Update the device information, and click the **Save** button.

10. **(Optional)** To delete a device from the unrestricted device list, select the check box next to the device, and click the **Delete** button.
11. Under **Filter Type**, select one of the following options to control website access for all other connected devices:

Allow List	Only the specified domains are accessible. All other websites are blocked.
Block List	The specified domains are blocked. All other websites are accessible.

12. **(Optional)** If you select the **Allow List** radio button, to add the domains you want to block, do the following:
    a. In the **Allow Domains** section, click the **Add** button.
    The Add domain in Allowlist pop-up page displays.
    b. In the **Domain Name** field, enter a domain.
    c. Click the **Save** button.
    The domain displays in the Allow Domains table.
13. **(Optional)** If you select the **Block List** radio button, to add the domains you want to block, do the following:
    a. In the **Block Domains** section, click the **Add** button.
    The Add domain in Blocklist pop-up page displays.
    b. In the **Domain Name** field, enter a domain.
    c. Click the **Save** button.
    The domain displays in the Block Domains table.
14. Click the **Save** button.
    Your settings are saved.
    The unrestricted device list and website filtering settings are saved and applied to all devices assigned to the mobile profile. Devices on the unrestricted list can access all websites, while all other connected devices are subject to the configured filter rules.

View connected devices

You can view devices assigned to a mobile profile. Each device on the list inherits the profile's configuration including WiFi settings, cellular parameters, access controls, schedules, and port filtering rules ensuring uniform policy enforcement across the group. Adding a device enrolls it under the profile's settings while removing it detaches it from those policies. Reviewing the Device List before making changes helps you understand the scope and impact across your deployed devices.

To view connected devices:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select **Mobile Hotspot > Mobile Profiles**.
   The Mobile Profiles page displays.
5. Select the check box of the mobile profile you want to edit.
6. Click the **Edit** button.
   The Edit Mobile Profile page displays.
7. Click the **Device List** tab.
   The Device List displays.

Clone a mobile profile

You can clone an existing mobile profile to a different organization or location.

To clone a mobile profile:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select **Mobile Hotspot > Mobile Profiles**.
   The Mobile Profile page displays.
5. Select the check box of a mobile profile, and click the **Clone** button.
   The Clone Configuration pop-up window displays.
6. From the **Organization** menu, select an organization to which you want to clone the mobile profile.
7. From the **Location** menu, select a location to which you want to clone the mobile profile.

8. In the **Name of the Clone** field, enter a name for the cloned profile.
   The name can be a maximum of 50 characters.
9. Click the **Clone** button.
   Your settings are saved.

Delete a mobile profile

You can delete an existing mobile profile from a location that is no longer required.

To delete a mobile profile:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select **Mobile Hotspot > Mobile Profiles**.
   The Mobile Profiles page displays.
5. Select the check box of a mobile profile, and click the **Delete** button.
   The Delete Mobile Profile pop-up window displays.
6. Click the **Delete** button.
   A pop-up window displays to advise that the mobile profile is successfully deleted.

Configure a geofence for a mobile hotspot

Geofencing lets you define a circular geographic boundary for a location. The geofencing logic runs in the cloud and mobile hotspot devices report only their GPS coordinates (latitude and longitude). When a device crosses the boundary in either direction, an alarm is raised and a notification is sent in the Insight Cloud Portal.

Activating geo-fencing for a location automatically activates GPS on all mobile hotspot devices in that location. Deactivating geo-fencing deactivates GPS on all devices in the location. Only one geofence is supported in a location.

NOTE: By default, GPS and geo-fencing are deactivated. Activating this feature requires your explicit consent to the collection and processing of location data in accordance with GDPR requirements. Location data is encrypted using AES-256 and transmitted over TLS 1.2 or higher.

NOTE: Geofencing is supported on the following models only: Nighthawk M6 Pro, Nighthawk M7, Nighthawk M7 Pro, Nighthawk M8 Pro, and Nighthawk NS300.

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Click **Mobile Hotspot > Geofence.**
   The Geofence page displays. The map supports zoom and pan to help you locate the exact position of the geo-fence.
5. Read the terms and conditions, select the check box, and click the **Enable Geofence** button.
   The Enable Device GPS pop-up window displays.
6. Click the **Yes** button.
   The Geofence map displays.
7. To set the geofence location, click the **Edit** button
8. Choose a location address or enter a latitude longitude.
9. Click the **Confirm** button.

10. **(Optional)** To configure custom geofence settings, click the **Geofence Settings** button.
    a. In the **Geofence Name** field, update the name.
    b. To configure the fence color, select a color.
    c. In the **Radius** section, adjust the radius size (between 100m – 100000m).
    d. Activate or deactivate the **Notify when device enters geofence** toggle.
    e. Activate or deactivate the **Notify when device leaves geofence** toggle.
    f. To delete the geo-fence, click the **Delete Fence** link, and click the **Delete** button.
    g. Click the **Save** button.

    Geofencing is now active for the location. GPS is activated on all mobile hotspot devices in the location, and an alert is raised in the Insight Cloud Portal whenever a device crosses the geo-fence boundary in either direction. You can click any device on the map to view its name, serial number, device status, and signal strength.

Next steps

To deactivate geofencing at any time, return to **Settings > Geo-fencing**, and deactivate the **Enable Geofencing** toggle. GPS deactivates on all mobile hotspot devices in the location.

Manage port forwarding

By default, a mobile hotspot acts as a firewall. Unsolicited incoming traffic from the internet cannot reach devices on the local network. Port forwarding creates an exception to this rule by mapping a defined port or port range on the hotspot's public facing interface to a specific internal IP address. This allows external clients to initiate connections to a chosen device behind the hotspot.

Common use cases include hosting a game server, enabling remote desktop access, running a local web server, or supporting peer-to-peer applications that require inbound connections.

Administration Mode

You can manage port forwarding rules by the following access modes:

Admin Enforced: Only the device administrator can create, edit, or delete port forwarding rules. Connected users cannot modify the configuration. This is the recommended setting for managed or enterprise deployments.

Customization Allowed: Connected users can define their own port forwarding rules in addition to any rules set by the administrator.

Security Considerations

Opening ports through port forwarding exposes the target device directly to internet traffic on those ports. To minimize security risk, only forward ports that are strictly necessary, ensure the target device runs up to date software, and use Admin Enforced mode to prevent unauthorized rule changes.

Activate port forwarding

Before you begin

You must have administrator credentials for the device. Ensure you have identified the local IP address of the device you want to receive forwarded traffic before enabling this feature.

Activate port forwarding on the hotspot to allow inbound internet traffic to reach devices on the local network.

To activate port forwarding:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the mobile hotspot.
   The mobile hotspot dashboard displays.
5. Click **Settings > Port Forwarding**.
6. Do one of the following:
   - Select the **Admin Enforced** radio button to restrict rule management to administrators only.
   - Select the **Customization Allowed** radio button to permit connected users to add their own rules.
7. Click the **Enable Port Forwarding** toggle so that it is positioned to the right.
   Port forwarding is now active. Any rules defined in the port forwarding rule table are applied to incoming traffic. If no rules have been configured, no traffic is forwarded until rules are added.

Add a port forwarding rule

Create a port forwarding rule to direct incoming traffic on a specified port range to a chosen device on the local network.

To add a port forwarding rule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.

2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the mobile hotspot.
   The mobile hotspot dashboard displays.
5. Click **Settings > Port Forwarding > Add**.
   The Add Service pop-up page displays.
6. In the **Rule Name** field, enter a name for the rule.
7. In the **IP Address** field, enter the local IP address of the target device.
8. From the **Port Type** menu, select one of the following:
   - **Single**: Forward a single external port to a single internal port on the target device.
   - **Range**: Forward a range of external ports to a corresponding range of internal ports on the target device.
9. In the **Port** field, enter the port number or the first port in the range to be forwarded.
10. From the **Protocol** menu, select one of the following protocols:
    - TCP: For connection-based traffic such as web browsing and remote desktop.
    - UDP: For low-latency traffic such as gaming or streaming.
11. Click the **Save** button.
    Your settings are saved.

    The new rule displays in the port forwarding rule table and becomes active immediately, provided port forwarding is activated. Incoming traffic on the specified port range is directed to the target device on the local network.

Manage access control

In the Insight Cloud Portal, access control for mobile hotspots determines whether the administrator centrally manages device settings or allows end users to make their own customizations.

You can manage access control by the following access control modes:

> **Admin Enforced**
>
> The Insight administrator controls the mobile hotspot configuration. Settings pushed from Insight take precedence and cannot be changed by the device user. This mode is recommended when consistent, centrally managed hotspot behavior is required across the organization. Select this access control mode to ensure all managed mobile hotspots maintain

a consistent configuration.

Customization Allowed

Device users are permitted to adjust hotspot settings on their own. The administrator may still configure defaults through Insight, but users can override them locally. This mode is suited to deployments where end-user flexibility is acceptable. Select this access control mode only when individual users need the ability to modify their own hotspot settings.

Configure access control settings

Before you begin

The Access Control mode must be set to **Admin Enforced** to manage these settings centrally. Use the access control settings to enable access control, select a filter type, and manage the services on your allow or block list.

To configure access control settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the mobile hotspot.
   The mobile hotspot dashboard displays.
5. Click **Settings > Access Control**.
6. Select one of the following access control modes radio buttons:
   - **Admin Enforced**: The administrator controls all access control settings. Device users cannot make changes.
   - **Customization Allowed**: Device users can modify access control settings locally.
7. Do one of the following:
   - Click the **Enable Access Control** toggle so that it is activated and positioned to the right.
   - Click the **Enable Access Control** toggle so that it is deactivated and positioned to the left.

   Traffic is filtered according to the selected filter type and configured services.

8. In the **Filter Type** section, select one of the following radio buttons to determine how traffic is filtered:
   - **Allow List**: Only traffic matching services on the list is permitted. All other traffic is blocked.
   - **Block List**: Traffic matching services on the list is blocked. All other traffic is permitted.

   Access control is configured for the mobile hotspot. Traffic is filtered according to the selected filter type and service list.

Add a device to an access control list

Add a device to the access control list for a mobile hotspot managed in NETGEAR Insight to control whether that device can access the network through the hotspot.

When access control is activated on a mobile hotspot, network access is determined by the filter type in use. If the filter type is set to Allow List, only devices you explicitly add are permitted to connect. If it is set to Block List, added devices are denied access while all others can connect. Each device entry requires a name and a MAC address to identify it on the network.

To add an access control rule:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the mobile hotspot.
   The mobile hotspot dashboard displays.
5. Click **Settings > Access Control**.
6. Configure the access control settings. For more information, see Configure access control settings on page 242.
7. Click the **Add Device** button.
   The Add Device pop-up page displays.
8. In the **Device Name** field, enter a name for the device.
9. In the **MAC Address** field, enter the MAC address of the device.
10. Click the **Save** button.
    Your settings are saved.

Edit device details in an access control list

You can update the details of a device in an access control list for a mobile hotspot at any time.

To edit device details in an access control list:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the mobile hotspot.
   The mobile hotspot dashboard displays.
5. Click **Settings > Access Control**.
6. From the table, select the check box of a device, and click the **Edit** button.
   The Edit Device pop-up page displays.
7. Update the device details.
8. Click the **Save** button.
   Your settings are saved.

Delete a device from an access control list

You can remove a device from an access control list when it is no longer needed.

To delete a device from an access control list:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the mobile hotspot.
   The mobile hotspot dashboard displays.
5. Click **Settings > Access Control**.

6. From the table, select the check box of a device, and click the **Delete** button.
   The Delete Device pop-up window displays.
7. Click the **Delete** button.
   The device is deleted and no longer displays in the table.

View LAN IP Settings

View the LAN IP address and subnet mask assigned to a mobile hotspot managed in NETGEAR Insight.

To view LAN IP settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the mobile hotspot.
   The mobile hotspot dashboard displays.
5. Click **Settings > LAN IP**.
   The LAN IP address page displays the view-only IP address and the subnet mask information.

View WiFi network settings

View the WiFi network settings and configured SSIDs for a mobile hotspot managed in Insight Cloud Portal.

To view WiFi network settings:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the mobile hotspot.
   The mobile hotspot dashboard displays.

5. Click **Settings > WiFi Networks**.
   The Settings section displays the following information:
   WiFi Mode: The frequency bands the hotspot broadcasts on, for example, Dual Band (2.4 GHz & 5 GHz).
   WiFi Range: The broadcast range setting of the hotspot, for example, Short.
   WPS: Whether WiFi Protected Setup is activated or deactivated.
   The WiFi networks table lists each configured SSID with the following details:

Column	Description
SSID Name	The name of the WiFi network.
Security	The security protocol applied to the network, for example, wpa2Wpa3.
Band	The frequency band the SSID broadcasts on, either 2.4 GHz or 5 GHz.
Channel	The wireless channel in use, for example, Auto.
Channel Width	The channel width used by the network, for example, VHT80 or HT20.

Diagnostics and Troubleshooting

You can access a set of diagnostic utilities in the Insight Cloud Portal. Whether you are checking basic connectivity, measuring internet speed, or tracing the path packets take across the network, each tool is designed to help you pinpoint the problem.

This chapter contains the following sections:

- Check the Internet speed
- Ping the IP address or domain name of a device or network location
- Look up a DNS domain name or IP address
- Trace a route
- Activate Secure Diagnostics mode

Check the Internet speed

To check the Internet speed:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Select **Troubleshooting > Ookla Speedtest**.
   The Speed Test page displays.
6. Click the **Run Speedtest** button.
   The Privacy Policy pop-up window displays.
7. Click the **Agree** button.
   After a short delay, the page displays the measured latency (delay) in ms, download speed in Mbps, and upload speed in Mbps.
8. Optional: To stop the speed test, click the **Stop** button.

NOTE: If you do not stop the speed test, the test stops by itself after about 40 seconds, and the name of the Test Speed button changes to Test Again.

9. To view the test history, click the **View History** link.
   A table shows the results of previous tests.

Ping the IP address or domain name of a device or network location

The ping test measures how long it takes for packets to travel from the device to a target IP address or hostname and back. Use this tool to confirm that a remote host is reachable and to measure round-trip latency.

Before running the test, you can adjust the following settings: ping count, packet size (in bytes), ping timeout (in seconds), and ping interval (in seconds).

To ping the IP address or domain name of a device or network location:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Select **Troubleshooting > Ping Test**.
   The Ping Test page displays.
6. In the **IP Address or Hostname** field, enter the device IP address.
7. Use the default settings or specify the settings in the following table.

Setting	Description
Ping Count	The number of pings that the router must send. The range is from 1 to 1024. The default number is 16.
Packet Size	The size of each ping packet. The range is from 4 to 1024. The default size is 64 bytes.
Ping Interval	The interval between pings. The range is from 1 to 10. The default interval is 1 second.
Ping Timeout	The period after which a ping times out. The range is from 1 to 300. The default period is 3 seconds.

8. Click the **Run Test** button.
9. To stop the ping test before the ping count is reached or if the ping times out, click the **Stop** button.
   The Ping Result section displays the results of your query.

Look up a DNS domain name or IP address

The DNS lookup tool queries the Domain Name System to resolve a hostname to its corresponding IP address. Use this tool to verify that DNS resolution is working correctly on your network and to identify misconfigured name server settings.

To look up a DNS domain name or IP address:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Select **Troubleshooting > DNS Lookup**.
   The DNS Lookup page displays.
6. In the **Domain Host** field, type the domain name or IP address for which you want to look up the DNS translation.
7. Click the **Test Now** button.
   The DNS Lookup Result section displays the results of your query.

Trace a route

Traceroute maps the network path that packets take from your device to a target host, showing each hop along the route and the latency at each step. Use this tool to identify exactly where delays or failures occur between your network and a remote destination.

To trace a route:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.

2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Select **Troubleshooting > Traceroute**.
   The Traceroute page displays.
6. In the **IP Address or Hostname** field, type the domain name or IP address for which you want to trace the route.
7. Click the **Run Test** button.
   The Traceroute Result section displays the results of your query.

Activate Secure Diagnostics mode

The Secure Diagnostics mode gives the NETGEAR Support team controlled, encrypted access to device logs and performance data. Activating this mode speeds up issue resolution without compromising network security.

To activate secure diagnostics mode:

1. Access the Insight Cloud Portal.
   For more information, see Log in to the Insight Cloud Portal on page 9.
   The Insight Dashboard displays.
2. Click the **Locations** tab.
   The Locations page displays.
3. Select the name of a location.
   The location dashboard displays.
4. Select the **Devices** tab, and click the name of the router.
   The router dashboard displays.
5. Select **Troubleshooting > Diagnostic**.
   The Diagnostic page displays.
6. Activate the **Secure Diagnostics Mode** toggle so it is positioned to right.
   The Secure Diagnostic Mode pop-up window displays.
7. Click the **OK** button.
8. Click the **Save** button.
   Your settings are saved.

Dashboard information and supported countries

This topic provides an overview of your device model, including key device information and the list of countries and regions where it is supported.

This chapter contains the following sections:

- Organization dashboard
- Location dashboard
- Device dashboards
- Countries with Insight support
- Supported Devices

Organization dashboard

Organizations list

To access organizations dashboards, select the **Organizations** tab and select an organization.

The Organizations list displays all organizations in your Insight environment. The following columns display for each organization:

Column	Description
Name	The name of the organization. Select the name to open the organization dashboard.
Total Devices	The total number of devices in the organization. Select the number to view individual devices.
Online Devices	The number of devices in the organization that are connected to Insight. Select the number to view individual devices.
Offline Devices	The number of devices in the organization that are disconnected from Insight. Select the number to view individual devices.
Subscription Status	The subscription status for the organization. Displays **All Active** when all subscriptions are current, or the number of expired subscriptions when one or more have lapsed.

The following controls are available in the Organizations list:

Control	Description
Add	Creates a new organization.

Control	Description
Edit	Edits the selected organization.
Delete	Deletes the selected organization.
Search	Filters the list by organization name.
Three-dot menu	Displays options to edit or delete an individual organization.

General summary section

The **General Summary** section of the organization dashboard displays the following tiles:

Field	Values	Description
Devices	Online, Offline, Critical	Shows the overall status of all devices in the organization. Select the number in the **Online**, **Offline**, or **Critical** field to view the status of individual devices. A device displays as Critical when it is offline or experiencing issues like connectivity, restarting, IP conflicts, or similar problems.
Subscription Status	Active, In Grace, Expiring Soon	Shows the overall status of all subscriptions in the organization. For more information about Insight subscriptions, see Manage Insight subscriptions on page 24.
Alarms	Online, Warning	Shows the overall alarm status for the organization. Select the number in the **Online** or **Warning** field to view individual alarms.

Device summary section

The **Device Summary** section displays the status of each device type in the organization:

Field	Values	Description
Switches	Online, Offline, Critical	Shows the total number of switches in the organization and their connection status.
Routers	Online, Offline, Critical	Shows the total number of routers in the organization and their connection status.
Access Points	Online, Offline, Critical	Shows the total number of access points in the organization and their connection status.
MHS	Online, Offline, Critical	Shows the total number of mobile hotspots in the organization and their

Wireless summary section

The **Wireless Summary** section displays the following views:

View	Description
WiFi Distribution	Displays a breakdown of connected wireless clients by frequency band: 2.4 GHz, 5 GHz, and 6 GHz.
Connected Client Trend	Displays a graph of connected client counts over time, broken down by frequency band. Use the menu in the top right of this section to filter by time period.
Top APs by Traffic	Lists access points in the organization ranked by traffic usage. Use the menu in the top right of this section to filter by time period.
Top SSIDs by Traffic	Displays a graph of SSID traffic usage over time, broken down by SSID. Use the menu in the top right of this section to filter by time period.

Wired summary section

The **Wired Summary** section displays the following views:

View	Description
Top Switches by Uplink Utilization	Lists switches in the organization ranked by uplink utilization. Use the menu in the top right of this section to filter by time period.
Top Switches by Port Errors	Lists switches in the organization ranked by the number of port errors. Use the menu in the top right of this section to filter by time period.
Top Switches by PoE Utilization	Lists switches in the organization ranked by PoE utilization. Use the menu in the top right of this section to filter by time period.
Top Routers by Traffic	Displays a graph of router traffic over time. Use the menu in the top right of this section to filter by time period.
Top Router WAN Port Errors	Lists routers in the organization ranked by the number of WAN port errors. Use the menu in the top right of this section to filter by time period.

Location summary section

The **Location Summary** section displays the following views:

View	Description
Top Locations by Traffic	Lists locations in the organization ranked by traffic usage. Use the menu in the top right of this section to filter by time period.

View	Description
Top Locations by Clients	Lists locations in the organization ranked by the number of connected clients. Use the menu in the top right of this section to filter by time period.
Top Locations by Critical Alarms	Lists locations in the organization ranked by the number of critical alarms. Use the menu in the top right of this section to filter by time period.
Top Locations by Devices	Lists locations in the organization ranked by total device count. Use the menu in the top right of this section to filter by time period.

Location dashboard

Locations list

To access location dashboards, select the **Locations** tab and select a location.

The Locations list displays all locations in your Insight environment.

The following table displays the information that displays on a location dashboard in the Insight Cloud Portal:

Field	Description
Name	The name of the location.
Organization	The organization that the location is part of.
Total Devices	The total number of devices in the location.
Online Devices	The total number of devices in the location that are connected to Insight.
Offline Devices	The total number of devices in the location that are disconnected from Insight.
Subscription Status	The status of the Insight subscription for the location.
Connected Clients	The total number of clients connected to Insight-managed devices in the location.

When select a location from the Locations table, the following sections display:

Network summary section

The Network Summary section of the dashboard displays the following tiles:

Field	Values	Description
Devices	Online, Offline, Critical	Shows the overall status of all

Field	Values	Description
		devices in the location. Select the number in the **Online**, **Offline**, or **Critical** field to view the status of individual devices. A device displays as Critical when it is offline or experiencing issues like connectivity, restarting, IP conflicts, or similar problems.
Subscription Status	Active, In Grace, Expiring Soon	Shows the overall status of all subscriptions in the location. For more information about Insight subscriptions, see Manage Insight subscriptions on page 24.
Alarms	Online, Warning	Shows the overall alarm status for the location. Select the number in the **Online** or **Warning** field to view individual alarms.

Device Summary section

The Device Summary section displays the status of each device type in the location:

Field	Values	Description
Switches	Online, Offline, Critical	Shows the total number of switches in the location and their connection status.
Routers	Online, Offline, Critical	Shows the total number of routers in the location and their connection status.
Access Points	Online, Offline, Critical	Shows the total number of access points in the location and their connection status.
MHS	Online, Offline, Critical	Shows the total number of mobile hotspots in the location and their connection status.

Wireless Summary section

The Wireless Summary section displays the following views:

View	Description
WiFi Distribution	Displays a breakdown of connected wireless clients by frequency band: 2.4 GHz, 5 GHz, and 6 GHz.
Connected Client Trend	Displays a graph of connected client counts over time, broken down by frequency band. Use the menu in the top right of this section to filter by time period.
Top APs by Traffic	Lists access points ranked by traffic usage. Use the menu in the top right of this section to filter by time period.
Top SSIDs by Traffic	Lists SSIDs ranked by traffic usage. Use the menu in the top right of this section to filter by time period.

Wired Summary section

The Wired Summary section displays the following views:

View	Description
Top Switches by Uplink Utilization	Lists switches ranked by uplink utilization. Use the menu in the top right of this section to filter by time period.
Top Switches by Port Errors	Lists switches ranked by the number of port errors. Use the menu in the top right of this section to filter by time period.
Top Switches by PoE Utilization	Lists switches ranked by PoE utilization. Use the menu in the top right of this section to filter by time period.
Top Routers by Traffic	Displays a graph of router traffic over time. Use the menu in the top right of this section to filter by time period.
Top Router WAN Port Error	Lists routers ranked by the number of WAN port errors. Use the menu in the top right of this section to filter by time period.

Device dashboards

When you add a device to the Insight Cloud Portal, you can view device details and their activity summary. To access device dashboards, select **Dashboard > Devices**, and select the individual device name.

Device Details

The following table displays individual device details as they are configured in the Insight Cloud Portal:

Field	Description
Device Name	The name of the device.
Serial Number	The serial number of the device. This is normally listed on the device label.
MAC Address	The MAC address of the device. This is normally listed on the device label.
Model	The model number of the device. This is normally listed on the device label.
Up Time	The duration that the device has been online without restarting.
Status	Displays if the device is online, offline, restarting, unmanaged, or waiting for first connect.
Last Seen	Displays when the most recent instance the device was detected on the network.
IP Address	The IP address of the device. This is normally listed on the device label.
Firmware Version	The firmware version of the device. To find the latest firmware version for the device, visit http://www.netgear.com/support.
Admin Password	Displays the device admin password. To view this password, click on the eye icon, enter your Insight password, and click **OK**. This determines if you have the necessary admin rights to view the device password.

Depending on device type, other fields may display in the Summary tab.

The following table displays these additional fields:

Table 4.Additional device fields

Field	Description
Routers:	
WAN Mode	Displays whether the device is using WAN1, WAN2, or Dual WAN.
WAN IP Address	Displays the public IP address your ISP assigns to the device.
Temperature	Displays the temperature of the device.
Fan	Displays the status of the fan.
VLAN	Displays the number of VLANs configured on the device.
Connected Clients	Displays the number of devices that are connected to your Insight environment. This count includes both wired and wireless clients and updates

Field	Description
	automatically as devices join or leave the network.
IPSec Site-to-Site	Displays the total number of active IPSec Site-to-Site VPN tunnels currently established on the device. This count reflects how many remote networks are securely connected using encrypted IPSec tunnels and updates automatically as connections go up or down.
IPSec Clients	Displays the number of active IPSec Client VPN connections currently established on the device. This count reflects how many individual users are securely connected using IPSec remote-access VPN and updates automatically as clients connect or disconnect.
OpenVPN Clients	Displays the number of active OpenVPN Client connections currently established on the device. This count represents how many remote users are securely connected using OpenVPN and updates automatically as clients connect or disconnect.
WireGuard Clients	Displays the number of active WireGuard Client connections currently established on the device. This count shows how many remote users are securely connected using the WireGuard VPN protocol and updates automatically as clients connect or disconnect.
Switches: Smart Switches and Fully Managed Switches (M4250 and M4350 only)	
Date & Time	Displays the date and time in the configured timezone of the switch.
Bootcode Version	Displays the bootcode version running on the switch.
Mobile Hotspots	
Battery Level	Displays the current battery charge level of the mobile hotspot.
Traffic	Displays the amount of mobile data used by the hotspot over a selected period, including upload and download usage.
Country/Region	Displays the country or region where the mobile hotspot is operating, which can affect available bands and roaming behavior.
Clients	Displays the total number of devices currently connected to the mobile hotspot across all connection types.
2.4 GHz Clients	Displays the number of clients connected to the 2.4 GHz WiFi band on the mobile hotspot.
5 GHz Clients	Displays the number of clients connected to the 5 GHz WiFi band on the mobile hotspot.
6 GHz Clients	Displays the number of clients connected to the 6 GHz WiFi band on the

Field	Description
	mobile hotspot, if supported.
Mobile Number	Displays the mobile phone number associated with the SIM card installed in the hotspot, if provided by the carrier.
USB Clients	Displays the number of devices connected to the mobile hotspot through the USB port.
LAN Clients	Displays the number of devices connected to the mobile hotspot through the LAN (Ethernet) port.

NOTE: Insight-managed access points do not display any additional fields.

Router Dashboard

In addition to the information listed in Device Details, the following information displays in the Summary tab of a router:

Field	Description
Ports	Displays what ports are connected and disconnected. For more information, click the individual port.
Traffic Summary	Displays an overview of the total data flowing through the router, including upload and download activity across all interfaces.
Speed	Displays the LAN port speeds in Kbps. You can filter this section by each LAN and duration.
Client	Displays connected clients to the router. You can filter this section by selecting the VLAN and duration.
Connectivity Status	Displays when the router is connected or disconnected. You can filter this by duration.
Network Usage	Displays the network usage on each WAN in Kbps.

Switch Dashboard

In addition to the information in the Device Details section, the following information displays in the Summary tab of a switch. This information is applicable to smart switches and fully managed switches (M4250 series and M4350 series only).

Field	Description
Ports	Displays port information. The following statuses can display on a switch port:

Field	Description
	Available, Assigned, Connected and Powered, Disabled, Error, PoE, PoE Disabled, LAG, 1G SFP Fiber Port, 10G SFP+ Fiber Port, 25G SFP28 Fiber Port, 100G QSFP Fiber Port, 50G QSFP Fiber Port, Uplink Port, Authorized, Unauthorized, Admin Down, Block, Auto-Trunk, Always Active, Protected port. For more information, click the individual port.
Statistics	• **Current Power Consumption**: Displays the available power and consumed power. • **Performance Stats**: Displays the temperature, CPU Usage, Fan State, Tx Data, and Rx Data.
Traffic	Displays the traffic usage of the switch. This details the amount of data flowing through the switch's ports, including both incoming and outgoing traffic.
VLANs in use	Displays the VLANs that are in use on the switch. The following fields display: Name, VLAN ID, Network Type, and Secondary VLAN Type. Click the three dots on the right of each VLAN to edit or delete the VLAN.

Access Point Dashboard

In addition to the information in the Device Details section, the following information displays in the Summary tab of an access point.

Field	Description
Client OS	Displays the status of each wireless radio, including whether it is enabled and the band in use.
Channel Utilization	Displays how much of the wireless channel capacity is currently in use. You can filter this by frequency and radio band.
Statistics	Displays an overview of all configured wireless networks, including the number of connected clients and their traffic usage.

Mobile Hotspot Dashboard

In addition to the information in the Device Details section, the following information displays in the Summary tab of a mobile hotspot.

Field	Description
Traffic	Displays an overview of monthly usage and daily or weekly traffic usage.
Device Details	Displays an overview of device functions.

Field	Description
Network	Displays the current cellular network status, signal measurements, radio band information, and network identification parameters.
Cellular Network	Displays information about the active cellular connection, including network status, operator details, cellular technology, connection uptime, transmitted and received data, and SIM identification information.

Countries with Insight support

This section lists the countries that are supported in Insight, including their ISO and country codes, wireless region, subscription payment availability, and LMS key support. If your country is not supported in Insight, you must purchase a subscription from a third party vendor.

Country Name	ISO Code	Country Code	Wireless Region	Annual subscriptions	Monthly subscriptions	LMS Key supported
Afghanistan	4	AF	No	No	No	Yes
Albania	8	AL	Yes	No	No	Yes
Algeria	12	DZ	Yes	No	No	Yes
Andorra	20	AD	No	No	No	Yes
Argentina	32	AR	Yes	No	No	Yes
Armenia	51	AM	Yes	No	No	Yes
Australia	5000	AU	Yes	Yes	Yes	Yes
Austria	40	AT	Yes	Yes	Yes	Yes
Azerbaijan	31	AZ	Yes	No	No	Yes
Bahamas	44	BS	No	No	No	Yes
Bahrain	48	BH	Yes	No	No	Yes
Bangladesh	50	BD	Yes	No	No	Yes
Belgium	56	BE	Yes	Yes	Yes	Yes
Belize	84	BZ	Yes	No	No	Yes
Belarus	112	BY	Yes	No	No	Yes
Bolivia	68	BO	Yes	No	No	Yes
Bosnia &	70	BA	Yes	No	No	Yes

Country Name	ISO Code	Country Code	Wireless Region	Annual subscriptions	Monthly subscriptions	LMS Key supported
Herzegovina						
Brazil	76	BR	Yes	No	No	Yes
Brunei Darussalam	96	BN	Yes	No	No	Yes
Bulgaria	100	BG	Yes	Yes	Yes	Yes
Canada	124	CA	Yes	Yes	Yes	Yes
Chile	152	CL	Yes	No	No	Yes
China	156	CN	Yes	No	No	Yes
Colombia	170	CO	Yes	No	No	Yes
Costa Rica	188	CR	Yes	No	No	Yes
Croatia	191	HR	Yes	Yes	Yes	Yes
Cyprus	196	CY	Yes	Yes	Yes	Yes
Czech Republic	203	CZ	Yes	Yes	Yes	Yes
Denmark	208	DK	Yes	Yes	Yes	Yes
Dominican Republic	214	DO	Yes	No	No	Yes
Ecuador	218	EC	Yes	No	No	Yes
Egypt	818	EG	Yes	No	No	Yes
El Salvador	222	SV	Yes	No	No	Yes
Estonia	233	EE	Yes	Yes	Yes	Yes
Finland	246	FI	Yes	Yes	Yes	Yes
France	250	FR	Yes	Yes	Yes	Yes
French Guiana	254	GF	No	No	No	Yes
French Polynesia	258	PF	No	No	No	Yes
Georgia	268	GE	Yes	No	No	Yes
Germany	276	DE	Yes	Yes	Yes	Yes

Country Name	ISO Code	Country Code	Wireless Region	Annual subscriptions	Monthly subscriptions	LMS Key supported
Gibraltar	292	GI	No	No	No	Yes
Greece	300	GR	Yes	Yes	Yes	Yes
Greenland	304	GL	No	No	No	Yes
Guadeloupe	312	GP	No	No	No	Yes
Guam	316	GU	No	No	No	Yes
Guatemala	320	GT	Yes	No	No	Yes
Honduras	340	HN	Yes	No	No	Yes
Hong Kong	344	HK	Yes	Yes	Yes	Yes
Hungary	348	HU	Yes	Yes	Yes	Yes
Iceland	352	IS	Yes	No	No	Yes
India	356	IN	Yes	No	No	Yes
Indonesia	360	ID	Yes	No	No	Yes
Iran	364	IR	No	No	No	Yes
Iraq	368	IQ	Yes	No	No	Yes
Ireland	372	IE	Yes	Yes	Yes	Yes
Israel	376	IL	Yes	No	No	Yes
Italy	380	IT	Yes	Yes	Yes	Yes
Jamaica	388	JM	Yes	No	No	Yes
Japan	392	JP	Yes	Yes	Yes	Yes
Jordan	400	JO	Yes	No	No	Yes
Kazakhstan	398	KZ	Yes	No	No	Yes
Kenya	404	KE	Yes	No	No	Yes
Korea Republic	410	KR	Yes	No	No	Yes
Kuwait	414	KW	Yes	No	No	Yes
Latvia	428	LV	Yes	Yes	Yes	Yes
Lebanon	422	LB	Yes	No	No	Yes

Country Name	ISO Code	Country Code	Wireless Region	Annual subscriptions	Monthly subscriptions	LMS Key supported
Liechtenstein	438	LI	No	No	No	Yes
Lithuania	440	LT	Yes	Yes	Yes	Yes
Luxembourg	442	LU	Yes	Yes	Yes	Yes
Macau	446	MO	Yes	No	No	Yes
Macedonia	807	MK	Yes	No	No	Yes
Malaysia	458	MY	Yes	No	No	Yes
Malta	470	MT	Yes	Yes	Yes	Yes
Martinique	474	MQ	No	No	No	Yes
Mauritius	480	MU	No	No	No	Yes
Mexico	484	MX	Yes	No	No	Yes
Moldova	498	MD	No	No	No	Yes
Monaco	492	MC	Yes	No	No	Yes
Montenegro	499	ME	Yes	No	No	Yes
Morocco	504	MA	Yes	No	No	Yes
Netherlands	528	NL	Yes	Yes	Yes	Yes
Netherlands Antilles	530	AN	Yes	No	No	Yes
New Zealand	554	NZ	Yes	Yes	Yes	Yes
Nicaragua	558	NI	Yes	No	No	Yes
Nigeria	566	NG	No	No	No	Yes
Norway	578	NO	Yes	Yes	Yes	Yes
Oman	512	OM	Yes	No	No	Yes
Pakistan	586	PK	No	No	No	Yes
Panama	591	PA	Yes	No	No	Yes
Paraguay	600	PY	Yes	No	No	Yes
Peru	604	PE	Yes	No	No	Yes
Philippines	608	PH	Yes	No	No	Yes

Country Name	ISO Code	Country Code	Wireless Region	Annual subscriptions	Monthly subscriptions	LMS Key supported
Poland	616	PL	Yes	Yes	Yes	Yes
Portugal	620	PT	Yes	Yes	Yes	Yes
Puerto Rico	630	PR	Yes	No	No	Yes
Qatar	634	QA	Yes	No	No	Yes
Reunion	638	RE	No	No	No	Yes
Romania	642	RO	Yes	Yes	Yes	Yes
Russia	643	RU	Yes	No	No	Yes
San Marino	674	SM	No	No	No	Yes
Saudi Arabia	682	SA	Yes	No	No	Yes
Serbia	688	RS	Yes	No	No	Yes
Singapore	702	SG	Yes	Yes	Yes	Yes
Slovakia	703	SK	Yes	Yes	Yes	Yes
Slovenia	705	SI	Yes	Yes	Yes	Yes
South Africa	710	ZA	Yes	Yes	Yes	Yes
Spain	724	ES	Yes	Yes	Yes	Yes
Sri Lanka	144	LK	Yes	No	No	Yes
Sweden	752	SE	Yes	Yes	Yes	Yes
Switzerland	756	CH	Yes	Yes	Yes	Yes
Taiwan	158	TW	Yes	No	No	Yes
Thailand	764	TH	Yes	No	No	Yes
Trinidad and Tobago	780	TT	Yes	No	No	Yes
Tunisia	788	TN	Yes	No	No	Yes
Turkey	792	TR	Yes	No	No	Yes
UAE	784	AE	Yes	No	No	Yes
Ukraine	804	UA	Yes	No	No	Yes
UK	826	GB	Yes	Yes	Yes	Yes

Country Name	ISO Code	Country Code	Wireless Region	Annual subscriptions	Monthly subscriptions	LMS Key supported
United States of America	840	US	Yes	Yes	Yes	Yes
United States Virgin Island	850	VI	No	No	No	Yes
Uruguay	858	UY	Yes	No	No	Yes
Vatican City	336	VA	No	No	No	Yes
Venezuela	862	VE	Yes	No	No	Yes
Vietnam	704	VN	Yes	No	No	Yes
Yemen	887	YE	Yes	No	No	Yes
Zimbabwe	716	ZW	Yes	No	No	Yes

Supported Devices

Table 5.Supported devices

Product	Set up	Manage	Monitor	Discover	Register
Insight-managed smart switches	X	X	X	X	X
Insight-managed access points	X	X	X	X	X
Insight-managed routers (PR60X, PR460X, PR30SG)*	X	X	X	X	X
Insight-managed mobile hotspots*	X	X	X	X	X
Orbi Pro WiFi systems	X	X	X	X	X
Easy Managed Smart Switches			X	X	X
Fully Managed Switches (M4250, M4350)	X	X	X	X	X
Fully Managed Switches (M4300, M4500)				X	X
Unmanaged switches					X

NOTE: For more information on supported mobile hotspot models, see Which mobile routers are compatible with NETGEAR Insight.

NOTE: If you manage a device in Insight, then it counts towards your total devices on your Insight subscription. If you can only discover, register, and monitor a device in Insight, it does not count toward your device total for your Insight subscription. For information about pricing, see insight.netgear.com and the NETGEAR Insight knowledge base at netgear.com/support/product/insight.